Beyond Isolated Fixes: A Comprehensive Survey on Hallucination Mitigation with a Three-Dimensional Taxonomy and Integrative Framework

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***Abstract*—Hallucinations in large language and vision–language models constitute a persistent and systemic failure mode that fundamentally undermines their reliability in high-stakes applications. Rather than framing hallucination as a localized generation error, this paper reconceptualizes it as a lifecycle phenomenon—emerging from upstream deficiencies in data collection, reinforced by training-time biases, and exacerbated by inference-time decoding pathologies. To address this challenge, we propose a unified framework for hallucination mitigation grounded in a structured three-dimensional taxonomy that spans mechanism-based strategies, phase-specific interventions, and cross-phase integrative approaches. This taxonomy elucidates how technical solutions, knowledge-enhanced methods, and framework-level designs can be systematically orchestrated across pre-, in-, and post-generation stages to support more robust, interpretable, and auditable model behavior. Within this paradigm, detection is positioned not as a terminal task, but as a diagnostic backbone that enables adaptive and multi-stage mitigation workflows. Building on this foundation, we introduce an integrative hybrid framework that consolidates previously fragmented techniques into coherent pipelines, demonstrating how cross-phase synergy and iterative refinement can enhance reliability in real-world deployment. This framework further reveals latent connections across diverse strands of research, facilitating the identification of underexplored intersections and the development of synergistic method combinations. By reframing hallucination mitigation as a system-level challenge rather than a set of isolated fixes, this work offers both a theoretical lens and a practical roadmap for advancing the trustworthiness of generative AI systems.**

Keywords*—*large language models, large vision–language models, AI hallucination, hallucination detection, hallucination mitigation, three-dimensional taxonomy, integrative framework, generative AI reliability.

I. Introduction

The emergence of large language models (LLMs) such as GPT, LLaMA, and Mistral has triggered a paradigm shift in the field of natural language processing (NLP), achieving unprecedented progress in question answering, summarization, dialogue, and complex reasoning generation. Their rapid deployment has reshaped digital ecosystems, transforming how individuals search for information, interact with intelligent systems, and even conduct scientific inquiry. As of August 2025, OpenAI announced that ChatGPT had surpassed 700 million weekly active users, marking a fourfold increase

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compared to the same period last year and solidifying its status as one of the fastest-adopted software products in history [1].

Yet, beneath this rapid expansion lies a critical vulnerability: LLMs, while linguistically fluent and contextually adaptive, frequently produce outputs that are factually inaccurate, unverifiable, or entirely fabricated—phenomena collectively referred to as *hallucinations* [2]. Unlike conventional software bugs, hallucinations are systemic manifestations of data bias, training misalignments, and reasoning deficits inherent to generative models [3], [4]. Their consequences extend far beyond academic debates. For instance, a recent empirical study reported that DeepSeek exhibited an alarming inaccuracy rate of 91.43% when generating scientific references [5], illustrating how misleading outputs can undermine trust in AI and exacerbate the risks of misinformation. These examples highlight a pressing tension: as LLMs become indispensable infrastructures of knowledge production, their susceptibility to hallucination threatens to erode both reliability and societal trust.

In response to these challenges, this paper provides a systematic and integrative review of hallucination mitigation in large language models (LLMs) and large vision-language models (LVLMs). We survey the landscape of hallucination research from approximately 2018 to 2025, covering both foundational models (GPT, LLaMA, Mistral) and multimodal architectures, with a focus on English-language systems while acknowledging multilingual considerations where relevant. Beyond outlining the underlying mechanisms and problems that give rise to hallucinations, we synthesize advances in detection and mitigation techniques and examine evaluation practices that benchmark their effectiveness. Central to our contribution is a three-dimensional taxonomy that organizes mitigation strategies by mechanism, temporal phase, and cross-phase integration. Building on this taxonomy, we further propose an integrative hybrid framework that demonstrates how diverse methods can be orchestrated to overcome the limitations of isolated approaches. By offering both a structured survey and a forward-looking synthesis, this study aims to guide researchers, practitioners, and policymakers toward more reliable and trustworthy deployment of generative AI.

To fully appreciate the urgency and complexity of the hallucination problem, we first examine concrete instances where these failures have manifested in critical applications.

A. Real-World Representative Cases of Hallucinations

While much of the discussion on hallucinations has been grounded in benchmark evaluations or controlled experiments, their deployment in real-world environments has revealed far more consequential risks. Across domains ranging from law and healthcare to finance and technical systems, AI-generated outputs that appear coherent yet are factually incorrect have resulted in professional sanctions, measurable economic losses, and, in some cases, direct threats to human well-being. These incidents demonstrate that hallucinations are not isolated anomalies but systemic vulnerabilities, exposing structural weaknesses in both technical architectures and institutional safeguards. Understanding their societal impact through representative cases is therefore essential for motivating the urgency of robust detection and mitigation strategies.

1) Technical Performance Degradation in Advanced Models: Despite rapid architectural advances, recent evidence suggests that improvements in reasoning capacity and efficiency do not necessarily translate into reduced hallucination rates. For instance, OpenAI's o3 and o4-mini models, released in April 2025 with enhanced reasoning mechanisms, still demonstrated hallucination rates of 33% and 48%, respectively [6], while DeepSeek-R1 reported a hallucination rate of 14.3%, substantially higher than the 3.9% observed in its predecessor DeepSeek-V3 [7]. Further empirical analysis shows that advanced models often augment otherwise accurate responses with unsupported or spurious details [8]. These demonstrate that increasing model sophistication can inadvertently diversify hallucination modalities rather than eliminate them, underscoring the need for mitigation strategies specifically designed to address emergent error patterns rather than relying solely on scaling or incremental architectural refinement.

2) Legal Field: The legal sector has experienced significant disruptions due to AI hallucinations, with multiple documented cases leading to professional sanctions and procedural reforms. In the United States District Court for Wyoming, attorneys from Morgan & Morgan submitted a brief containing nine non-existent case citations generated by an internal AI platform, prompting potential Rule 11 sanctions and firm-wide policy revisions on AI usage [9], [10]. Similarly, the Central District of California imposed $31,100 in monetary penalties on attorneys from Ellis George LLP and K&L Gates LLP after they submitted briefs with fabricated citations [11], while international incidents such as the Ko v. Li case in Toronto saw counsel face contempt proceedings for citing non-existent precedents [11]. Collectively, these cases reveal systemic vulnerabilities in the legal profession's adoption of AI, particularly the lack of robust verification protocols and accountability mechanisms. Given that legal practice fundamentally depends on the precision of precedent and citation, the profession is uniquely susceptible to the risks of AI-generated misinformation, making hallucination mitigation an urgent prerequisite for responsible adoption.

3) Economic Impact Assessment: AI hallucinations have generated substantial economic consequences across industrial sectors, establishing new precedents for corporate liability and operational risk. In Air Canada v. Moffatt, the Civil Resolution Tribunal ordered $812 CAD in damages after the airline's chatbot incorrectly promised retroactive bereavement fare discounts, a ruling that underscored corporate accountability for autonomous AI agent representations [12]. The financial services industry demonstrates particular vulnerability, as AI systems have been documented producing fabricated revenue figures, inaccurate stock prices, and even nonexistent regulatory requirements [13], while document processing platforms have misreported critical data such as stock split ratios, thereby creating significant compliance and decision-making risks [13]. At the aggregate level, industry surveys estimate that organizations spend an average of $14,200 annually per employee on hallucination detection and correction, with global productivity losses attributable to verification demands reaching $67.4 billion in 2024 [14]. These findings indicate that hallucinations impose not merely incidental operational costs but systemic inefficiencies that erode the anticipated productivity gains of AI adoption. Such inefficiencies directly contradict the expected productivity benefits of AI integration.

4) Medical and Healthcare Scenarios: The medical domain represents one of the highest-risk areas for AI hallucinations [15], with documented failures spanning transcription, diagnosis, and treatment recommendation systems. OpenAI's Whisper model, deployed through Nabla for medical conversation transcription, has exhibited systematic hallucination patterns such as fabricated diagnoses and misattributed statements [16], with error rates disproportionately affecting vulnerable patient populations and raising concerns regarding healthcare equity and patient safety [17]. Diagnostic and treatment recommendation systems have also demonstrated severe vulnerabilities: research conducted at Brigham and Women's Hospital reported ChatGPT producing contraindicated cancer treatment suggestions, including curative protocols for terminal conditions [18], thereby exposing patients to the risk of inappropriate therapeutic pathways during critical decision-making. Further analysis by researchers at Mendel and UMass Amherst identified so-called "faithfulness hallucinations" in medical documentation generated by GPT-4o and Llama-3 [19], in which inaccurate or fabricated details appeared across key sections of clinical notes—such as patient information, findings, and diagnoses—highlighting structural vulnerabilities and the urgent need for robust detection mechanisms [20]. While current ethical guidelines emphasize AI's role as a decision-support tool rather than an autonomous decision-maker, persistent gaps between normative frameworks and real-world implementation highlight the urgent need for robust safeguards in clinical AI deployment [21].

5) Cross-Domain Analysis and Implications: A persistent challenge arises from the overconfidence of LLMs, as they often produce factually incorrect responses with high confidence, making it difficult for users to distinguish between accurate and hallucinated information [22]. At the same time, domain-specific manifestations—such as fabricated citations in law, erroneous stock data in finance, or unsafe treatment recommendations in medicine—underscore that mitigation strategies cannot be one-size-fits-all but must instead be tailored to sector-specific requirements and risk profiles. Compounding this is the problem of *verification cost*: the resources required to detect and correct hallucinations often far exceed the cost of generating them, resulting in negative returns on AI investment in high-stakes contexts [23]. Collectively, these cross-domain patterns highlight hallucinations as a systemic rather than localized phenomenon, necessitating multi-layered approaches that integrate technical safeguards, institutional protocols, and regulatory oversight to ensure reliability and trust in real-world deployments.

These representative cases make clear that hallucinations are not isolated anomalies but systemic failures with tangible consequences across technical, legal, economic, and healthcare domains. Their recurrence across sectors reflects both shared structural vulnerabilities—such as confidence–competence mismatch, human–AI interaction failures, and verification cost asymmetries—and domain-specific risks that magnify their impact in high-stakes contexts. Addressing these challenges necessitates comprehensive, multi-faceted strategies that go beyond incremental technical improvements, integrating responsible implementation, ethical and societal considerations, and collaborative standard-setting to ensure safety, transparency, and public trust. As AI systems become further embedded in critical infrastructures and decision-making pipelines, a mechanistic understanding of how hallucinations arise is indispensable for designing sustainable solutions, motivating the turn to their underlying mechanisms and lifecycle vulnerabilities.

B. Main Challenges

To situate these real-world failures within a broader research context, it is necessary to recognize that hallucination mitigation is constrained by a set of deeply rooted challenges. These obstacles are distributed across the entire lifecycle of LLMs, arising from knowledge boundaries and maintenance challenges, limitations in pre-training and alignment processes, annotation and evaluation difficulties, complexities in knowledge editing and updating, and constraints inherent in decoding-time strategies [24]. Addressing these interconnected problems is essential to advancing from ad hoc fixes toward more robust, generalizable, and trustworthy generative models. The following subsections examine these challenges in greater depth, highlighting both their individual manifestations and their cumulative role in sustaining hallucination risks.

1) Knowledge and Data Challenges: Despite advances in knowledge-grounded LLMs, significant obstacles remain in ensuring factual reliability. Real-world repositories are often incomplete or noisy, creating discrepancies between retrieved evidence and generated outputs [25]. LLMs have been described as "causal parrots," reflecting correlations in training data rather than genuine causal reasoning [26], which limits their ability to audit facts in sparse or fast-changing domains [27]. Incorporating external knowledge faces barriers such as annotation variability, input length restrictions, and costly retraining or fine-tuning [28], while static adaptation leaves models outdated without dynamic updating mechanisms [29]. In LVLMs, continual instruction tuning reveals notable fragilities such as catastrophic forgetting, where sequentially finetuned models tend to overwrite previously acquired knowledge, highlighting the challenges of maintaining balanced multimodal alignment across evolving tasks [30]. Such biases cause models to generate plausible but incorrect content even without supporting visual evidence [31], while existing remedies such as data re-annotation, RLHF, or external verification modules remain costly, labor-intensive, and difficult to generalize across architectures [31], [32]. Collectively, these challenges highlight the fragility of knowledge- and data-driven approaches, underscoring the need for adaptive, dynamically updated, and generalizable solutions.

2) Training and Instruction-Tuning Challenges: Beyond knowledge acquisition, training objectives and adaptation methods contribute substantially to hallucination persistence. Conventional maximum likelihood fine-tuning enforces token-by-token replication, which can amplify small deviations into fluent but unfounded continuations [33]. Many mitigation techniques rely on additional fine-tuning with manually crafted prompts or domain-specific supervision, creating prohibitive demands for annotated data and human labor [34]. Instruction tuning for LVLMs introduces further obstacles: constructing high-quality, instruction-following datasets is far more resource-intensive than building standard input–output pairs, significantly raising the cost of adaptation [30]. Moreover, task-specific tuning risks overfitting and catastrophic forgetting, reducing the model's ability to generalize across diverse tasks and limiting transferability across architectures or domains [35]. Together, these limitations highlight the resource-intensive, fragile, and poor scalability of current fine-tuning and instruction-tuning approaches, underscoring the need for more data-efficient and robust alternatives.

3) Reliability and Adaptability Constraints: Progress in hallucination mitigation is further constrained by challenges in both evaluation and model adaptation. On the evaluation side, the absence of a universally accepted definition of correctness makes it difficult to annotate hallucinations consistently, placing a heavy burden on annotators and limiting the comparability of benchmarks [36]. Effective model editing remains challenging, as existing methods struggle to mitigate hallucinations without compromising the original capabilities of LLMs, and often face limitations in generalization to unseen data [37]. Together, these constraints highlight fundamental obstacles in reliably diagnosing hallucinations and sustainably adapting models, underscoring the need for standardized evaluation frameworks and more precise, architecture-agnostic editing techniques.

4) Inference-Time and Resource Constraints: Beyond training and evaluation, mitigation strategies deployed at inference also encounter critical constraints. Decoding-time interventions provide lightweight alternatives to retraining, aiming to mitigate hallucinations during inference without excessive computational overhead [38]. Contrastive decoding, for example, enhances factuality without relying on external knowledge [39], but its dependence on auxiliary "amateur" models can propagate incorrect knowledge and lacks fine-grained token-level control, limiting effectiveness for smaller LLMs [40]. Variants that manipulate output distributions through coarse subtraction or uncertain guidance may even introduce new errors [41], while uncertainty-based methods and heavyweight auxiliary models increase latency and remain imperfect in predictive reliability [42]. Reducing hallucinations of large vision–language models has been explored through adaptive focal-contrast decoding with fine-grained visual grounding [43]. However, existing approaches are often inefficient, costly, or constrained by training data and model-specific biases, and thus lack generalizability across different LVLM architectures [44]. Their reliance on architecture-specific designs and costly training further reduces generalizability [43], [44], underscoring the need for training-free, plug-and-play solutions that are both model-agnostic and capable of maintaining fluency while mitigating hallucinations comprehensively.

In summary, hallucination mitigation is hindered by interconnected challenges spanning knowledge incompleteness, training inefficiencies, evaluation bottlenecks, model editing difficulties, and inference-time resource constraints. These obstacles reveal that hallucinations are not artifacts of any single stage but the outcome of systemic limitations across the entire model lifecycle. Addressing them requires solutions that move beyond isolated fixes toward integrative approaches capable of balancing efficiency, scalability, and robustness. The next section, therefore, surveys existing solutions and emerging directions, highlighting how current research seeks to overcome these challenges and charting pathways toward more reliable and trustworthy generative AI.

C. Existing Solutions and Emerging Directions

Addressing the challenges outlined in Section I-B is far from straightforward, and in response to growing concerns over AI-generated hallucinations, a wide spectrum of mitigation strategies has emerged, spanning different paradigms and stages of the model lifecycle [45]. These methods range from training-time alignment techniques such as supervised fine-tuning, reinforcement learning with human feedback (RLHF), and direct preference optimization (DPO) [34], [46], [47] to inference-time approaches like retrieval-augmented generation (RAG) [48], knowledge graph grounding [49], prompt engineering [50], [51], and contrastive decoding [39]. Post-generation strategies such as consistency checks, self-refinement, and external validation further provide safeguards by detecting and revising unsupported outputs [8], [24], [52]. While each class offers distinct advantages, they also exhibit inherent limitations. The detailed review of these methods will be discussed in Section III and IV, where we present a concise overview of existing hallucination mitigation methods, highlighting both the challenges they address and the limitations that persist.

Prior surveys have attempted to bring order to this growing body of work. Luo et al. [53] categorize mitigation into knowledge-based, retrieval-based, and training-guided strategies; Bruno et al. [54] propose a task-oriented framework including fine-tuning, knowledge graphs, memory augmentation, prompts, and feedback; and Tonmoy [55] simplifies approaches into prompt engineering versus model development. While valuable, these taxonomies tend to collapse distinct dimensions, leaving open questions about how methods align with different lifecycle phases, interact with detection and abstention, or complement one another in integrated pipelines.

Hallucination mitigation workflows often incorporate detection as a supporting component, since identifying unsupported

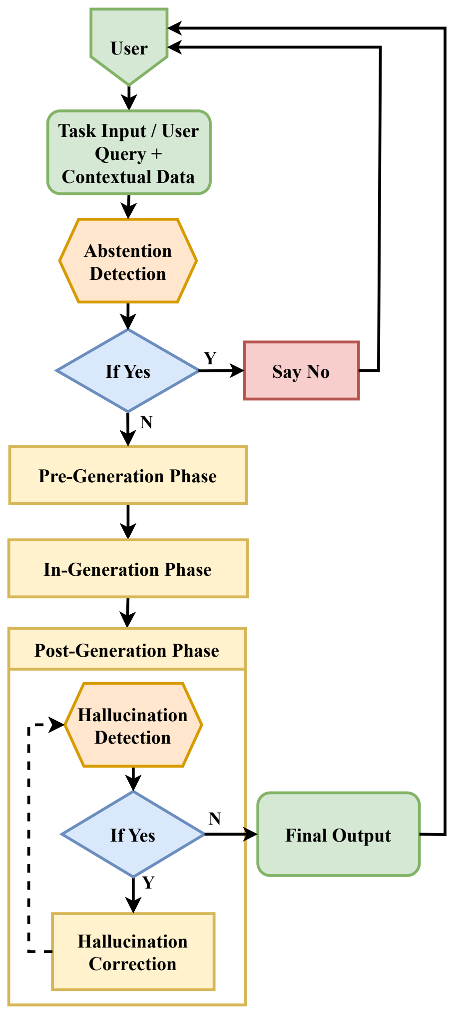

Fig. 1. Basic Pipeline for Hallucination Detection and Mitigation

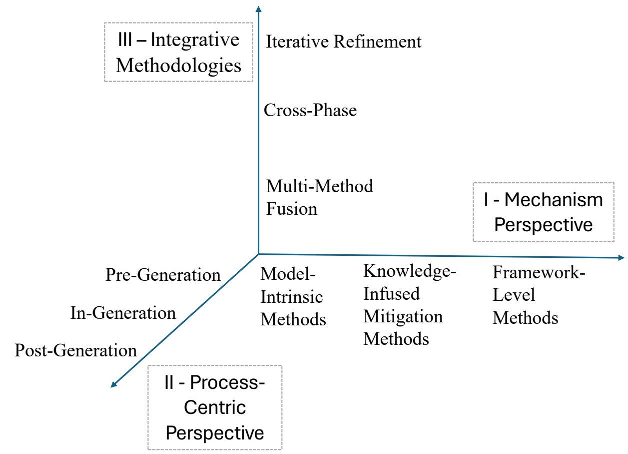

Fig. 2. Three-Dimensional Taxonomy of Hallucination Mitigation Strategies

or fabricated content is typically a prerequisite for effective correction [8]. Detection enables factual grounding, output revision, or abstention strategies in which models conservatively refuse uncertain queries (e.g., by responding "I do not know") [56], [57]. Abstention can be implemented through confidence thresholds, entropy-based signals, or conformal prediction [58], and has shown promise in reducing high-stakes hallucinations by enforcing a principle of conservative generation. While not sufficient on its own, detection strengthens mitigation at both ends of the pipeline: pre-generation, where retrieval or abstention mechanisms screen inputs and anticipate risks, and post-generation, where verification and correction safeguard final outputs. As illustrated in Fig. 1, mitigation itself spans a layered defense system—*pre-generation preparation*, *in-generation interventions*, and *post-generation refinement*—with detection mechanisms acting as gatekeepers before and after generation to ensure that interventions are timely and context-sensitive. Yet despite the breadth of available strategies, current efforts remain fragmented, with methods optimized for narrow scenarios, classified inconsistently across surveys, and seldom evaluated under unified benchmarks, underscoring the need for a more integrative taxonomy.

Beyond these established paradigms, several emerging directions are shaping the field. Causal learning approaches reinterpret generation as a structural causal process [25], [59], enabling interventions such as counterfactual reasoning [60], and causal discovery [27]. Consistency-based verification has been extended into iterative refinement loops such as Chain-of-Verification (CoVe) [61] and ReVISE [52], though these methods remain susceptible to inheriting model biases [62]. Abstention policies [57] and uncertainty-aware modeling [58] aim to make models safer by encouraging refusal over fabrication. Meanwhile, **agent-based systems** represent a promising frontier, empowering LLMs to act as reasoning agents capable of tool use, external API calls, and dynamic verification [63], [64], though their reliability ultimately depends on the quality and trustworthiness of external resources [65], [66].

In summary, hallucination mitigation research has produced a diverse but fragmented toolkit that spans training-time alignment, inference-time controls, post-generation safeguards, and emerging paradigms in causal reasoning, abstention, and agent-based verification. Yet despite this progress, methods remain siloed, inconsistently categorized, and limited in cross-phase integration. To move beyond isolated fixes, this paper introduces a **Three-Dimensional Taxonomy** that systematically organizes hallucination mitigation strategies by mechanism, lifecycle stage, and integrative role, offering a unified lens through which to analyze their interdependencies and chart future directions.

D. Three-Dimensional Taxonomy

The paper introduces a comprehensive three-dimensional taxonomy of hallucination mitigation strategies as illustrated in Fig. 2. Each dimension captures a distinct yet complementary perspective, enabling a systematic understanding of how methods are designed, when they are applied, and how they can be combined into coherent ecosystems.

The **first dimension** emphasizes the *mechanistic foundation* of both detection and mitigation methods. It differentiates strategies according to their underlying principles: (1) *Model-intrinsic* Methods, which intervene directly in the model's architecture, parameters, or decoding procedures to enhance robustness or extract internal signals for hallucination detection; (2) *Knowledge-Infused* Methods, which anchor generation in external factual sources, knowledge graphs, or retrieval modules, thereby addressing the limitations of the model's parametric memory while also enabling external verification; and (3) *Meta- and Framework-Based* Methods, which orchestrate higher-order structures such as multi-agent debate, self-reflection, or abstention policies, often combining multiple mechanisms under an overarching control paradigm. This mechanistic lens provides insight into what drives factual alignment, how errors can be diagnosed, and where structural innovations can emerge.

The **second dimension** adopts a *process-centric* perspective, situating methods along the generative pipeline and emphasizing that both detection and mitigation can be aligned with different temporal stages. In the *Pre-Generation* phase, detection seeks to manage risks through input screening, contextual alignment, or abstention answering, while mitigation complements this by proactively preparing the model via data curation, prompt optimization, and fine-tuning. During *In-Generation*, detection plays a monitoring role by tracking uncertainty signals or causal attention dynamics as outputs unfold, whereas mitigation operates in real time to steer or constrain generation through decoding control, guided sampling, or knowledge infusion. In the *Post-Generation* phase, detection becomes particularly critical, leveraging fact-checking against structured knowledge, consistency analysis, or human-in-the-loop evaluation, while mitigation responds with retrospective correction through external validators, re-ranking, or iterative self-refinement loops. By positioning detection and mitigation side by side across these phases, this process-centric perspective highlights the lifecycle nature of hallucination research, where the timing of intervention is as decisive as the underlying mechanism, and where complementary strategies must be orchestrated for maximal effectiveness.

The **third dimension** foregrounds *integrative methodologies* that transcend mechanistic boundaries and temporal silos. Rather than treating methods as isolated, this dimension captures the rise of adaptive, hybrid, and cross-phase architectures. It encompasses (1) *Multi-Method Fusion*, where heterogeneous techniques are synergistically combined to offset individual weaknesses; (2) *Cross-Phase Strategies*, which ensure continuity of factuality signals and risk controls across multiple stages of the generation pipeline, benefiting both detection and mitigation; and (3) *Iterative Refinement*, which embeds cyclical feedback loops of generation, verification, and correction, thereby progressively converging toward factual accuracy. These approaches embody an evolutionary shift from piecemeal defenses to system-level resilience.

Through this three-dimensional, process-aligned taxonomy, the study moves beyond flat categorizations and provides a holistic framework that not only situates existing methods but also reveals underexplored intersections. This structure offers a roadmap for guiding the design of next-generation hallucination research that integrates detection and mitigation in robust, adaptive, and scalable ways.

E. Significance and Contributions of This Study

Despite rapid advances, hallucination research remains fragmented: mitigation techniques are often developed in isolation, and detection methods are treated as separate rather than complementary processes. This hampers not only theoretical understanding but also the practical reliability of large language models (LLMs) and large vision–language models (LVLMs). Our study addresses these gaps by offering an integrative perspective that unifies detection and mitigation within a structured three-dimensional taxonomy, thereby transforming piecemeal solutions into a coherent framework with direct implications for both research and practice.

Specifically, the paper makes the following contributions:

- **Mechanism and Problems in Section II:** This section establishes the foundation for mitigation by systematically tracing how hallucinations emerge and evolve across the entire model lifecycle. We highlight that hallucinations originate from deficiencies in data collection and preprocessing, are reinforced during pretraining through statistical biases and attention pathologies, amplified in fine-tuning by exposure bias and catastrophic forgetting, and persist at inference through encoding errors and decoding traps. By reframing hallucinations as systemic outcomes rather than isolated anomalies, this analysis provides a mechanistic understanding of their lifecycle dynamics and motivates why effective mitigation must target multiple stages of development and deployment.
- **Detection Methods in Section III:** This section consolidates hallucination detection approaches and situates them within the three-dimensional taxonomy. We survey fact-based, knowledge-grounded, consistency-based, uncertainty-based, causal attention, dual/contrastive, and feedback-driven techniques, clarifying the diagnostic signals each provides. Rather than treating detection as an endpoint, we frame it as an enabling layer that transforms mechanistic insights into actionable triggers for mitigation—whether by filtering high-risk inputs pre-generation, verifying outputs post-generation, or supplying hybrid signals for integrative pipelines. This perspective highlights detection not as an isolated safeguard but as the diagnostic backbone of mitigation workflows.
- **Mitigation Methods in Section IV:** This section spans a diverse space of hallucination mitigation strategies that operate at different mechanisms and phases of the generation pipeline. Our three-dimensional taxonomy highlights (i) mechanism-based methods—model-intrinsic, knowledge-infused, and meta/framework-based; (ii) phase-based interventions—pre-, in-, and post-generation; and (iii) cross-phase and integrative approaches that link these layers through multi-stage, multi-

method, and iterative pipelines. Together, this framework shows that effective mitigation requires not isolated fixes but coordinated, system-level designs that embed prevention, regulation, and correction throughout the lifecycle of generative models.
- **An Integrative Hybrid Framework in Section V:** We propose an integrative hybrid framework that addresses systemic weaknesses such as phase isolation, resource intensity, and limited generalization. The framework is built on three principles: cross-phase continuity, multi-method synergy, and iterative refinement. Through a detailed medical case study, we demonstrate how this framework transforms hallucination mitigation from fragmented interventions into a coherent, adaptive, and auditable pipeline for real-world deployment.

In summary, this study contributes (1) a unified theoretical framework for organizing hallucination detection and mitigation, and (2) a systematic mapping of hybrid mitigation strategies with practical implementation pathways. Together, these contributions provide both a conceptual foundation and a practical roadmap for advancing robust, generalizable, and trustworthy hallucination mitigation in LLMs and LVLMs.

F. Paper Organization

The remainder of this paper is organized as follows. Section II establishes the foundational understanding by tracing how hallucinations emerge across the model lifecycle. Section III surveys detection methods that serve as diagnostic tools for identifying hallucinations. Section IV presents our three-dimensional taxonomy of mitigation strategies. Section V proposes an integrative hybrid framework based on our analysis. Section VI discusses limitations and future directions, and Section VII concludes the paper.

II. Mechanism and Problems

While the central aim of this paper is to examine strategies for mitigating hallucinations, a careful analysis of their underlying mechanisms is a necessary starting point. Hallucinations in large language and multimodal models are not random artifacts but systemic outcomes rooted in vulnerabilities that span the entire lifecycle of model development and deployment. These lifecycle-dependent mechanisms translate into concrete problems that complicate both detection and mitigation. By structuring the analysis around the data–pretraining–training–inference pipeline, this section highlights how hallucinations propagate through different phases of development and why effective mitigation must ultimately be grounded in a mechanistic understanding of these dynamics.

A. Evolution of Hallucination in Large Language Models and Large Vision-Language Models

The understanding of hallucination in LLMs has undergone a profound transformation since the advent of transformer-based foundation models. Early research characterized hallucinations simply as factual inconsistencies in generated text [8], treating them as isolated instances of incorrect information. As the field matured, however, researchers developed more nuanced taxonomies that distinguish between two fundamental categories: *factuality hallucinations*, which conflict with established world knowledge, and *faithfulness hallucinations*, which deviate from the provided input context [24]. This refinement marked a turning point, reframing hallucination not as a superficial bug to be patched, but as an intrinsic computational phenomenon arising from the probabilistic nature of neural language generation. Under this paradigm, optimization objectives inherently favor fluency and coherence, often at the expense of factual accuracy.

Building on this foundation, subsequent research introduced a paradigm shift by conceptualizing hallucination through a dual lens [8]. On the one hand, the generative flexibility of LLMs enables creativity and emergent reasoning capabilities; on the other, this same flexibility inevitably produces systematic factual errors [67], [68]. Mechanistic studies have since identified convergent sources of error: **attention mechanism limitations** [69], **knowledge boundary effects in parametric memory** [70], and **optimization biases** that systematically privilege linguistic coherence over factual consistency [71]. These findings underscore the need for mitigation approaches that address root causes rather than superficial outputs.

The most recent evolutionary phase emerged with the transition to Large Vision-Language Models (LVLMs), which amplified the complexity of hallucination phenomena beyond text-only settings. Multimodality introduced fundamentally new challenges, including **modality alignment** and **unique challenges** absent in unimodal architectures [72]. Here, hallucinations manifest as **object hallucination** (perceiving non-existent entities), **attribute misassignment**, **spatial distortions**, and **cross-modal inconsistencies** where generated text contradicts visual evidence. Temporal frameworks such as VidHal expose vulnerabilities in video-grounded tasks [73]. Mechanistic studies have further implicated architectural bottlenecks, such as vision-aware head divergence [74], while specialized detection frameworks (e.g., AMBER, M-HalDetect) have been proposed to handle multimodal-specific errors [75], [76]. Accordingly, this transition constitutes a qualitative shift rather than a marginal difficulty increase, underscoring the need for improved alignment training, stronger visual grounding and vision–language alignment, and specialized, context-aware mitigation strategies to enhance reliability [77].

B. Lifecycle of Hallucination

1) Hallucination Begins with the Data: The hallucination lifecycle originates at the earliest stages of model development—specifically during data collection and pre-processing. The design, composition, and quality of training corpora establish the foundational boundaries of a model's knowledge, and hence, shape its propensity to hallucinate.

a) Data Collection and Knowledge Boundaries: LLMs derive their factual capacity entirely from the corpora on which they are pretrained [78]Consequently, the scope of these corpora defines an implicit knowledge boundary [79]. Content that is absent, outdated, or underrepresented in the training data becomes a blind spot, leaving the model prone to generating outputs based on statistically inferred patterns rather than grounded knowledge [80].

A key contributor to these gaps is temporal staleness, as models encode information only up to the cutoff point of their pretraining [24], [80]–[82]. This renders them incapable of accounting for real-world developments thereafter [24], [81]. Moreover, restrictions related to copyright and privacy further limit access to high-quality data sources, exacerbating informational incompleteness [80], [82].

Despite the immense parameter budgets of contemporary LLMs, their capacity remains insufficient to memorize the long tail of human knowledge. When presented with out-of-domain or sparsely represented queries, the model generates high-probability but epistemically uncertain outputs, often resulting in fluent yet hallucinatory responses [83], [84].

b) Data Processing and Quality: While data collection defines what a model can potentially know, the pre-processing pipeline determines how that knowledge is encoded. Poorly curated pre-processing, such as insufficient filtering, inaccurate annotations, or unbalanced distributions, can introduce latent biases and structural deficiencies that propagate hallucinations [85], [86].

In supervised settings, the use of hard labels, where only one "correct" token is reinforced, limits the model's flexibility to handle ambiguity [85]. This practice, though computationally efficient, encourages brittle decision-making and can result in confident but incorrect outputs when multiple valid completions exist [85].

Additionally, the persistence of conflictual information within training data can yield unstable generation behaviors [86]. Contradictory facts within the same knowledge cluster lead the model to produce inconsistent or context-dependent outputs, depending on how the prompt is phrased [86].

Even after optimal processing, hallucinations may stem from intrinsic deficiencies in data quality. These manifest in several distinct ways:

- **Data Accuracy.** Much of the pretraining corpus comprises unstructured web data, which, while abundant, is often rife with errors, outdated content, and systemic bias [80], [87]. This noisy input is directly encoded into the model's parametric memory. Further degradation arises from compression-based storage techniques, which, as shown by Yin et al., reduce fidelity and downstream performance as compression ratios increase [88].
- **Term Frequency Imbalance.** Disproportionate token frequencies can cause overemphasis on common entities while underrepresenting rare or domain-specific terms [60], [89]. This imbalance creates a phenomenon known as **knowledge overshadowing**, where dominant tokens override relevant but less frequent alternatives during generation [89].
- **Insufficient Diversity.** Hallucinations are especially prevalent in prompts involving negation, counterfactual reasoning, or underrepresented linguistic structures [60], [90], [91]. These failure modes arise because such forms are sparsely represented in most standard training sets [90], [91]. Empirical work shows that explicitly integrating counterfactual reasoning during training can significantly mitigate these hallucination patterns [60].
  In multimodal domains, diversity limitations are even more pronounced. For image and video tasks, the available datasets are often repetitive and lack sufficient variety in object representations [92], due to the high cost of collecting high-quality data pairs [93].

2) Hallucination continues during Pretraining: The pre-training stage, in which LLMs learn next-token prediction over large corpora, endows them with factual knowledge alongside statistical priors, structural biases, and implicit generative tendencies [94]. These latent characteristics often become the origin of hallucination behaviors, as outlined below.

a) Statistical Bias and Memorization: During next-token prediction training, LLMs encode both token-level co-occurrence patterns and global distributional statistics [94]–[97]. This encoding leads to two primary forms of hallucination-inducing bias:

- **Frequency bias.** LLMs tend to favor high-frequency patterns from the training corpus. McKenna et al. [94] demonstrate that the memorization of token frequencies causes models to overproduce statistically common entities, even when these are irrelevant or incorrect in context. This bias particularly affects long-tail knowledge, where rare entities or facts, despite being more accurate, are suppressed during inference [24], [98].
- **Attestation Bias.** During pre-training, the model learns to use named entities as "indices" into its propositional memory [94] Once the "indices" have been attested anywhere in the corpus, the model tends to affirm the statement, regardless of the new context or contradictory premise [94]. Combined with teacher-forcing, this results in **stochastic parroting—**where the model replicates previously seen continuations instead of reasoning over entailment or contradiction [28], [99]. This mechanism contributes to the model's difficulty with low-resource knowledge domains and underrepresented perspectives.

b) Attention Pathologies: Hallucinations in LLMs can emerge from intrinsic limitations in the self-attention mechanism [100], the foundational architecture of transformers, which serves as the backbone of most modern large language models. Since attention governs how input tokens are weighted and contextualized [101], its failure directly impacts factual fidelity. Two primary failure modes are frequently observed: unexpected attention to outlier tokens [84], [95], [102], [103] and inadequate attention to relevant content [96], [97].

- **Unexpected Attention Distribution.** Attention heads occasionally assign disproportionate weights to irrelevant or outlier tokens, allowing semantically uninformative elements to dominate the output, thereby distorting factual grounding [95]. Yu et al. [84] demonstrate that some multi-head attention modules hallucinate due to

the failure of choosing the correct object attribute in upper transformer layers, where false-premise attention heads override other accurate retrievals, resulting in the factually incorrect selections and distorting output accuracy [102], [103].
- **Lost Attention.** Transformers are inherently challenged by long input sequences [104]. As context length increases, attention becomes diluted, and the model struggles to maintain focus on salient tokens. Hahn [96] and Chiang et al. [97] independently confirm that transformers fail to model hierarchical dependencies and exhibit degraded performance on long-range relationships. This limitation is exemplified in the widely observed ”lost in the middle” effect, wherein the model disproportionately attends to the beginning and end of an input, neglecting central content [105], [106].

c) Cross-Modal Biases: In multimodal architectures such as LVLMs, hallucination risks are inherited and compounded across modalities—arising from limitations in both visual encoders and language modules, as well as from the bottlenecks introduced by alignment architectures [107], [108].

- **Vision Encoder Limitations.** The inability of vision encoders to accurately ground images stems from their tendency to capture low-resolution and poor visual granularity [109], with an overemphasis on visually salient objects while neglecting fine-grained relationships or background context [110].
- **Language Priors.** Due to the underperformance of visual representations, LVLMs increasingly depend on language priors to generate outputs [74], [111], [112]. When visual evidence is ambiguous, models default to statistically likely text that may be wrong, leading to language-driven hallucinations [74], [93], [112].
- **Alignment Module Bottlenecks.** Lightweight alignment modules (e.g., linear projections or Q-Formers) serve to bridge visual and language features, but these components can also act as semantic bottlenecks, limiting the effective bandwidth and accuracy of cross-modal interactions [107], [108]. Inadequate alignment leads to mismatched associations and hallucinated object references in generated text [113].

3) Hallucination Extends Through Training: Even beyond data and pretraining, hallucination continues to manifest during subsequent training stages, particularly through misaligned uncertainty modeling [22] and limitations in task-specific fine-tuning [37], [114]. The following factors further entrench the model's tendency to generate confident but factually unsupported outputs.

a) Inadequate Handling of Uncertainty and Output Overconfidence: A persistent source of hallucination lies in the disconnect between a model's internal epistemic uncertainty and its externally observed confidence [115]. Most LLMs are not explicitly trained to express doubt or abstain from answering when information is insufficient. As a result, they often produce fluent, overconfident completions even in the absence of reliable knowledge.

Tupsakhare et al. [22] argue that standard pretraining and fine-tuning pipelines fail to teach models how to recognize and respect knowledge boundaries. This misalignment manifests in two principal ways:

- **Inability to Reject.** Rather than abstaining from uncertain or unanswerable questions, models tend to fabricate plausible-sounding responses. This overproduction bias systematically favors generation over abstention, even when doing so compromises factual integrity [116]. Techniques that train models to say “I don't know” [117] or encourage abstention in high-uncertainty cases [56] have shown empirical improvements in mitigating hallucinations.
- **Consistency over Accuracy.** Language models are trained to maximize coherence and fluency, which next-token objectives reward, even when this departs from factual truth [87], [118]. This fluency bias fosters hallucination snowballing, wherein initial fabricated facts are compounded by subsequent tokens that maintain internal narrative consistency rather than factual accuracy [118].

The culmination of these failures is the generation of overconfident output [24], masking their factual unreliability and complicating downstream evaluation.

b) Fine-tuning Alignment: Although fine-tuning is widely used to adapt pretrained models to downstream tasks and reduce hallucination [119], it can paradoxically introduce new forms of factual inconsistency—particularly under domain shift conditions, where the fine-tuning data distribution diverges from pretraining [114].

One central mechanism of risk is exposure bias [120], which arises from the mismatch between training (via teacher forcing) and inference (via autoregressive generation). This discrepancy induces compounding error drift, especially when models are required to generate long, coherent outputs without intermediate supervision [120], [121].

Additionally, Gekhman et al. [114] clarify that LLMs actually struggle to learn new knowledge during the fine-tuning stage, and any unknown knowledge they are exposed to will instead become a trigger of hallucination, appearing as a form of training overfitting to novel data.

Fine-tuning can also trigger **catastrophic forgetting**, whereby the model's previously acquired knowledge is partially overwritten or distorted by the new alignment objective [37]. The *quality*, *diversity*, and *factual consistency* of the fine-tuning corpus thus play a critical role in determining whether alignment reduces or amplifies hallucination [122].

4) Hallucination Persists Throughout Inference and Deployment: Even when a model is pretrained on high-quality data, aligned through careful fine-tuning, and equipped with robust internal representations, hallucinations may still arise at inference time. Inference-time hallucinations typically fall into two categories: (1) failures in content encoding [102], [111], and (2) failures in decoding strategies that transform internal representations into surface-level text [24], [81].

a) Content Encoding Failures: Rather than achieving a truly logical and semantic understanding of inputs, LLMs rely on shallow linguistic and object co-occurrence statistics learned during the training process [102], [111]:

- **Task-Training Misalignment.** Real-world downstream tasks often require model behaviors that diverge from training objectives, creating model encoding dissonance [123]. This statistical approach to language processing leads to failure in handling unexpected inputs and nuanced user needs, for example, the requirement of generating counterfactual reasoning [91] and handling chit-chat dialogue [124].
- **Fragility to Prompt Variation.** Models often fail to produce consistent factual outputs across semantically equivalent paraphrases, indicating an insufficient grasp of deeper contextual relationships [83].
- **Deficits in Contextual Awareness.** LLMs often struggle with maintaining consistent attention over long contexts, leading to contextual misinterpretation [22]. Response strategies such as context-aware decoding [125] and contextual positional double encoding [126] have been proposed to address these weaknesses and successfully demonstrated empirical reductions in hallucination rates.
- **Context Length Extremes.** While models with short context windows are limited in reasoning scope [127], models operating on very long contexts are susceptible to semantic phase transitions and instability in output fidelity, increasing the hallucination risk [128].

b) Decoding Traps: The process of selecting output tokens, typically governed by decoding strategies, represents another critical juncture where hallucinations can manifest, regardless of the model's underlying knowledge [87], [129]–[131].

- **Likelihood Trap in Deterministic Decoding.** Traditional decoding algorithms like greedy decoding and beam search [132] prioritize the most probable token at each step. While effective at generating coherent text, these methods tend to fall into the likelihood trap, where sequences with high model probability diverge from human-judged factuality or truthfulness [129], [130]. This leads to confident yet unfounded assertions [131].
- **Uncertainty Amplification in Stochastic Decoding.** In contrast, modern LLMs often employ top-k or top-p sampling [133] to introduce diversity and reduce determinism. However, these stochastic methods can inadvertently increase output volatility, as randomness in token selection introduces higher variance and uncertainty in final outputs [87]. Empirical studies confirm that hallucination rates tend to rise in proportion to decoding entropy [24], [81].

Together, these decoding traps highlight a structural limitation of autoregressive generation: the inability to consistently balance linguistic fluency, factual reliability, and task alignment without explicit hallucination mitigation mechanisms in the inference process.

C. Summary of Mechanism and Problems

Hallucination in large language models (LLMs) and large vision-language models (LVLMs) is best understood as a lifecycle-wide phenomenon rather than a single-point defect. It originates in data deficiencies, where omissions, noise, and imbalances establish implicit knowledge boundaries; becomes entrenched during pretraining through inductive biases, memorization patterns, and attention pathologies; is further amplified during fine-tuning by exposure bias, domain shift, and catastrophic forgetting; and ultimately manifests at inference through encoding errors and decoding traps that distort the model's internal knowledge state.

This systemic trajectory highlights a central challenge: hallucination cannot be eradicated through piecemeal fixes, as vulnerabilities propagate and interact across phases. Addressing it instead requires detection mechanisms that expose both latent and surface-level inconsistencies, and mitigation strategies that intervene throughout the pipeline—from data curation and architectural redesign to inference-time controls and external verification. Building on this mechanistic perspective, the following sections turn to detection frameworks (Section III) and mitigation methodologies (Section IV), organized along temporal phases and technical dimensions

III. Detection Methods for Hallucination

Building on the preceding analysis of mechanisms and lifecycle vulnerabilities, it becomes evident that hallucinations in large language and vision–language models are not incidental anomalies but systemic phenomena that pervade data, pretraining, training, and inference. Therefore, reliable detection is a necessary prerequisite for effective mitigation. Without it, downstream interventions risk either undercorrecting, leaving residual errors unaddressed, or overcorrecting, thereby suppressing the generative flexibility essential for creativity and reasoning. Thus, detection serves as the diagnostic backbone of the mitigation pipeline, transforming mechanistic insights into actionable signals. Effective detection consequently requires both high precision and high recall, which is sensitive enough to capture subtle errors, yet robust enough to avoid excessive false alarms that would unnecessarily constrain generative capacity.

Viewed through the lens of our three-dimensional taxonomy, hallucination detection constitutes a diverse but coherent landscape. Along the **mechanism dimension**, approaches range from fact verification and knowledge-grounded reasoning to consistency analysis, uncertainty estimation, causal attention probing, dual or contrastive model comparison, and feedback-driven evaluation. Along the **phase dimension**, detection may occur at multiple checkpoints: **front-end abstention detection** proactively filters high-risk or unanswerable queries before generation, while **back-end hallucination detection** evaluates completed outputs to identify and correct factual inconsistencies, with dynamic in-generation checks offering intermediate safeguards. Finally, the **integrative dimension** highlights opportunities for hybridization, where heterogeneous signals—uncertainty, factual grounding, or inter-model contrast—can be orchestrated into iterative or cross-phase workflows. In this sense, detection is not an isolated safeguard but a structurally embedded diagnostic layer that enables effective, phase-aware, and integrative mitigation.

A. Mechanism-Based Detection

The first dimension of our taxonomy focuses on the underlying *mechanisms* through which hallucinations are detected. This perspective addresses the question of how diagnostic signals are extracted, ranging from explicit fact and knowledge verification to internal consistency checks, uncertainty estimation, causal probing of attention mechanisms, dual or contrastive model comparisons, and feedback-driven evaluation. Each mechanism embodies distinct assumptions about what constitutes reliable evidence of hallucination—whether grounded in external factual references, internal probability distributions, inter-model divergences, or evaluative feedback. While methodologically diverse, these approaches collectively establish the foundations of hallucination detection by defining the signals on which subsequent phase- and integration-based strategies can operate. As illustrated in Fig. 3, mechanism-based detection can be systematically organized into seven major categories—fact-based, knowledge-based, consistency-based, dual model, uncertainty-based, causal attention-based, and feedback-based—each of which further decomposes into representative sub-methods that highlight their unique diagnostic principles.

1) Fact-based Detection: Fact-based detection constitutes the most straightforward approach to identifying hallucinations, focusing on whether generated outputs can be validated against explicitly verifiable facts. These methods operate by extracting factual units—such as entities, relations, or claims—from model text and systematically testing their correctness against reference evidence or entailment frameworks. By targeting explicit factual validity, fact-based detection provides fine-grained and interpretable signals of inconsistency, making it particularly suitable for tasks where correctness must be demonstrable, such as summarization, question answering, or knowledge-intensive generation. Fact-based methods target isolated factual claims, whereas knowledge-based methods provide systematic reasoning across structured or external repositories.

a) Named Entity Recognition (NER) and Entity Relationship: approaches detect hallucinations in LLMs by extracting named entities and analyzing their semantic or factual relationships to uncover inconsistencies or fabricated content [42]. Here, we adopt the term Entity Relations (ER) to emphasize the modeling of links between entities, which is more consistent with the literature on knowledge representation and information extraction. NER identifies key entities in generated text, while ER modeling evaluates whether their relationships align with plausible world knowledge. For example, outputs such as “Einstein discovered penicillin” are flagged because the NER step extracts “Einstein” and “penicillin,” and ER analysis recognizes that the asserted relation contradicts established facts. Hallucinations are thus flagged based on relational

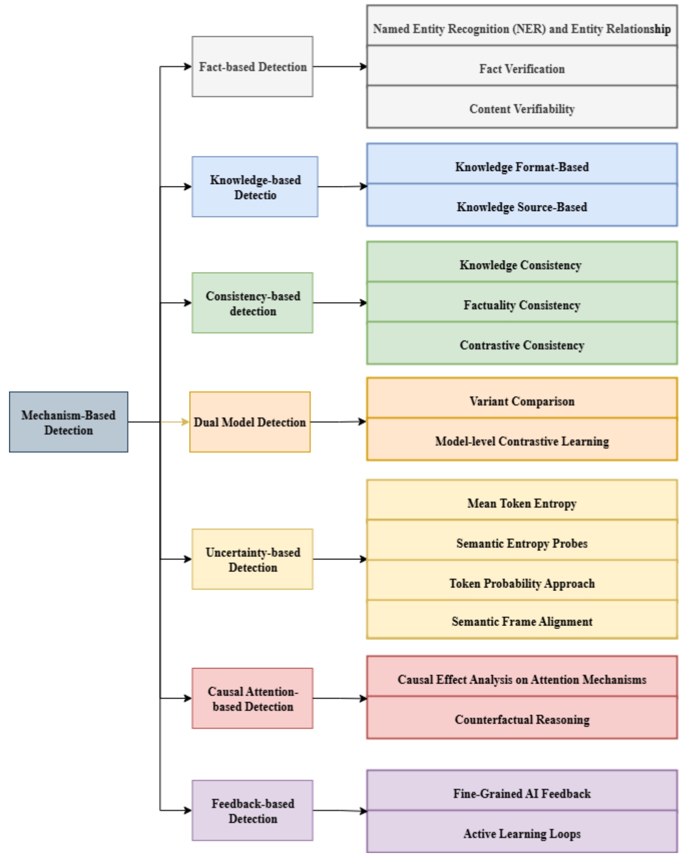

Fig. 3. Detection Methods for Hallucination

discrepancies, enabling targeted identification of problematic spans. Once detected, these errors can be mitigated through prompt engineering, where the LLM itself is leveraged to refine and correct outputs for improved factual reliability [42].

Recent work extends this paradigm by integrating NER with complementary techniques to capture more nuanced hallucination patterns. Wang et al. [134] propose a decision-tree framework that combines NER, Natural Language Inference (NLI), and Span-Based Detection (SBD), where NLI evaluates whether claims entail or contradict retrieved evidence, and errors are refined through a rewriting mechanism balancing precision, latency, and computational cost. Voznyuk et al. [135] further introduce AdvaCheck, which couples NER with Retrieval-Augmented Generation (RAG) and LLM-based verification: entities are extracted, corroborating evidence is retrieved, and an LLM verifier assesses factual consistency. This pipeline detects not only gross factual hallucinations but also fine-grained errors such as name misspellings or title distortions, which are challenging even for human annotators.

b) Fact Verification: methods detect hallucinations by extracting explicit factual units from generated text and systematically testing them against reference evidence or entailment frameworks. Unlike surface-level plausibility checks, these approaches aim to identify precise factual claims and determine whether they are supported, contradicted, or unverifiable. *FACTOID (FACtual enTAILment fOr hallucInation*

Detection) exemplifies this paradigm by extending textual entailment to hallucination detection through the notion of *Factual Entailment (FE)*, not only classifying claims but also highlighting the specific spans responsible for violations, thereby improving interpretability [136].

Building on this foundation, subsequent work adapts fact verification into retrieval-augmented pipelines. LettuceDetect [137] employs a lightweight verifier to compare outputs against retrieved passages, reducing computational overhead in large-scale applications. FRANQ [138] further distinguishes *faithfulness* (alignment with the given context) from *factuality* (consistency with external truth), introducing an uncertainty-aware framework for finer discrimination. Factored Verification achieves state-of-the-art hallucination detection in summarization by first decomposing summaries into claims and then verifying each claim with an LLM-as-verifier, assigning correctness probabilities to aggregate factual reliability [139]. Collectively, these approaches demonstrate that fact verification can function both as an independent mechanism and as an integral component of retrieval-augmented architectures.

c) Content Verifiability: assesses whether model-generated content can be explicitly traced back to supporting evidence in the source context or external references [36], [140]. By enforcing demonstrable grounding, it flags unsupported or fabricated claims as hallucinations. This property makes verifiability particularly valuable in domains where correctness must be auditable, such as summarization, question answering, generative search, and long-form generation. However, these approaches face inherent challenges: they struggle with subjective or creative outputs lacking referenceable evidence, novel insights for which no prior sources exist, and cases where the underlying references themselves may be incomplete or erroneous.

In practice, verifiability has been operationalized through automatic scoring and reference-checking systems. In summary, the HADAS framework introduces a verifiability score that measures whether each unit of summarized content can be directly located in the input text [36]. In retrieval-augmented systems, it also underpins the detection of reference hallucinations, where models fabricate nonexistent books, articles, or authors. Agrawal et al. [140] demonstrate that genuine references yield stable responses across repeated queries, while fabricated ones produce inconsistent or contradictory answers. Such *consistency patterns* serve as a lightweight detection signal, illustrating how verifiability can move beyond direct evidence matching to exploit behavioral regularities. Overall, content verifiability functions primarily as a *diagnostic detection signal*—highlighting unsupported information—while its corrective role typically depends on integration with downstream abstention, rewriting, or feedback mechanisms.

2) Knowledge-based Detection: Knowledge-based detection methods identify hallucinations by anchoring model outputs to explicit forms or sources of knowledge. Unlike fact-based approaches that directly verify individual claims, knowledge-based methods leverage structured representations and knowledge repositories to provide systematic and often more scalable validation. This category encompasses two complementary directions. The first is **knowledge format-based detection**, which exploits the representational structure of knowledge—such as in-context exemplars, dependency parses, or knowledge graphs—to expose unsupported or inconsistent outputs. The second is **knowledge source-based detection**, which grounds model responses in stable repositories of information, ranging from the parametric knowledge encoded in model weights to external resources such as curated databases or encyclopedias. Together, these approaches extend hallucination detection from isolated fact-checking to knowledge-grounded reasoning, thereby enhancing robustness and interpretability across diverse application domains.

a) Knowledge Format-Based: builds on the structural properties of knowledge representations rather than specific content repositories. By leveraging formats such as in-context exemplars, syntactic dependency structures, and symbolic graphs, these methods expose unsupported or inconsistent outputs through representational alignment. Unlike fact-based detection, which verifies isolated claims against reference evidence, format-based approaches emphasize the organization and relational structure of information, enabling scalable and interpretable detection across diverse tasks.

- **In-Context Learning (Context Aligner with In-Prompt Instructions)** [141] enables large language models to perform hallucination detection by conditioning on carefully designed prompts that illustrate factual versus non-factual responses. Rather than retraining the model, ICL leverages pretrained knowledge and adapts dynamically by recognizing input–output relationships provided in the prompt [142]. By comparing current model outputs with in-context exemplars, inconsistencies or unsupported claims can be flagged as potential hallucinations. This makes ICL a flexible and reference-free approach for scalable factuality checking. Ji et al. [143] apply in-context instruction learning to design a customized, reference-free factuality scorer. The model evaluates its own generated knowledge, assigns a factuality score, and self-reflects by refining outputs when the score falls below a threshold.

  Few-shot and *Zero-shot* learning are widely used ICL strategies for hallucination detection. By framing detection as a prompt-based classification task [144], the model can label outputs as "Hallucination" or "Not Hallucination" without requiring additional fine-tuning. In the zero-shot setting, natural language instructions define the hallucination concept directly. These predictions are then leveraged to construct a small but diverse set of labeled examples for few-shot prompting, where the model is re-prompted with demonstrations derived from the zero-shot stage [144]. To further account for model uncertainty, outputs are sampled multiple times under varying decoding temperatures, and majority voting is applied to estimate hallucination likelihood [144].

  While ICL offers a lightweight and flexible mechanism for hallucination detection, its effectiveness is highly sensitive to the design of contextual demonstrations and prompt structures [142], [145]. Minor changes in wording, ordering, or domain coverage can significantly impact performance. Consequently, auxiliary strategies such as pruning irrelevant or noisy segments of the input context have been proposed to reduce distraction and exposure bias, thereby improving factual alignment [146]. These findings emphasize that ICL should not be treated as a static prompting technique but as a context-sensitive process in which careful input curation plays a central role.

Although sometimes discussed alongside ICL, *Sentence Trimming* is more accurately categorized as a structural detection method. Shi et al. [146] propose a dependency-parse-based technique that compares an input graph with the dependency parse tree (DPT) of the draft response. The process involves three steps: first, constructing the DPT of the generated sentence; second, decomposing the input graph into entities and locating their corresponding positions within the DPT; and third, identifying entities that fall outside these aligned positions. Entities on the left of the leftmost cited node or on the right of the rightmost cited node in the DPT lack grounding in the input graph, and are therefore flagged as potential hallucinations [146]. This method highlights the value of syntactic and graph-based constraints for detecting informational inconsistencies, complementing but differing from in-context learning approaches.

- **Knowledge Graph Reasoning** provides a structured pathway for hallucination detection by grounding model outputs against curated, symbolic knowledge representations. Unlike surface-level text comparison, KG reasoning explicitly links generated entities and relations to canonical entries in a graph, enabling systematic validation of factual claims and fine-grained identification of inconsistencies. This structured approach directly addresses a core challenge of hallucination detection: distinguishing between plausible but unsupported statements and those that can be validated against established knowledge [147]. Recent methods have leveraged this principle to develop explainable and robust detection frameworks. GraphEval [148] represents model outputs as knowledge graph structures and checks the consistency of triples, thereby providing interpretable signals of where hallucinations occur. Similarly, CoKGLM [149] integrates textual and structural information, enhancing robustness and transparency by cross-examining LLM outputs with KG-based reasoning.

  The KnowHalu framework [150] exemplifies the integration of structured and unstructured sources. It first decomposes model outputs into individual factual claims, then aligns the entities and relations within those claims to entries in external knowledge graphs such as Wikidata [150]. To extend coverage, it also retrieves supporting or refuting evidence from unstructured corpora such as Wikipedia [150]. Each claim is scored for factual consistency, and hallucinations are flagged when the aggregated evidence indicates contradiction, lack of support, or unverifiability [150]. By combining the symbolic precision of KGs with the adaptability of text-based retrieval, KnowHalu demonstrates how KG reasoning can form a hybrid evidence-checking pipeline for effective hallucination detection.

  Beyond KG-based detection frameworks, in safety-critical healthcare, Gilbert et al. (2024) extend KG-grounded reasoning to a full curation pipeline: pairing LLMs with KG-based RAG as a verifiable "model of truth", adding human-in-the-loop quality control and reliability labeling to temper hallucination and non-determinism [151]. They further envision patient-level digital twins—persistent KGs enabling low-cost, auditable verification over time—and argue for hybrid, partially automated KG construction, contrasting KG-grounded and embedding-only retrieval [151]

b) Knowledge Source-Based: hallucination detection methods operate by grounding model outputs against established repositories of knowledge. The central principle is that hallucinations become apparent when generated claims contradict stable, verifiable knowledge representations. In this context, internal knowledge refers to the parametric knowledge encoded within the pretrained weights of an LLM [78], while external knowledge refers to structured or unstructured sources that are explicitly retrieved at inference time, such as knowledge graphs, databases, or encyclopedias [152], [153]. This distinction is important because internal knowledge can provide rapid, low-cost validation signals but is limited by coverage and potential training biases, whereas external knowledge offers authoritative grounding but introduces retrieval complexity and potential noise [8].

A representative approach is knowledge graph-based retrofitting, which proceeds through a structured pipeline [49]. First, a secondary LLM decomposes the generated response into atomic factual claims, ensuring that complex sentences are reduced into verifiable units. Next, entities mentioned in these claims are recognized, and a relevant subgraph is retrieved from the underlying knowledge graph to capture contextual relations. From this subgraph, salient triples are selected to represent the most pertinent factual information, while extraneous or weakly connected knowledge is filtered out. Finally, the extracted claims are systematically compared against the selected triples, and any misalignment, contradiction, or lack of support is flagged as a hallucination [49]. This pipeline illustrates how grounding against external knowledge sources can provide fine-grained, interpretable detection of hallucinations, complementing the implicit signals derived from a model's internal knowledge.

3) Consistency-based detection: rests on the idea that hallucinations often manifest as inconsistencies—either with external knowledge or with the model's own internal reasoning. These approaches evaluate whether generated outputs remain coherent across knowledge sources, inference steps, and perturbations. For example, a model may produce different answers to semantically equivalent questions, or provide reasoning chains that contradict its final conclusion, both of which serve as indicators of hallucination. While this paradigm offers a lightweight and reference-agnostic means of identifying errors, it also entails trade-offs: models can still generate hallucinations that are internally consistent yet factually incorrect, thereby escaping detection. This limitation highlights the need to combine consistency checks with complementary strategies that directly verify factual grounding.

a) Knowledge Consistency: is one of the important principles for hallucination detection, ensuring the consistency of knowledge, which evaluates whether generated content aligns with established background knowledge. Unlike self-evaluation approaches that rely solely on internal introspection, knowledge-consistency methods explicitly identify potentially unreliable spans and validate them against external sources. This external grounding enables more systematic detection of factual errors while preserving interpretability.

KnowHalu extends this paradigm by leveraging multi-form knowledge and introducing a multi-step factual validation pipeline [150]. The method decomposes the original query into step-wise reasoning and sub-queries, retrieves relevant evidence from both unstructured text and structured triples, and then aggregates the results across sources to produce a consolidated verdict. By combining structured and unstructured evidence, KnowHalu provides a more robust framework for identifying knowledge inconsistencies.

Varshney et al. [154] propose another approach that focuses on low-confidence regions of generated text, under the assumption that hallucinations are more likely to arise in uncertain spans. Their method first pinpoints tokens with low probability, then requires the model to verify or justify these spans by retrieving external knowledge [154]. If the retrieved evidence fails to substantiate the generated claim, the content is flagged as hallucinated [154]. This strategy highlights how uncertainty signals can be integrated with knowledge consistency checks to achieve fine-grained hallucination detection.

b) Factuality Consistency: in hallucination detection refers to the principle that models should actively assess whether their generated outputs remain logically coherent and aligned with their internal knowledge. Unlike knowledge-consistency methods that validate outputs against external sources, factuality-consistency approaches rely on the model's introspective ability to scrutinize and refine its own responses.

Two representative paradigms illustrate this principle: **Interactive Self-Reflection** and **Self-Evaluation for Factuality Alignment**. Both exploit internal self-checking mechanisms to enhance factual accuracy, but they differ in their strategies—one emphasizes dynamic iterative refinement at inference time [143], while the other integrates systematic self-evaluation into training to improve long-term alignment [155].

- **Interactive Self-Reflection Method** employs a feedback loop in which the model generates an initial response and then iteratively questions its own factual accuracy through self-generated prompts [143]. These prompts are designed to surface contradictions or unsupported claims by probing consistency against the model's internal knowledge and contextual cues. The approach follows a generate–score–refine cycle, supported by CTRLEval, an unsupervised and task-agnostic evaluation metric that computes consistency scores for answers. When scores fall below a threshold, the model introspects, self-corrects, and revises its outputs. Through this iterative refinement, the method effectively enhances factual reliability without relying on external resources [143].
- **Self-Evaluation for Factuality Alignment** detects hallucinations by prompting the model to validate the factual correctness of its own responses through structured self-assessment [155]. It introduces Self-Eval, where the model inspects its outputs for inaccuracies, supported by Self-Knowledge Tuning (SK-Tuning) to calibrate confidence and strengthen internal consistency. Discrepancies or unsupported claims identified during this process are used as signals for Direct Preference Optimization, enabling fine-tuning that systematically suppresses hallucinations and improves factual precision. By relying exclusively on internal resources rather than external annotations, this method provides a scalable pathway to align models toward truthfulness [155].

c) Contrastive Consistency: detection leverages the principle that hallucinations can be surfaced by contrasting model behaviors under perturbed conditions. The approach works by deliberately modifying the input or the decoding process and then observing discrepancies in predictions. The underlying intuition is that hallucination-prone outputs are more likely to fluctuate disproportionately under perturbations, whereas content that is well-grounded in knowledge should, in principle, exhibit greater stability [33], [156]. However, this assumption must be qualified: even legitimate and factual responses can display sensitivity to perturbations, particularly in cases of complex reasoning or ambiguous contexts [142], [145]. Thus, contrastive learning provides a useful but not exhaustive diagnostic signal.

In practice, contrastive detection methods differ in the type of contrastive signal they exploit—ranging from lexical and semantic perturbations of the input, to stochastic variations in decoding, to comparisons across multiple generations [156]. Despite their diversity, these methods share a common philosophy: they treat discrepancies as indicators of hallucination risk, often without requiring external knowledge or additional retraining. This makes contrastive learning a lightweight yet versatile strategy for hallucination detection, particularly when integrated with knowledge-grounded verification pipelines or consistency-based self-reflection mechanisms [33], [156].

- **Original vs. Disturbance instruction prompts.** The core idea of this strategy is that hallucinations often arise from spurious co-occurrence priors or shallow correlations, which can be stressed by introducing small disturbances into the instruction. By contrasting model predictions under the original versus a perturbed instruction, unstable tokens can be surfaced as hallucination-prone.

- **Instruction Contrastive Decoding (ICD)** operationalizes this principle by injecting small disturbance role prefixes into the instruction to deliberately stress instruction-aware fusion [32]. At each decoding step, the model generates two token-level distributions—one conditioned on the original instruction and one on a perturbed version—and compares them [32]. Tokens whose probabilities are disproportionately boosted under perturbation are flagged as spurious signals indicative of hallucination [32]. To suppress these signals, ICD subtracts a scaled disturbed-logit vector from the base logits before sampling, creating an online identify-and-suppress detector during decoding [32]. This contrastive signal is particularly effective against co-occurrence-driven errors: disturbances amplify priors misaligned with evidence while leaving grounded tokens relatively stable [32]. As a result, ICD enables sentence-level, model-internal hallucination detection without requiring auxiliary models, additional training, or external knowledge [32].

- **Adding Gaussian Noise.** Adding controlled noise to model inputs is a long-standing strategy in machine learning for probing robustness and exposing hidden failure modes. In the context of hallucination detection, introducing Gaussian noise serves as a perturbation mechanism that selectively degrades genuine evidence while leaving spurious correlations more exposed. This allows contrastive frameworks to differentiate tokens grounded in reliable signals from those supported mainly by co-occurrence priors or language biases.
  - **Visual Contrastive Decoding (VCD)** addresses object hallucinations by contrasting a model's output distributions for the original image with those for a deliberately distorted version, where Gaussian noise is progressively injected through a forward-diffusion process [31]. The added noise elevates visual uncertainty and weakens genuine visual cues. Consequently, tokens that become more confident under the distorted view are flagged as hallucination-prone, since their support derives from co-occurrence priors or unimodal language bias rather than authentic visual evidence [31]. Concretely, VCD performs decoding under both conditions at each step and compares discrepancies in token logits and probabilities; disproportionate gains in the distorted case signal non-visual or spurious concepts [31]. This detection signal is entirely model-internal, requires no additional training, and avoids reliance on auxiliary detectors or external knowledge [31].

- **Mix of True & False Inputs.** The central principle of mixed-input hallucination detection is that hallucinations can be revealed by explicitly contrasting model behavior on true versus false inputs. By introducing carefully constructed counterfactual or incorrect knowledge alongside verified ground truth [33], these methods probe the degree to which a model's outputs are grounded in accurate evidence versus spurious correlations.
  Two representative methods illustrate this idea: **Mixed Contrastive Learning (MixCL)** and **Source-Contrastive Decoding**. Both exploit the contrast between true and false inputs, but they differ in scope and technical implementation. MixCL operates at the training stage, producing model-internal detection signals without requiring external evaluators [33], while Source-Contrastive Decoding provides a lightweight inference-time check of input–output alignment in generative tasks [157].
  - **Mixed Contrastive Learning** (MixCL) constructs hard negative knowledge for each dialogue context by combining two types of negatives: (i) retrieved negatives that are topically related but irrelevant or non-factual, and (ii) model-generated negatives obtained through bootstrapping hallucinated cases [33]. A natural language inference-based disjointness check ensures these negatives either contradict or fail to entail the positives, guaranteeing they represent spurious content [33]. To localize hallucinations, MixCL performs span-level extraction of entities and constituents from both positives and negatives, and then contrasts token-level preferences across these spans. Tokens whose support derives disproportionately from negative spans are flagged as hallucination-prone, providing a model-internal detection signal that supervises training without auxiliary evaluators [33].
  - **Source-Contrastive Decoding** detects hallucinations in machine translation by contrasting translation probabilities under correct source inputs versus artificially corrupted ones, such as randomly replaced or distorted segments [157]. If a translation remains probable under false inputs, it is likely hallucination-driven, reflecting reliance on language priors rather than accurate source grounding. By leveraging such counterfactual contrasts, Source-Contrastive Decoding exposes spurious dependencies that would otherwise remain undetected [157].

- **Cross Layer Contrast (Logic vectors from different layers).** The principle of cross-layer contrast is that lower and intermediate layers of LLMs tend to encode shallow syntactic or correlation-based patterns, while upper layers increasingly capture semantic and factual representations. By comparing discrepancies across these layers, models can expose tokens whose support relies disproportionately on non-factual patterns, thereby signaling hallucination-prone content.
  Two representative methods are **Lower Layers Matter (LOL)** and **Inter-Layer Contrastive Decoding (DiLa)**. Both exploit inter-layer disagreement as an internal hallucination signal, but they differ in scope: LOL incorporates

an external "amateur" model and emphasizes multi-layer fusion, while DiLa remains self-contained and adaptively selects the most divergent hidden layer at each decoding step for fine-grained detection.

- **Lower Layers Matter** (LOL) contrasts the base LLM with a deliberately trained "amateur" variant and computes contrast signals not only at the final layer but also at early-exit layers [41]. By fusing evidence across layers, LOL highlights divergences amplified in lower layers, localizing tokens whose support comes from shallow, non-factual priors [41]. The amateur model is fine-tuned on non-factual data to induce hallucination tendencies, so the base–amateur gap becomes a proxy for spurious content. During decoding, the method monitors both final- and lower-layer contrasts, flagging tokens disproportionately supported by the amateur side as hallucination-prone. To further sharpen this signal, LOL integrates a truthfulness-refocused module that injects instruction guidance, emphasizing factual keywords and generating auxiliary contrast cues. Discrepancies between these cues and model predictions indicate non-truthful continuations [41]. Together, the fused multi-layer contrasts and instruction-based evidence act as a model-internal detector for hallucination-prone tokens.
- **Inter-Layer Contrastive Decoding** (DiLa) detects hallucination-prone tokens by contrasting the next-token distributions of the final layer with those from an internal hidden layer dynamically chosen for maximal divergence at each step [40]. Specifically, it computes a log-probability ratio between final- and hidden-layer distributions, leveraging the notion that higher layers encode factual semantics while earlier layers are biased toward syntactic regularities. When hidden-layer confidence outweighs final-layer support, the token is flagged as non-grounded [40]. To mitigate false positives, DiLa applies an adaptive plausibility constraint that restricts contrastive checks to tokens with sufficient support under the final-layer distribution. At each step, the hidden layer is re-selected using Jensen Shannon divergence to ensure the most informative perspective is chosen, enabling fine-grained, step-wise localization of unstable tokens [40]. In essence, inter-layer disagreement serves as a reliable internal signal: when confidence inflates in lower or intermediate layers relative to the final layer, the token is marked as relying on shallow priors rather than factual knowledge.

• **Input Removal & Corruption (Video/Text manipulations).** Input perturbation-based detection operates by systematically removing or corrupting parts of the input to test whether token predictions remain stable and grounded. If predictions shift disproportionately under such manipulations, the affected tokens are flagged as hallucination-prone.
Two representative methods, **Language-Contrastive Decoding (LCD)** and **HALC**, share this philosophy of deliberately weakening or distorting one modality to test grounding robustness. Both are lightweight, training-free, and operate online, but they differ in their focus: LCD suppresses the visual modality to reveal language-driven errors [44], whereas HALC perturbs the visual input itself to assess the stability of multimodal grounding [43].

- **Language-Contrastive Decoding** (LCD) detects hallucination risk by contrasting an LVLM's next-token probabilities with those of its underlying LLM when run on the same text prefix without the image [44]. If the language-only model assigns significantly higher confidence to a candidate token than the image-conditioned LVLM, this discrepancy is treated as evidence that the prediction is driven by linguistic priors rather than visual evidence. To stabilize the signal, LCD first applies adaptive plausibility, restricting attention to tokens already deemed plausible under the LVLM distribution. The contrast's weight is further modulated by the LLM's conditional entropy: when the language model is confident, its signal dominates; when uncertain, its influence is reduced [44]. This design yields a dynamic, model-internal hallucination indicator without requiring auxiliary models or retraining.
- **HALC** probes hallucination risk by testing the stability of token predictions across perturbed visual contexts during generation [43]. It first localizes token-specific regions using a zero-shot object detector (e.g., Grounding DINO or OWLv2), then samples multiple fields-of-view (FOVs) around these regions to collect diverse visual evidence. At each step, HALC selects pairs of FOVs that maximally disagree in their effect on token probabilities, using this contrast to expose unstable, non-grounded support. Hallucination-prone tokens typically exhibit noisy or fluctuating likelihoods across different crops—for instance, spurious objects may peak under mismatched views—whereas truly grounded tokens remain stable [43]. By treating these cross-context discrepancies as adaptive grounding signals, HALC flags tokens lacking robust visual support, providing an internal, online detection cue without auxiliary supervision or training.

4) Dual Model Detection: The dual model approach detects hallucinations by maintaining two model variants that differ in their susceptibility to hallucination and then contrasting their outputs or internal representations. The central intuition is that fabricated or unsupported content manifests as systematic discrepancies between a truth-preserving model and a hallucination-prone counterpart. By comparing token distributions or representation-level signals across these models, hallucination-prone spans can be identified and filtered. Unlike knowledge-grounded methods, dual model detection relies on the relative divergence between model variants rather than external fact-checking.

a) Variant Comparison: constructs multiple fine-tuned variants of the same base model—typically an "amateur" model deliberately trained on hallucination-inducing data and a more stable variant that preserves linguistic fidelity. By comparing logits or decoding behaviors across these special copies, hallucination-prone tokens can be revealed as divergences between stable and unstable variants [158].

- **Freeze-CD** applies a contrastive decoding framework with local freezing training to expose hallucinations in LLMs [158]. It constructs two amateur variants fine-tuned on hallucination-inducing datasets: (i) a fully fine-tuned model that amplifies hallucinations and (ii) a locally frozen model that preserves lower-layer semantic and grammatical features while still allowing partial hallucination. During inference, hallucination detection is achieved through logit subtraction: discrepancies between the original model and the amateur models highlight potentially fabricated information. The merged output can then suppress inconsistent or non-factual elements, improving truthfulness without requiring external resources [158].

b) Model-level Contrastive Learning: creates a pair of LLMs explicitly optimized in opposite directions: one fine-tuned on factual datasets to reinforce truthfulness, and another fine-tuned on hallucination-prone data to exaggerate erroneous patterns. By iteratively contrasting their representation layers, hallucinations can be detected as signals that align with the "bad" model but diverge from the "good" one [37].

- **Iter-AHMCL** introduces an iterative model-level contrastive learning framework, contrasting representations from a "positive" model (trained on factual, hallucination-free datasets) against a "negative" model (trained on hallucination-inducing datasets) [37]. By fine-tuning the same base LLM along these opposing directions, the method constructs a contrastive signal at the representation layer: discrepancies highlight inconsistent or erroneous information associated with hallucinations. Iterative optimization refines this separation, allowing hallucinated spans to be suppressed and factual accuracy enhanced during inference—again without requiring external knowledge or additional retrieval [37].

5) Uncertainty-based Detection: Uncertainty-based hallucination detection methods operate on the principle that hallucinations are more likely to arise when the model is uncertain about its predictions. Uncertainty is typically reflected in the probability distribution over tokens or in the semantic space of generated outputs. By quantifying this uncertainty through entropy, probability features, or semantic alignment, such methods aim to identify responses that are less reliable [159]–[161]. However, this principle has inherent limitations: models can still produce hallucinations with high confidence, while being uncertain about correct information. This mismatch suggests that uncertainty alone is an imperfect proxy and must often be combined with complementary signals for robust detection.

Despite this caveat, existing methods share the goal of leveraging uncertainty but differ in the source of their signals and the granularity with which they operate. Token-level metrics capture local uncertainty by analyzing the sharpness or flatness of token probability distributions [159], [160]. Semantic entropy approaches approximate the uncertainty of global meaning by comparing multiple generations sampled in the semantic embedding space [161]. Frame alignment methods go further by assessing whether the generated outputs preserve the intended semantic structure of the input, thereby linking uncertainty to the preservation of meaning [36], [162]. Together, these techniques provide complementary perspectives, offering both fine-grained and holistic views of uncertainty-based hallucination detection.

a) Mean Token Entropy: is a fundamental uncertainty metric that quantifies the average entropy across generated tokens. With the entropy H of a single token X defined as $H(X) = -\sum_{x \in V} p(x) \log p(x)$, where V denotes the model's vocabulary, entropy reaches its maximum when $p(x)$ is uniform over all tokens and drops to its minimum when the distribution is fully concentrated on a single token (i.e., one token has probability 1 and all others 0) [159]. A higher mean token entropy reflects greater uncertainty in the model's predictions, making it a useful signal for identifying potential hallucinations in generated text [159].

b) Semantic Entropy Probes: (SEP) is a lightweight and effective method for hallucination detection in LLMs based on the concept of semantic entropy, which measures model uncertainty over the meaning of generated outputs [161]. Instead of clustering multiple responses for the same query by semantic equivalence, SEP trains a linear probe on hidden states from a single model generation to predict semantic entropy scores, thereby reducing computational cost while maintaining strong detection performance [161].

c) Token Probability Approach: have long been employed to signal uncertainty in sequential generation tasks [160]. A token probability approach for hallucination detection extracts four key features from a generator LLM_G and an evaluator LLM_E: (i) Minimum Token Probability, (ii) Average Token Probability, (iii) Maximum LLM_E Probability Deviation, and (iv) Minimum LLM_E Probability Spread [160]. These numerical features, closely related to average and maximum entropy, serve as indicators of model uncertainty and can be leveraged to flag hallucination-prone outputs [160].

d) Semantic Frame Alignment: evaluates whether generated content preserves the factual and semantic structure of the source input [162]. Hallucinations frequently manifest as confabulations, where outputs deviate semantically from the input despite maintaining surface-level plausibility [162]. Unlike Semantic Entropy, which captures response diversity but may overlook deeper inconsistencies, frame alignment explicitly tests for factual and semantic consistency between source and output.

Entailment-based models such as FactKB [36] operationalize this principle by computing a semantic frame score for document–summary pairs, estimating the probability of factual alignment with the source document. This enables highlighting of unsupported or fabricated claims without requiring extensive external annotation [36].

Further, Semantic Divergence Metrics (SDM) [162] extend the approach by measuring semantic misalignment not only across multiple responses but also across paraphrases of the same prompt. By clustering sentence embeddings into a shared topic space and computing information-theoretic divergence, SDM provides a fine-grained quantification of confabulation and semantic drift [162].

6) Causal Attention-based Detection: Causal attention-based detection methods aim to move beyond purely correlational signals by estimating the causal influence of inputs and attention components on generated outputs. The central intuition is that if an output is primarily driven by irrelevant or insufficient input evidence, it can be flagged as a hallucination. This causal perspective is particularly relevant for multimodal models, where spurious cross-modal correlations often induce hallucinations.

Two main approaches embody this principle. First, causal effect analysis on attention mechanisms explicitly models the causal role of attention layers using Structural Causal Models (SCMs) and performs interventions on attention distributions [163]. Second, counterfactual reasoning manipulates the input directly, removing or altering key components, to observe how outputs change under alternative scenarios [25]. Together, these approaches offer complementary perspectives on grounding and attribution.

a) Causal Effect Analysis on Attention Mechanisms: seeks to quantify how variations in attention distributions causally influence model outputs [163]. This involves constructing an SCM to capture how attention mediates the influence of inputs on outputs. Interventions are then applied to attention layers (e.g., perturbing or reweighting attention distributions) to simulate alternative causal scenarios and compare the resulting outputs. If model predictions prove highly sensitive to such interventions, this indicates that responses may be driven by misaligned or spurious attention patterns, thus signaling potential hallucination [163].

b) Counterfactual Reasoning: measures the causal contribution of input components to outputs [25]. By generating counterfactual scenarios where key elements of the input are removed or altered, and comparing the corresponding outputs with the factual baseline, the method estimates the degree of grounding. Outputs that remain largely unchanged despite substantial input removal are considered insufficiently grounded and hence hallucination-prone. In this way, counterfactual reasoning favors outputs that are causally dependent on the correct input information [25].

7) Feedback-based Detection: Feedback-based detection methods build on the principle that hallucinations can be effectively identified and mitigated when explicit evaluative feedback is incorporated into the generation pipeline. These approaches introduce an auxiliary evaluation layer—often powered by large models or iterative loops—that assesses outputs and generates structured signals regarding their factuality, type, and severity. Such signals not only facilitate reliable detection but also provide actionable guidance for correction and iterative model improvement.

a) Fine-Grained AI Feedback: leverages powerful models such as GPT-4 and GPT-4V to provide detailed annotations of hallucinations in generated responses [164]. Given a hallucination-annotated dataset, the system examines each sentence in the model's output relative to a specific input and prompt, producing a structured six-tuple. This tuple encodes the sentence, hallucination type, reason for classification, severity score, and justification [164]. By distinguishing between minor inaccuracies and critical factual errors, fine-grained feedback enables precise categorization and prioritization of hallucinations, making it possible to design targeted mitigation strategies [164].

b) Active Learning Loops: establish a dynamic interaction between detection and model improvement. Instead of relying solely on static benchmark datasets, active learning continuously identifies the most informative or error-prone samples for feedback, thereby expanding coverage of diverse hallucination patterns and avoiding overfitting to narrow error types. This closed-loop process allows detection systems to become progressively more robust and generalizable across domains.

For instance, HADAS operationalizes this paradigm by evaluating outputs along three distinct dimensions of hallucination [36]. Each dimension is assessed with a specialized metric, producing a feedback score that is subsequently normalized and aggregated into a unified hallucination score representing overall reliability [36]. To broaden detection coverage, HADAS employs a diversity-aware sampling strategy within the active learning loop, ensuring that the feedback process systematically captures heterogeneous hallucination types rather than disproportionately focusing on a single error class [36].

B. Phase-Based Detection: Front-End and Back-End Strategies

Within the phase-based dimension of our taxonomy, detection mechanisms can be situated at distinct checkpoints of the generation pipeline. Two positions are of particular importance: *front-end abstention detection*, which intervenes before generation begins, and *back-end hallucination detection*, which operates after an output has been produced. Together, they define a complementary architecture that balances preventive refusal with corrective verification.

*1) **Front-End Abstention Detection**:* Abstention detection functions as a proactive safeguard by identifying high-risk, ill-posed, or knowledge-sparse user queries prior to generation. Rather than allowing the model to proceed with a potentially unreliable answer, abstention detection enables the system to withhold responses (*say-no strategy*) or redirect them to alternative workflows. At this stage, detection relies on lightweight but decisive signals such as *uncertainty estimation*—where low-confidence token distributions are flagged as unreliable [159], [161]—and *consistency analysis* across paraphrased or perturbed queries [143]. In addition, *knowledge-based introspection* can provide rapid assessments of whether a query lies outside the model's representational scope [78]. By filtering unanswerable or adversarial prompts in advance, abstention detection reduces the burden on downstream corrective mechanisms and lowers the overall risk of spurious outputs.

Refusal / Abstention Strategies ("Say No") constitute the primary operational paradigm for front-end detection, reframing hallucination mitigation as an alignment problem where models are empowered to strategically decline uncertain tasks rather than attempt speculative generations [58]. Technically, refusal can be implemented through internal uncertainty quantification and multi-sample consistency checks, where divergence across candidate responses triggers a non-committal answer [56]. Beyond reactive safeguards, refusal behaviors can be actively shaped during supervised fine-tuning or reinforcement learning. For instance, HonestAI introduces abstention-oriented supervision to encourage cautious answers in ambiguous scenarios, significantly reducing spurious generations without compromising utility [117]. Similarly, Kang et al. [165] demonstrate that labeling unfamiliar training examples with abstention signals (e.g., "I don't know") enables models to generalize refusal behavior to unseen queries. Collectively, these findings highlight abstention detection not as a fallback but as a principled alignment strategy that institutionalizes uncertainty-aware non-response, transforming "not answering" into a valid and safe operational choice.

2) Back-End Hallucination Detection: Whereas abstention detection prevents unreliable generations from being attempted, back-end hallucination detection focuses on verifying and correcting outputs that have already been produced. Here, more resource-intensive strategies can be deployed, including *fact-based verification* through entity recognition and relation analysis [42], [136], *knowledge-grounded reasoning* via retrieval or knowledge graph alignment [148], [150], and *content verifiability scoring* for summarization and generative QA tasks [36]. Complementary signals are also provided by *causal attention analysis*, which probes whether outputs are grounded in relevant inputs [163], as well as *dual- or contrastive-model comparisons*, which expose hallucinations by contrasting stable and hallucination-prone variants [37], [158]. Finally, *feedback-driven evaluation* can be integrated into this stage, where powerful models or active learning loops generate structured annotations that guide both detection and iterative correction [164].

Taken together, abstention detection and hallucination detection illustrate how phase-aware strategies balance efficiency and accuracy: the former avoids unnecessary generation by intervening early with lightweight checks, while the latter enables fine-grained verification and correction once outputs are produced. Both are indispensable to a holistic mitigation pipeline and align naturally with the broader three-dimensional framework.

C. Integrative Detection Strategies: Cross-Phase and Multi-Signal Pipelines

Beyond mechanism-specific detectors and phase-specific checkpoints, a third dimension of hallucination detection highlights the system-level integration of heterogeneous signals. Rather than introducing yet another isolated detector, this perspective asks how to compose, orchestrate, and iterate diverse cues: factual verification, uncertainty estimates, consistency checks, causal attributions, and feedback signals, which form into coherent pipelines that are explainable, auditable, and robust across modalities. In practice, explicit integrative approaches remain comparatively scarce; most existing methods still operate within a single mechanism or phase. We therefore conceptualize this third dimension as a guiding framework, under which otherwise localized techniques may be reinterpreted and extended toward integrative deployments. Three prototypical paradigms are illustrative.

1) Multi-Signal Fusion: Detection reliability improves when orthogonal signals are combined rather than applied in isolation. For example, factual verification can be fused with token-level uncertainty scores, or verifiability assessments can be coupled with consistency probes, yielding more fine-grained judgments than any single metric alone [136], [150], [160]. Even methods such as dual-model contrast [158], though often framed as mechanism-specific, can be reinterpreted as fusion pipelines that combine comparative signals with grounding or verifiability modules. We argue that future designs should move beyond raw score aggregation toward calibrated fusion, where detector weights are dynamically adjusted to task, modality, and context.

2) Cross-Phase Orchestration: Hallucination detection benefits from coordination across pre-, in-, and post-generation stages. In principle, lightweight abstention mechanisms should intercept high-risk queries before generation; online probes such as contrastive decoding or multimodal perturbations should monitor instability during decoding; and back-end verifiers should finalize judgments through claim decomposition and grounding. Although most current approaches address only a single stage, we contend that their greatest value lies in cross-phase orchestration: risk signals flagged upstream can condition thresholds downstream, while post-hoc verification outcomes can be propagated backward to adjust abstention or in-generation gates. For LVLMs in particular, such orchestration should enforce cross-modal grounding by requiring stability under both visual and textual perturbations.

3) Feedback-Driven Looping: Detection systems become progressively stronger when evaluative feedback is reintegrated into the pipeline. Fine-grained annotations—whether from LLM-as-judge or human evaluators—can be used to adjust prompts, recalibrate thresholds, or select challenging cases for active learning [36], [164]. Even introspective methods such as self-reflection and self-evaluation [143], [155], though originally proposed as mechanism-level techniques, can be situated within this paradigm as lightweight loop signals that close the gap between generation and verification. We argue that the long-term challenge is to formalize such loops into continuous quality pipelines, where detector outputs are systematically logged, audited, and recycled as training or prompting signals.

Recent work illustrates how all three paradigms can converge in system-level architectures. Gosmar and Dahl [65] present a multi-agent pipeline where generation, review, refinement, and KPI-based evaluation agents communicate via structured JSON envelopes, passing evidence such as flagged spans, explanations, and disclaimers. This design simultaneously fuses heterogeneous signals, orchestrates checks across phases, and closes the loop through KPI-driven feedback—demonstrating that integrative detection is not only a theoretical axis but a realizable system-level practice. Yet such examples remain rare, underscoring the need for further research into cross-method, cross-dimension, and loop-based detection as a foundational complement to existing mechanism- and phase-specific strategies.

The third dimension reframes hallucination detection as an architectural challenge: to govern when, where, and how multiple diagnostic signals are mobilized. While existing methods provide the building blocks, their integrative recombination into multi-signal, cross-phase, and loop-based pipelines remains underexplored. This gap presents both a conceptual opportunity and a practical imperative for advancing hallucination detection beyond isolated fixes toward robust, adaptive, and deployable systems.

D. Summary of Detection Methods

Taken together, hallucination detection emerges as a multi-dimensional yet still fragmented field. Along the **mechanism dimension**, methods range from fact- and knowledge-based verification to consistency analysis, uncertainty estimation, causal attention probing, dual/contrastive model comparison, and feedback-driven evaluation, each offering complementary diagnostic signals. Along the **phase dimension**, detection can intervene before or after generation, balancing lightweight abstention with dynamic probes and post-hoc verification. Finally, along the **integrative dimension**, a few emerging systems illustrate the potential of multi-signal fusion, cross-phase orchestration, and feedback-driven looping, though such designs remain rare. Collectively, these approaches demonstrate that detection is not merely an auxiliary safeguard but the enabling layer that diagnoses error modes, calibrates generative reliability, and provides actionable triggers for mitigation. At the same time, the scarcity of unified and scalable detection pipelines underscores a critical research gap, highlighting the need for integrative architectures that can reconcile precision and recall, support cross-modal robustness, and sustain auditable deployment. Building on this foundation, the next section turns to mitigation strategies, examining how detection signals can be operationalized into systematic interventions for trustworthy generative AI.

IV. Mitigation Methods for Hallucination

This section presents a structured synthesis of hallucination mitigation methods grounded in a three-dimensional taxonomy that spans *mechanistic foundations*, *temporal intervention phases*, and *integrative, cross-phase strategies*. As illustrated in Fig. 4, this framework situates existing techniques along two orthogonal axes—methodological mechanism and generative phase—while also accounting for hybrid designs that transcend individual stages. By aligning these dimensions, the taxonomy provides a coherent lens through which to analyze, compare, and extend current approaches, facilitating both granular classification and system-level optimization of hallucination mitigation.

A. First Dimension: Mechanism-Based Taxonomy of Hallucination Mitigation Methods

Following the introductory discussion of the mitigation framework, this subsection initiates a detailed examination of the first dimension within our three-dimensional taxonomy of hallucination mitigation strategies. In this taxonomy, mitigation methods are classified into three overarching categories: (1) Model-intrinsic Methods, (2) Knowledge-Infused Mitigation Methods, and (3) Meta- and Framework-Based Methods.

As shown in Fig. 5, **Model-intrinsic Methods** encompass interventions directly embedded within the model's architecture or decoding process, such as fine-tuning, prompt engineering, and advanced decoding algorithms. These approaches primarily aim to refine the model's internal reasoning mechanisms and exert fine-grained control over generation dynamics. As shown in Fig. 6, **Knowledge-Infused Mitigation Methods** anchor model outputs in verifiable factual evidence by leveraging external repositories, including Retrieval-Augmented Generation (RAG) systems, knowledge graph integration, and consistency-guided verification pipelines. As shown in Fig. 7, **Meta- and Framework-Based Methods** operate at a higher conceptual level, orchestrating multi-layered strategies such as self-regulation mechanisms, refusal and abstention-based policies, multi-model coordination frameworks, and iterative feedback loops. These approaches facilitate cross-verification, adaptive learning, and dynamic alignment with factual knowledge across contexts.

Each of these categories targets hallucinations through distinct mechanisms of intervention. In the following subsections, each category will be systematically introduced, with emphasis placed on representative techniques, their methodological foundations, and practical instantiations.

1) Model-intrinsic Methods: Model-intrinsic Methods constitute the first category of our mechanism-based taxonomy, referring to interventions that integrate directly into a model's core architecture and decoding pipeline. Operating at the intrinsic level of the generation process, these methods influence how tokens are selected, sequences are constructed, and contextual information is internally processed. By targeting the foundational reasoning and decision-making dynamics of large language models, Model-intrinsic Methods form the backbone of many mitigation strategies.

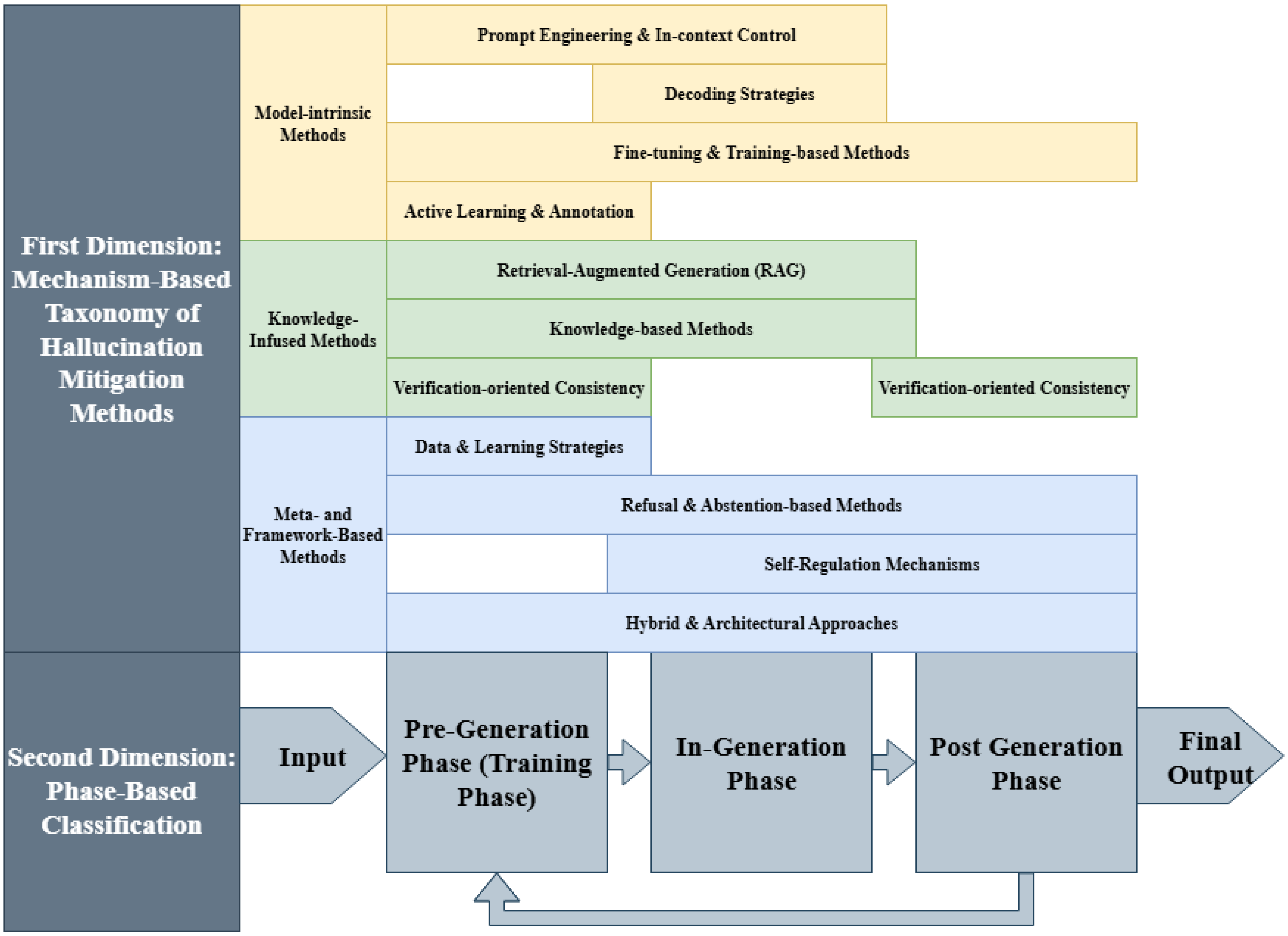

Fig. 4. Overall Integration (First and Second Dimension)

As illustrated in Fig. 5, this category can be further subdivided into four principal modes of intervention: (1) *Prompt Engineering & In-Context Control*, which designs and structures inputs to constrain the model's reasoning trajectory and minimize semantic drift. (2) *Decoding Strategies*, which refine the search and selection process during inference to balance accuracy, diversity, and factual grounding. (3) *Fine-Tuning & Training-Based Methods*, which recalibrate model parameters or adapt task-specific objectives to enhance factual alignment and reduce systematic error propagation. (4) *Active Learning & Annotation*, which improves data quality and coverage through targeted human or automated annotation, enabling models to learn from more reliable supervision signals.

In the subsequent subsections, each of these subgroups will be explored in detail, beginning with their conceptual motivations and extending to representative methodological implementations.

a) Prompt Engineering & In-Context Control: Prompt Engineering and In-Context Control represent one of the most direct and widely adopted strategies for hallucination mitigation. These approaches intervene at the input level of the model by shaping how information is presented, guiding the model's reasoning trajectory, and constraining potential divergence from factual or contextual truth [166], [167]. Prompt engineering focuses on the design and structuring of instructions, enabling models to better interpret task requirements and reduce ambiguity in generation. In parallel, uncertainty- or confidence-based prompting introduces adaptive control mechanisms, where prompt structures dynamically adjust based on the model's self-assessed reliability [166], [168]. Extending further, In-Context Learning (ICL) enhances model alignment by embedding demonstrations, reasoning chains, or reference exemplars within the input sequence [167], [169]. Subvariants such as Chain-of-Thought (CoT) prompting encourage explicit intermediate reasoning steps, while few-shot and zero-shot prompting exploit contextual exemplars to steer model predictions without requiring parameter updates [50], [141], [170].

- **Prompt Engineering** constitutes a foundational approach to hallucination mitigation by strategically crafting and structuring input instructions to optimize model behavior. Rather than altering model parameters, prompt engineering leverages the conditioning power of natural language instructions to constrain reasoning trajectories, enhance factual alignment, and mitigate the tendency to generate unsupported content [171]. These methods

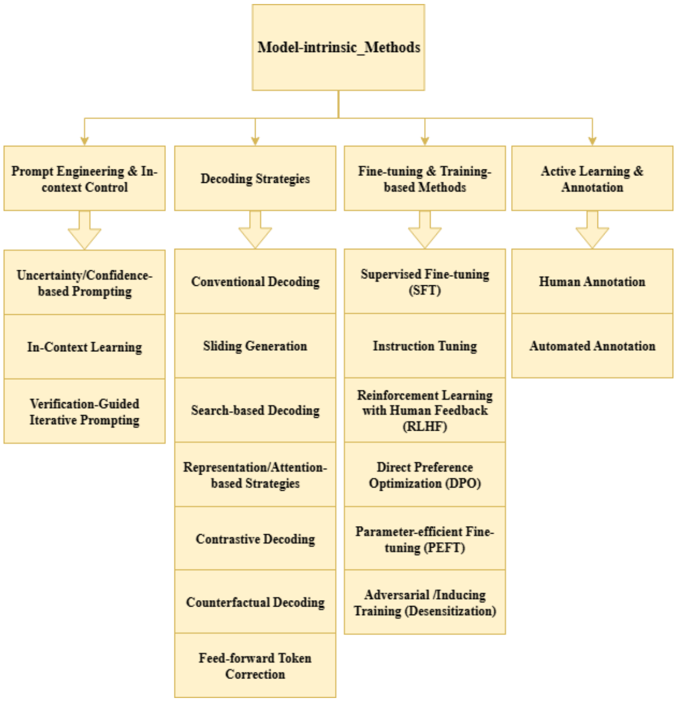

Fig. 5. Model-intrinsic Methods

often incorporate external knowledge cues, counterfactual interventions, or domain-specific grounding to reduce over-reliance on the model's parametric memory, which may be outdated or biased.

A variety of strategies have been proposed under this paradigm. For example, prompt-based methods structure queries with contextual evidence or verification checkpoints, sometimes embedding step-by-step reasoning directives that encourage the model to articulate intermediate logic rather than producing unsubstantiated conclusions [172], [173]. More advanced frameworks integrate search-enhanced prompting, such as Monte Carlo Tree Search (MCTS) with dynamic prompt adjustments, balancing exploration and exploitation to achieve more robust factual grounding [173]. Additionally, specialized prompt designs employ counterfactual keyword implantation or hierarchical representations to override misleading priors in pre-trained knowledge and align outputs with contextual truth [60], [174].

As a concrete instance of context-conditioned prompting, we next describe a tag-augmented scheme that enforces source-grounded generation and competence-aware abstention at inference time [175]. In this scheme, task context is augmented with embedded source tags (e.g., "(source 3626)") and an instruction to include sources, enabling recognition and flagging of out-of-domain behavior and reducing hallucinations without modifying model parameters [175]. Hallucination is based on no-context prompts using generated-URL validity as a proxy [175]. Supplying context markedly reduces fabricated URLs across engines (from 2,445 total, 1,605 invalid, to 48 total), and with tagged context, only 1 of 244 responses is unverifiable (approximately 99.88% reduction) [175]. When context and question are mismatched, models frequently state the mismatch rather than invent sources [175]. Limitations include susceptibility to "disregard context" instructions and to contaminated context, shifting the challenge to search/context evaluation [175]. Beyond textual conditioning, multimodal prompt engineering extends this principle to visual–textual integration, embedding cross-modal evidence to reinforce factual consistency in vision-language tasks [35]. Such techniques have demonstrated effectiveness in high-stakes domains, including scientific reasoning, financial forecasting, and clinical decision support, where hallucination carries disproportionate risk. By refining the interpretability and precision of model responses without requiring retraining, prompt engineering offers a lightweight, scalable, and task-adaptive mechanism for enhancing trustworthiness across both proprietary and open-weight large language models [47].

- **Uncertainty/Confidence-based Strategies** refers to adaptive input strategies that incorporate a model's self-assessed reliability into framework design, thereby reducing overconfident hallucinations and guiding corrective behaviors [176], [177]. Instead of relying solely on static instructions, these methods exploit uncertainty signals—derived from token probabilities, semantic entropy, or external confidence estimators—to trigger re-prompting, abstention, or supplementary evidence retrieval [177], [178]. For example, dynamic supplementation frameworks integrate real-time federated search results when uncertainty is detected, enabling more factually grounded outputs [174]. Similarly, Interactive DualChecker monitors model confidence in teacher–student distillation, re-prompting low-confidence teachers with enriched context and refining borderline student predictions via rationales [28].

Further innovations demonstrate the versatility of this paradigm across modalities and reasoning settings. In multimodal models, MemVR's "look-twice" mechanism reinjects visual tokens at points of high uncertainty, emulating human re-checking behavior to improve factual alignment [179]. For textual LLMs, studies of false premise hallucinations reveal that low-confidence predictions correlate with disruptions in specific attention heads, enabling prompt-based interventions to suppress such errors [103]. Confidence-guided debate frameworks like DebUnc further show that incorporating uncertainty into multi-agent prompting prevents convergence on confidently wrong answers [159], while abstention strategies use statistical or in-dialogue uncertainty thresholds to withhold unreliable responses, reducing hallucinations in unanswerable queries [58]. Collectively, these approaches transform prompt control into a dynamic, reliability-aware mechanism, enabling LLMs to self-correct or abstain in high-risk scenarios.

Another effective strategy is to directly mitigate the underlying factors that contribute to hallucinations. Paper [180] demonstrates that object hallucinations frequently arise because models are overly sensitive to both high- and low-frequency image features, causing them to mis-detect objects even in the absence of genuine visual cues. By explicitly addressing these root causes—namely redundant frequency-domain signals and spurious correlations—and suppressing unnecessary high- and low-frequency features during inference, hallucination rates can be significantly reduced [180]. The Multi-Frequency Perturbations (MFP) method has proven effective across multiple benchmarks, including CHAIR and POPE, and across diverse architectures such as CLIP, SigLIP, Vicuna, and LLaMA [180]. Moreover, when combined with inference-time techniques, it achieves state-of-the-art performance [180].

- **In-context Learning (ICL)** refers to a powerful paradigm in which large language models (LLMs) perform tasks by conditioning on examples or instructions provided in the prompt, without any parameter updates. This strategy leverages the LLM's pretrained latent knowledge, guiding output behavior through pattern-based inference rather than explicit training. Empirical evidence [142], [181] suggests that ICL allows models to compose internal task knowledge on the fly, effectively treating prompts as signals to retrieve or instantiate latent capabilities.

  – **Chain-of-Thought (CoT) Prompting** compels models to articulate intermediate reasoning steps prior to producing final outputs, thereby enhancing transparency and factual alignment. This decomposition of reasoning pathways reduces semantic drift and makes inconsistencies observable, enabling correction before they manifest as hallucinations [50], [182]. For example, Liang et al. introduce THAMES, a modular prompting framework that integrates CoT with verification layers to ensure consistency across multi-step reasoning [172]. Chang et al. propose a unified framework in which CoT decomposes complex tasks into sub-queries, aligning generation with external evidence [183]. Similarly, Ding et al. demonstrate that coupling CoT with retrieval-based augmentation enables reasoning chains to be explicitly grounded in document evidence, thereby curbing unsupported claims [184]. Collectively, these works illustrate how CoT transforms generation from a single-step output process into an iterative reasoning pipeline, enhancing factual fidelity and interpretability.

  – **Few-shot and zero-shot prompting** provide effective strategies to improve model reliability under low-resource settings [185]. Few-shot prompting introduces a handful of exemplars into the input prompt, enabling in-context learning without modifying model parameters, thereby significantly reducing the dependence on task-specific training data [141], [185]. In contrast, zero-shot prompting relies on carefully crafted instructions to guide model behavior when no exemplars are available [186].

- **Verification-Guided Iterative Prompting** introduces formal verification feedback into the prompting loop to iteratively steer LLMs toward factually consistent outputs [187]. In this paradigm, the LLM is first treated as a possibly hallucinating oracle whose outputs are automatically checked by a formal verifier against correctness specifications. Counterexamples or error explanations from the verifier are then appended to the next prompt, guiding the model away from previously invalid reasoning trajectories. Through repeated counterexample-guided refinements, the LLM converges toward a response that satisfies the verifier, effectively "dehallucinating" the model without altering its parameters. This approach demonstrates particular promise in safety-critical domains such as automated planning, where iterative correction is essential for reliability [187].

  Both approaches complement **Chain-of-Thought (CoT)** reasoning by embedding demonstrations or explicit task instructions directly into the prompt space. Empirical studies show that exemplar-driven instructions enhance factual grounding even under resource-constrained conditions [172], while refined zero-shot prompting with explicit topical constraints mitigates hallucinations in tasks such as topic modeling by reducing duplication and fabricated outputs [47]. These findings confirm that prompt design itself functions as a structural constraint on generative space, making ICL a lightweight yet versatile mechanism for hallucination control across domains.

  Another **context-based** hallucination mitigation method is to incorporate external context into the post-training supervision process [188]. The post-training supervision mechanism in Context-Informed Grounding Supervision (CINGS) appends external context during learning while masking *context tokens* from the loss [188]. In this way, CINGS shifts the model's reliance from internal parametric knowledge to external evidence, computes loss only over the *response tokens*, and prevents rote memorization of context, thus improving grounding in both text and vision-language domains [188].

  More broadly, this fits into the Context-Engineering framework, where Parameter-Efficient (PE) fine-tuning and Context Control strategies serve as lightweight mechanisms to steer model behavior without full retraining [188]. Context Control regulates how evidence is introduced and weighted during inference [188]. Within this continuum, CINGS acts as a post-training learning objective, providing a direct supervision signal that complements inference-time context engineering [188]. Together, these approaches form a cohesive strategy to strengthen grounding and mitigate hallucinations [188].

b) Decoding Strategies: form a class of hallucination mitigation techniques that intervene during the inference phase, directly shaping token selection and sequence construction without altering model parameters [41], [189], [190]. Unlike prompting, which conditions input design, decoding methods regulate generation dynamics in real time, making them particularly effective for efficient and adaptable deployment [41], [190]. By constraining candidate outputs, introducing verification mechanisms, or reweighting internal representations, these strategies promote factually grounded and coherent text [40], [191], [192]. Approaches range from *conventional decoding* methods such as greedy, beam, and sampling, to advanced techniques like sliding-window generation [193], search-based frameworks (e.g., MCTS, Tree-of-Thought), attention-guided decoding, contrastive and counterfactual reasoning, and feed-forward token correction [173].

- **Conventional decoding** algorithms constitute the foundational mechanisms by which large language models transform probabilistic token distributions into coherent sequences [41], [194]. Among the most widely adopted are greedy decoding [41], which selects the highest-probability token at each step to ensure determinism and efficiency, and beam search [43], which expands multiple candidate paths simultaneously to balance optimality and diversity in the generated sequence. In contrast, sampling-based approaches, including top-k [189] and nucleus (top-p) sampling, introduce stochasticity to the selection process, thereby enhancing creativity and reducing repetitive outputs. While these methods are not explicitly designed for hallucination mitigation, their influence on sequence exploration has direct implications for factual reliability [194]. Greedy decoding, for instance, minimizes randomness but risks amplifying biases embedded in parametric memory, whereas beam search can produce overly generic responses that lack grounding [41]. Sampling methods, though capable of generating diverse outputs, often increase the likelihood of factual deviations if not carefully constrained [190]. Consequently, conventional decoding forms the baseline upon which more advanced hallucination-aware strategies such as contrastive decoding or attention-guided interventions have been developed, highlighting the trade-offs between determinism, diversity, and factual alignment in sequence generation [190].
- **Sliding Generation** proposed by Harrington, is a structured decoding strategy specifically designed to mitigate hallucinations in long-sequence generation by segmenting the input into overlapping windows, each processed independently [195]. This segmentation constrains the model's focus to smaller, contextually coherent fragments, thereby alleviating cognitive overload and improving memory retention—two factors that often degrade performance in extended reasoning tasks [195]. Once partial outputs are produced for each window, the overlapping regions are aligned and merged to maintain discourse continuity and semantic consistency across the entire sequence [195]. By decomposing complex prompts into manageable sub-units, Sliding Generation reduces context dilution, promotes local coherence, and anchors generation in factual information [195]. These properties make it particularly effective for long-form narrative generation, multi-hop reasoning, and domains where maintaining logical fidelity across extended spans of text is critical [191], [195].
- **Search-based Decoding** approaches, such as Monte Carlo Tree Search (MCTS), Tree-of-Thoughts (ToT), and Uncertainty-Aware Beam Search (UABS), extend the decoding process into structured, exploratory search spaces to enhance multi-step reasoning and reduce hallucinations [173], [196], [197]. Unlike conventional decoding methods that operate greedily on token probabilities, these techniques reframe generation as a search problem over reasoning trajectories, allowing the model to evaluate, revise, and backtrack across multiple candidate paths before finalizing an output [173]. This structured exploration provides stronger guarantees of factual grounding and logical consistency, particularly in long-form text generation, multi-hop question answering, and planning-intensive tasks [196]. **Monte Carlo Tree Search (MCTS)** combines probabilistic sampling with a four-stage search procedure (selection, expansion, simulation, and backpropagation) to explore possible continuations of text [173]. Each node in the search tree corresponds to a partial generation, with simulations estimating downstream utility to guide the search toward factually robust and coherent outputs [173]. By systematically balancing exploration and exploitation, MCTS enables the model to converge on responses that are both fluent and trustworthy [173]. **Tree-of-Thought (ToT)** proposed by Shunyu Yao et al., decomposes the reasoning process into discrete steps of intermediate "thoughts," which are organized into a tree structure representing the evolution of reasoning states [196]. Each node in the tree corresponds to a state $s = [x, z_1, \cdots, z_i]$ where x is the original input and $z_1 \cdots z_i$ are the thoughts generated thus far [196]. The model proposes multiple continuations for each thought step, allowing the tree to branch into diverse reasoning directions [196]. Together, MCTS [173] and ToT [196] demonstrate how search-based decoding reframes inference as a structured reasoning problem, offering a principled means to mitigate hallucinations while enhancing transparency and robustness in complex generation tasks. Those methods demonstrate how search-based decoding reframes inference as a structured reasoning problem, offering a principled means to mitigate hallucinations while enhancing transparency and robustness in complex generation tasks.
- **Representation/Attention-based Strategies** help reduce the hallucination rate through Inference-Time Intervention [198]–[200]. These methods operate directly on a model's internal dynamics: representation editing mod-

ifies hidden states to amplify truthful signals or suppress misleading patterns [198], [199], while attention-based strategies regulate how focus is distributed across context tokens to ensure fact-critical information is prioritized [200]. By intervening inside the model rather than at the input or output level, they provide fine-grained, inference-time control that complements external grounding techniques and directly curbs hallucination at its source [198]–[200].

Representation-based Methods intervene directly in the hidden representations of LLMs to refine internal information before token prediction [199].

For example, TruthX introduces an auto-encoder to disentangle hidden representations into semantic and truthful latent spaces [198]. By applying contrastive learning to identify a "truthful editing direction" within this space, the method can intervene during inference by nudging hidden states toward more truthful regions [198]. Experiments on 13 advanced LLMs demonstrate an average 20% improvement in truthfulness on the TruthfulQA benchmark, and further analyses reveal that even editing a single vector is sufficient to steer outputs between truthful and hallucinatory responses [198].

Building on this direction, Expanding before Inferring (PLI) introduces Premature Layers Interpolation, a training-free and plug-and-play framework that constructs new premature layers by interpolating hidden states across adjacent layers [199]. Inspired by diffusion and sampling, this approach extends the depth of internal information processing, enhancing factual coherence without fine-tuning overhead [199]. Unlike input- or output-level interventions, PLI directly strengthens internal representation refinement, mitigating hallucinations at their source [199]. Experiments on four widely used benchmarks show that PLI outperforms baselines in most cases and can be combined with other mitigation techniques [199]. Further analysis indicates that its success is closely tied to LLMs' internal mechanisms, making PLI a lightweight yet effective strategy for hallucination reduction [199].

Attention-based Methods leverage the internal attention maps of transformer models, which are visualizations of how focus is distributed across tokens in an input sequence, to analyze and regulate the model's information processing during generation [200]. Attention maps expose salience attribution by indicating which tokens are considered most relevant at each decoding step, thereby offering insight into how syntactic and semantic dependencies are captured across layers [200].

Recent studies demonstrate that hallucinations often arise from misalignments in these patterns, such as over-attention to peripheral context or under-attention to fact-critical tokens [201]. By directly intervening at the attention level—through re-weighting, scaling, or dynamically reallocating focus—models can be guided to prioritize factually salient information while suppressing irrelevant signals [163], [201]. For example, attention map optimization techniques adjust weights during inference to strengthen the influence of crucial context tokens and mitigate noise, resulting in outputs that exhibit improved factual consistency, contextual grounding, and semantic coherence [200].

This fine-grained control over the model's focus makes attention-based decoding strategies particularly effective for knowledge-intensive tasks where accurate attention allocation is essential to suppress hallucinations and ensure faithful adherence to input context [163], [200], [201].Beyond attention re-weighting, architectural modifications such as *Concentric Causal Attention (CCA)* reorganize visual token positions and redesign causal masks to counteract the long-term decay of rotary position encoding (RoPE), thereby strengthening cross-modal alignment and significantly reducing object hallucinations in LVLMs [202].In addition, *OPERA* introduces an over-trust penalty on decoding logits combined with a retrospection-allocation mechanism, which detects attention aggregation patterns that correlate with hallucinations and dynamically re-routes generation to suppress them, offering a training-free and generalizable decoding-time mitigation strategy [203].

- **Contrastive Decoding (CD)** is a decoding-time strategy that leverages the comparative strengths of two language models of differing capacities to enhance factual reliability and reduce hallucinations [157], [204]. The underlying intuition is that smaller models often surface generic or hallucination-prone tokens, whereas larger models tend to capture more semantically consistent and factually grounded continuations [37], [158]. CD operationalizes this contrast by dynamically reweighting token probabilities: outputs disproportionately favored by the smaller model are penalized, while those preferred by the larger model are promoted [39]. This rebalancing mechanism allows the decoding process to suppress low-quality continuations without modifying the model's architecture or retraining [39]. Extensions such as Induce-then-Contrast Decoding (ICD) refine this principle by deliberately inducing a hallucination-prone variant of the base model to act as a contrastive signal, further amplifying the suppression of unreliable outputs [205]. By enforcing this fine-grained competition, CD yields more precise, coherent, and trustworthy generations in an inference-efficient manner.

  Similarly, Active Layer-Contrastive Decoding (ActLCD) extends the contrastive decoding paradigm by treating generation as a sequential decision-making problem [206]. Whereas prior methods like DoLa statically contrast shallow and deep layers at every token, ActLCD employs a reinforcement-learned policy guided by a reward-aware classifier to decide when to activate layer-wise contrastive signals dynamically [206]. This adaptive mechanism leverages factual representations encoded in deeper layers while avoiding unnecessary "overthinking" for simpler tokens, thereby mitigating the hallucination snowballing from autoregressive error propagation [206]. Unlike standard CD or ICD, ActLCD requires no auxiliary models, induced variants, or multiple samplings, operating instead in a single forward pass using only model-internal information [206]. Empirical results across five benchmarks—including TruthfulQA, LongFact, StrategyQA, GSM8k, and a domain-specific software hallucination suite—show consistent factuality improvements over state-of-the-art baselines, demonstrating robust gains from sentence-level answers to long-form and specialized domains [206].

While these methods focus on inference-time contrastive mechanisms, similar principles have also inspired training-time alignment strategies [207]. Atomic Consistency Preference Optimization (ACPO) introduces a self-supervised alignment framework that leverages contrastive signals across stochastic generations [207]. Instead of relying on external models or curated datasets, ACPO identifies atomic consistency signals—the degree to which individual factual units remain stable across multiple sampled responses—and automatically constructs contrastive pairs of high-quality (consistent) versus low-quality (inconsistent) data [207]. These pairs are then used for preference optimization, aligning the model toward factual outputs without external supervision [207]. Empirical evaluations show that ACPO outperforms strong supervised baselines such as FactAlign, achieving gains of +1.95 points on both the LongFact and BioGen benchmarks [207]. By grounding contrastive learning in self-consistency rather than external labels, ACPO provides a scalable, cost-efficient, and effective pathway to reduce factoid hallucinations, complementing inference-time contrastive decoding strategies with a training-time alignment perspective [207].

- **Counterfactual Decoding** introduces a causal inference perspective into the decoding process to distinguish contextually entailed knowledge from spurious associations, thereby reducing hallucination-prone outputs [25]. The approach operates by generating counterfactual variants of the original input through semantic perturbations such as negation, entity substitution, or discourse restructuring while keeping other factors constant [60]. By contrasting the model's predicted logits across true and counterfactual contexts, the method subtracts the latter from the former, effectively downweighting unsupported continuations and amplifying tokens that are causally grounded in the given input [25], [60], [163]. This counterfactual subtraction mechanism ensures that the generation process adheres closely to prompt-specific information, mitigating hallucinations driven by overgeneralization or parametric bias [60]. Extending beyond unimodal applications, multimodal adaptations construct counterfactuals by perturbing visual-semantic alignments—for example, masking critical image regions or mismatching cross-modal features—thereby addressing modality-prior-induced hallucinations and strengthening intermodal factual grounding [163].
- **Feed-forward Token Correction (FFNs)** exploits the structural and functional properties of feed-forward networks (FFNs) within transformer architectures to revise or re-rank token predictions during decoding, thereby enhancing factual alignment [208], [209]. In transformer models, FFNs function as nonlinear composition layers that can be conceptualized as key–value memory modules: keys are tuned to specific input patterns, while values shape the output distribution over tokens [192]. Lower-layer FFNs primarily capture syntactic or surface-level regularities, whereas upper layers encode higher-order semantic abstractions that critically influence final token selection [210]. Token correction methods intervene in these upper layers to refine intermediate representations according to contextual salience or factual grounding constraints. For example, the *Look-Twice* mechanism [179] introduces a two-pass decoding procedure in which preliminary outputs are projected back into FFN memory space and then reinterpreted through a secondary decoding pass. This iterative refinement allows the model to identify and correct semantically inconsistent or unsupported tokens in real time, without necessitating retraining or architectural modification [179]. Consequently, FFN-based correction represents a lightweight yet powerful strategy for hallucination mitigation, operating deep within the generative pipeline to ensure more reliable and semantically coherent outputs [179]. Beyond token-level interventions, recent work also explores **hidden state manipulation** through activation engineering, which leverages directional offsets in the model's hidden representation space to steer outputs away from hallucinated responses and toward factual ones [211]. Unlike contrastive decoding, which constrains generation by comparing probability distributions across models or decoding paths, this approach directly operates on internal hidden states, providing a complementary and model-intrinsic pathway for hallucination mitigation.

c) Fine-Tuning and Training-based Methods: represent a fundamental class of hallucination mitigation strategies that intervene at the level of model parameters, embedding factual consistency, task alignment, and human-preferred behaviors directly into the underlying probability distributions of large language models (LLMs). In contrast to inference-time interventions such as decoding strategies, which adjust outputs dynamically without altering the model's core, fine-tuning reshapes the model's internal representations through targeted optimization [166], [167].

This process is typically grounded in high-quality, domain-specific corpora or preference-based annotations, enabling the model to gradually internalize truth-grounded patterns and robust decision-making heuristics [47], [164], [212], [213]. Depending on the approach, fine-tuning may involve comprehensive full-parameter updates, selective adaptation through parameter-efficient methods, or adversarial training designed to desensitize models against hallucination-inducing prompts [212], [213]. By systematically aligning internal knowledge structures with external truth signals, fine-tuning provides a durable and generalizable mechanism for mitigating hallucinations, albeit at the cost of greater computational and data requirements compared to lighter inference-time techniques [212], [213].

- **Supervised Fine-Tuning (SFT)** is a foundational approach in the alignment of large language models, designed to calibrate model behavior against human expectations by training on carefully curated, annotated datasets containing factually accurate examples [198]. Often serving as the initial stage in alignment pipelines such as RLHF, SFT provides direct exposure to truth-grounded responses and desirable output structures, thereby establishing a baseline of factual reliability that enhances the efficacy of subsequent preference- or reward-based optimization [198], [214]. In the context of hallucination mitigation, SFT has proven effective in shaping the model's generative distribution toward verifiable content [198]. For example, Xia et al. fine-tune on a diversity-aware dataset explicitly constructed to capture hallucination-prone instances across varying factual densities, which enables models to generalize more robustly in real-world generation tasks [36]. Similarly, Qu et al. employ SFT on factual–counterfactual data pairings, strengthening the model's capacity to differentiate truth from fabrication and ensuring that generated outputs remain anchored in verifiable knowledge [215].
- **Faithfulness-based Loss Function** represents a training-oriented approach that augments conventional optimization with loss terms explicitly penalizing factually inconsistent or unfaithful generations. By directly encoding fidelity constraints into the objective function, these methods steer models toward producing outputs that remain more accurate and aligned with the source material. Such formulations complement supervised fine-tuning and preference optimization by embedding factual consistency as a first-class optimization target, strengthening reliability across diverse tasks [55].
- **Instruction tuning** refines language models by fine-tuning them on carefully constructed instruction–response pairs, thereby enhancing their capacity to follow user directives while mitigating hallucinations [212], [213]. Unlike generic supervised fine-tuning, this approach explicitly incorporates both positive and negative examples, enabling models to recognize and avoid fabricated or unsupported content [213], [216]. By leveraging robust datasets that span diverse tasks and include semantically challenging negative samples, instruction tuning helps counteract representational biases and strengthen alignment with factual or visual inputs [212]. Moreover, techniques such as adversarial augmentation further improve model robustness, ensuring greater factual accuracy in dialogic exchanges and other core applications [212]. This scalability makes instruction tuning a critical strategy for reliable deployment in domains such as open-domain querying and vision–language reasoning [212].
- **Hallucination-Induced Optimization (HIO)** introduces a contrastive optimization framework that leverages a fine-tuned Contrary Bradley–Terry Model to amplify the margin between hallucinated and correct tokens. By explicitly penalizing hallucination-prone generations and rewarding faithful ones, HIO strengthens contrastive decoding signals and substantially reduces hallucinations in large vision–language models. Positioned at the intersection of preference optimization and contrastive learning, this method extends existing fine-tuning strategies with a targeted mechanism for factual alignment in multimodal settings [217]
- **Reinforcement Learning with Human Feedback (RLHF)** represents a multi-stage fine-tuning paradigm that integrates human evaluative signals into the training loop to enhance factual alignment and reliability [34], [218]. Building on the foundation of Supervised Fine-Tuning (SFT), RLHF first constructs a reward model trained on human-annotated pairwise comparisons of model outputs [46]. This reward model serves as a proxy for human judgment, guiding reinforcement learning updates that optimize the language model's policy toward generating outputs judged more helpful, truthful, and aligned with user intent [46], [219], [220]. While RLHF has been instrumental in advancing the factuality and safety of large language models, it is also associated with significant computational costs, instability in the presence of poorly specified reward functions, and high sensitivity to hyperparameter tuning [34], [218]. These challenges have motivated research into more efficient alternatives, such as Direct Preference Optimization (DPO) [219]. Within the specific context of hallucination mitigation, RLHF has proven effective in reducing factually inconsistent outputs by incorporating reward signals that explicitly penalize hallucinations [34], [218]. For example, Wang et al. apply RLHF to align models with human factuality preferences across diverse domains, demonstrating that reinforcement-driven policy refinement can substantially suppress hallucination-prone behaviors [221].
- **Direct Preference Optimization (DPO)** is a preference-based fine-tuning approach that aligns large language models with human judgments while bypassing the complexities of reinforcement learning and explicit reward modeling [219]. By reframing preference learning as a binary classification problem, DPO optimizes model behavior through cross-entropy loss, directly increasing the likelihood of preferred responses relative to dispreferred ones [219]. Compared to RLHF, DPO offers greater simplicity, stability, and computational efficiency, making it a scalable alternative for preference alignment [222]. In the domain of hallucination mitigation, DPO has shown

particular promise in enhancing factual grounding without incurring the instability of reward-based optimization [222]. For instance, Mu employs DPO to address topic-specific hallucinations by systematically correcting outputs that deviate from factual granularity [47], while Xiao integrates DPO with hallucination detection signals, demonstrating its adaptability under noisy or imperfect supervision [164].

- **Parameter Efficient Fine-Tuning (PEFT)** mitigates hallucinations by adapting large language models through minimal parameter updates, such as adapter layers, prefix-tuning, or low-rank transformations [223]. By restricting optimization to select submodules, PEFT significantly reduces computational overhead while embedding factual constraints into the model's representations with minimal disruption to its general capabilities [223]. This makes PEFT particularly attractive in resource-constrained or domain-specific contexts, where full fine-tuning is prohibitively expensive [172]. Nonetheless, its effectiveness critically depends on the availability of high-quality supervision and the careful design of parameter-efficient modules, as poorly configured adaptations may fail to capture the nuanced signals required for robust hallucination suppression [223]. A representative example is mmT5, which extends parameter-efficient principles through a modular multilingual design that introduces language-specific bottleneck modules and partially frozen decoder parameters [224]. By structurally separating language-specific from language-agnostic representations, mmT5 mitigates representation drift during fine-tuning and effectively suppresses source-language hallucinations, achieving stable cross-lingual generalization in zero-shot generation [224].
- **Adversarial/Inducing Training (Desensitization)** builds on the principle of adversarial robustness by systematically exposing language models to adversarial or deliberately perturbed examples, thereby reducing their susceptibility to hallucination-inducing inputs [212], [213]. The core idea is to train models under adversarial pressure so that they learn to discriminate between factual and misleading cues, effectively desensitizing them to spurious correlations [225].
  In the hallucination mitigation context, Park et al. apply adversarial augmentation to dialogue systems by integrating semantically challenging negative examples, enabling the model to recognize and suppress fabricated content while enhancing its resilience to misleading conversational cues [213]. Similarly, Zhang et al. introduce a contrastive adversarial fine-tuning approach that induces controlled hallucinations and then penalizes them through preference optimization, strengthening the model's ability to reject unsupported continuations [205]. By combining adversarial exposure with desensitization training, these methods foster more robust alignment with factuality, particularly in domains where models are vulnerable to subtle semantic distortions.

d) Active Learning & Annotation: constitute upstream strategies for hallucination mitigation, aimed at strengthening the supervision signals that guide training or fine-tuning of large language models [226], [227]. By curating informative, diverse, and high-uncertainty examples, these approaches expose model deficiencies and enable more targeted updates, ultimately improving factual grounding and generalization [36]. Such methods are particularly valuable in resource-constrained or noisy data environments, where the efficiency and quality of supervision critically shape alignment outcomes [226], [227].

- **Human Annotation** remains the gold standard for high-quality supervision. Leveraging expert or crowd-sourced evaluation, it enables fine-grained identification of factual inaccuracies and nuanced misalignments across diverse tasks and domains [228]. Xia et al. emphasize that human-labeled examples play a pivotal role in calibrating hallucination detection models and establishing reliable ground truth for supervised fine-tuning and active learning pipelines [36]. However, despite its precision, human annotation faces challenges of high cost, labor intensity, and inter-annotator variability, which constrain its scalability [228].
- **Automated Annotation** provides scalable alternatives by employing model-driven heuristics, rule-based verification, or weak supervision techniques to generate labels across large corpora [229]. These systems detect inconsistencies, factual mismatches, or low-confidence responses, enabling rapid dataset construction for downstream training [229], [230]. Nevertheless, automated pipelines risk introducing label noise, necessitating post-processing steps such as filtering, confidence reweighting, or hybrid human-in-the-loop mechanisms to ensure reliability and effectiveness [230].

In sum, Model-intrinsic Methods provide direct interventions at the level of prompts, decoding, parameter optimization, and data supervision, thereby shaping the intrinsic generative dynamics of large language models. While these approaches enhance factual alignment from within the model's architecture and training pipeline, their effectiveness is often constrained by the limits of parametric knowledge and internal reasoning [33], [48].

To mitigate these limitations, increasing attention has been focused on *Knowledge-Infused Mitigation methods*, wherein model outputs are systematically grounded in external information sources to improve the accuracy of facts and reduce hallucinations. By leveraging retrieval systems, structured knowledge graphs, and verification mechanisms, these methods complement technical strategies by supplying reliable, up-to-date, and domain-specific factual signals that transcend what the model has memorized during pretraining [35], [48], [231].

2) Knowledge-Infused Mitigation Methods: Knowledge-Infused Mitigation Methods constitute the second category of our mechanism-based taxonomy, focusing on strategies that explicitly anchor large language models to external sources of factuality and structured knowledge. Operating at the extrinsic

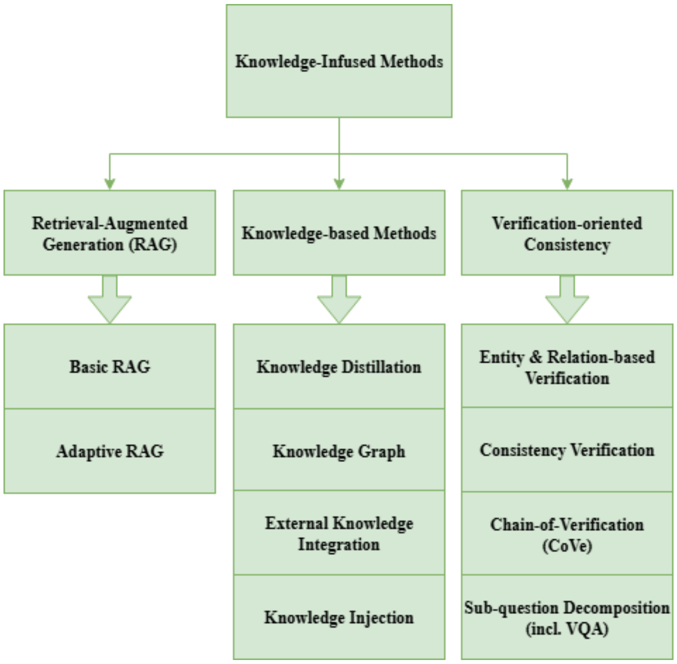

Fig. 6. Resource_Knowledge-based Methods

level of the generation process, these methods complement intrinsic adjustments by ensuring that outputs remain verifiable against reliable references [231], [232]. Rather than relying solely on parametric memory, they ground model reasoning in curated resources, thereby reducing semantic drift and enhancing factual robustness [8]

As illustrated in Fig. 6, this category can be further divided into three principal modes of intervention: (1) *Retrieval-Augmented Generation (RAG)*, which enriches the generative process by integrating relevant documents or passages retrieved from external corpora, with adaptive variants tailoring retrieval scope to task complexity [48], [184], [231]. (2) *Knowledge-grounding Methods*, which embed factual constraints into model reasoning by leveraging structured resources such as knowledge graphs, external knowledge base integration, and mechanisms of knowledge injection or distillation [233]. (3) *Consistency-guided Verification*, which validates outputs through entity- and relation-based checks, factual consistency verification, and multi-step procedures like Chain-of-Verification (CoVe) or sub-question decomposition, ensuring that final outputs remain logically coherent and factually grounded [61], [234].

In the subsequent subsections, each of these subgroups will be examined in detail, beginning with their conceptual underpinnings and extending to representative methodological implementations.

a) Retrieval-Augmented Generation (RAG): has emerged as a cornerstone technique for mitigating hallucinations in large language models (LLMs) by directly addressing the limitations of parametric memory. Since LLMs are constrained by the scope and temporal boundaries of their training data, they often produce outdated or fabricated content when confronted with knowledge-intensive tasks [48], [231]. RAG alleviates this problem by coupling generative models with a retrieval module that dynamically accesses external knowledge bases or document corpora, thereby enriching the generation process with timely, verifiable, and contextually relevant information [127]. Through this hybrid design, RAG extends the reasoning capacity of LLMs beyond static parameters, grounding outputs in external evidence while preserving fluency and adaptability [231], [235], [236]. Importantly, the choice of knowledge sources, ranging from open-domain corpora to specialized domain repositories as well as the integration stage within the inference pipeline, gives rise to distinct variants of RAG [237], [238]. These include *basic retrieval-augmented methods*, which provide general factual reinforcement, and *adaptive RAG approaches*, which dynamically modulate retrieval strategies according to task requirements [184]. The following subsections examine these branches in detail, highlighting their methodological innovations and implications for hallucination mitigation.

- **Basic RAG** represents the foundational paradigm in retrieval-based mitigation, wherein a retriever component dynamically supplements a large language model (LLM) with external knowledge at inference time. The retriever first encodes the input query and searches a pre-indexed corpus—commonly large-scale resources such as Wikipedia or domain-specific databases—for the top-k relevant documents [231], [236]. The retrieved content is then integrated with the input prompt and provided to the generator, which conditions its response on both the internal parametric knowledge and the external evidence [231], [236]. This architecture enables models to overcome the static limitations of pretraining corpora by incorporating factual, up-to-date, and contextually grounded information at generation time [236]. One of the core strengths of Basic RAG lies in its ability to reduce hallucinations that stem from the probabilistic nature of LLM decoding and incomplete memorization of facts [236]. By grounding responses in retrieved passages, Basic RAG improves factual precision, recall, and coherence, particularly in knowledge-intensive tasks such as question answering and knowledge-grounded dialogue [232]. Furthermore, empirical studies demonstrate that even simple implementations—where retrieval operates in a one-pass manner without adaptive control or iterative refinement—can substantially mitigate the generation of fabricated or outdated content, enhancing model trustworthiness in real-world applications [27], [174]. As the canonical form of retrieval-augmented architectures, Basic RAG establishes the methodological baseline upon which more advanced variants, such as adaptive retrieval or consistency-guided verification, have been developed [49], [184]. It thus provides both a practical framework for reducing hallucinations and a conceptual foundation for subsequent innovations in retrieval-anchored generation systems.

- **Adaptive RAG** refines the conventional RAG framework by dynamically deciding when and how to incorporate external knowledge, thereby balancing factual grounding with computational efficiency [184], [239], [240]. Unlike standard RAG, which indiscriminately performs retrieval for all queries, Adaptive RAG introduces a decision module that estimates whether the model's internal knowledge suffices or whether external evidence is required [184]. When internal predictions are judged unreliable—often through uncertainty estimation, factuality scoring, or hallucination detection signals—the system selectively retrieves and integrates supporting documents to reinforce the generation process [184]. This conditional retrieval not only suppresses hallucinations but also reduces latency and redundancy by avoiding unnecessary database queries [184]. Recent advances further extend this paradigm with adaptive confidence calibration and reinforcement-guided retrieval policies, enabling models to dynamically allocate retrieval resources across diverse tasks and domains [240]. For instance, the COFT framework introduces a coarse-to-fine highlighting mechanism that leverages knowledge graphs and contextual weighting to dynamically emphasize key lexical units, significantly improving hallucination mitigation in long and noisy contexts [241]. Collectively, Adaptive RAG demonstrates that hallucination mitigation can be achieved more effectively by integrating retrieval as a context-sensitive intervention, rather than as a uniform pre-processing step [184], [239].

b) Knowledge-based Methods: constitute a central strand of hallucination mitigation strategies that ground large language models (LLMs) in structured or semi-structured factual resources [242]. Unlike retrieval-augmented generation, which dynamically supplements model outputs with external documents at inference time, knowledge-based approaches integrate curated repositories more intrinsically into the reasoning and generation process [233]. By anchoring outputs to explicit semantic structures or verifiable factual stores, these methods constrain the model's generative space and suppress unsupported or fabricated continuations.

This category can be further divided into four major subtypes: (1) *Knowledge Graphs* [243], which encode entities and relations as structured graphs to enforce logical and factual consistency; (2) *External Knowledge Base Integration* [236], [244], which links models to domain-specific or general-purpose repositories such as encyclopedias, scientific corpora, or medical databases; (3) *Knowledge Injection* [233], [245], which embeds factual cues directly into model representations through lexical, syntactic, or embedding-level augmentation; and (4) *Knowledge Distillation* [246], which transfers factual grounding from stronger or specialized teacher models to student models via supervised training. Collectively, these approaches emphasize the role of structured factual grounding as a complement to probabilistic generation, providing reliable mechanisms for mitigating hallucinations across diverse tasks and application domains.

- **Knowledge Distillation** (KD) is a transfer learning paradigm in which a compact student model is trained to emulate the predictive behaviors or intermediate representations of a more capable teacher model [246]. While originally introduced for model compression, KD has increasingly evolved into a crucial method for hallucination mitigation [246]. By transferring factual grounding, consistency, and reasoning strategies from teacher to student, KD enables lightweight models to benefit from the robustness of large-scale LLMs without requiring direct access to proprietary corpora or costly reinforcement learning pipelines [246]. In this sense, KD serves as both an efficiency mechanism and an alignment strategy, embedding verifiable knowledge into models that are more computationally accessible [246].
  Recent works have extended KD into diverse settings for hallucination reduction. Sharma et al. adapt KD to zero-resource environments, showing that teacher-generated rationales and structured signals allow students to internalize verifiable reasoning even in the absence of labeled data [247]. McDonald et al. demonstrate that transferring calibrated probability distributions from robust teachers enables student models to filter out overconfident but factually incorrect outputs, enhancing factual alignment under noisy conditions [248]. Complementing these approaches, Nguyen et al. propose smoothing-based distillation, where entropy-constrained teacher signals reduce spurious correlations and strengthen factual precision in low-resource environments [85]. Collectively, these advances underscore KD as a scalable pathway for embedding structured teacher knowledge into student models, thereby improving factual reliability and mitigating hallucinations across diverse deployment scenarios.
- **Knowledge Graph** is a multi-relational structure that encodes factual knowledge as triples of entities and relations, providing a semantically rich and machine-readable framework for reasoning Unlike unstructured text corpora, KGs impose a structured representation where entities are represented as nodes and relations as typed edges, thereby enabling systematic inferencing, entity disambiguation, and relational grounding [243], [249]. This relational architecture makes KGs particularly well-suited for hallucination mitigation, as they offer explicit mappings between concepts and their attributes that LLMs can leverage for factual consistency. For example, in SPARQL-based question answering, hallucinations often arise when models attempt to generate Uniform Resource Identifiers (URIs) from parametric memory, leading to non-existent or incorrect entity identifiers. By integrating external KG-based retrieval or alignment modules, recent work demonstrates that grounding model generations in structured knowledge repositories substantially reduces hallucinated outputs and strengthens factual reliability [249].

Building on this foundation, recent studies have explored various integration strategies. Guan et al. [49] propose a dual-phase alignment mechanism that incorporates KG entities to constrain semantic drift during generation. Similarly, Niu et al. [240] introduce a causal alignment approach, where KG reasoning paths are explicitly modeled to ensure output faithfulness. Yu et al. [25] extend this paradigm by leveraging cause–effect look-alike reasoning to detect and suppress hallucinations that arise from spurious correlations, showing how KG structures can be used not only for grounding but also for counterfactual validation of model outputs. Collectively, these approaches highlight KGs as both a factual anchor and a reasoning substrate, capable of transforming hallucination mitigation from post-hoc detection into proactive semantic alignment. Consequently, knowledge graph integration has emerged as one of the most effective strategies for addressing hallucinations in LLMs and LVLMs, especially in tasks requiring precise entity disambiguation and multi-hop reasoning

- **External Knowledge Integration** extends the generative capacity of language models by supplying them with accurate, up-to-date, and domain-specific information from external repositories. Unlike knowledge graphs that rely on structured relational reasoning, external sources such as Wikipedia, curated databases, and domain-specific corpora provide rich unstructured evidence that can be dynamically retrieved and embedded into the generation process. This augmentation enhances the model's ability to handle knowledge-intensive tasks while mitigating hallucinations caused by outdated or incomplete internal representations [232], [236].

  Recent advances move beyond static retrieval pipelines by introducing dynamic and adaptive mechanisms. For example, Chen et al. propose a context-aware dynamic supplementation framework that identifies hallucination-prone responses and selectively enriches them with retrieved evidence, thereby improving factual alignment without excessive overhead [174]. By flexibly integrating external evidence only when needed, these methods balance efficiency with accuracy, ensuring that external knowledge enhances the reliability of model outputs rather than overwhelming the inference process. Collectively, external knowledge integration demonstrates strong potential to ground large language models in verifiable information and to adapt seamlessly across evolving domains.
- **Knowledge Injection** (KI) represents a targeted approach to bridging the gap between structured symbolic knowledge and language model generation. Unlike retrieval-augmented methods that rely on external document search, KI directly incorporates curated knowledge elements—often sourced from knowledge graphs or curated databases—into the model's input space [245]. Typically, this involves transforming relevant entity or relation triples into natural language or embedding-compatible formats and inserting them into prompts, thereby ensuring that the model is exposed to domain-specific, factual signals during inference. This mechanism is particularly effective for grounding responses in contexts where the pretraining corpus lacks sufficient coverage, such as medical or legal domains [29].

  Recent research highlights multiple design paradigms of KI. For instance, HALO demonstrates that integrating structured knowledge with internal consistency checks can significantly suppress hallucinations in dialogue systems [156]. Similarly, Capellini et al. formalize KI as a modular pipeline in which factual triples from knowledge bases are aligned with model attention layers to guide reasoning, showing improvements in both faithfulness and contextual coherence across diverse tasks [29]. Compared to broader retrieval-based paradigms, KI provides a more precise and semantically interpretable channel of supervision, ensuring that model outputs remain anchored in verifiable context rather than heuristic approximations. As such, Knowledge Injection stands as a key strategy in knowledge-based hallucination mitigation, offering both transparency and robustness in high-stakes applications.

c) Verification-oriented consistency: represents a class of hallucination mitigation methods that emphasize cross-checking generated outputs against internal or external signals of coherence and factuality. Unlike resource-based retrieval or knowledge injection, which enrich the model's input with additional information, consistency-driven approaches operate primarily at the validation layer, systematically verifying whether the model's outputs align with known entities, established relations, or logically coherent structures. By leveraging the principle that reliable outputs must remain consistent across multiple perspectives, contexts, or decomposed subtasks, these methods suppress spurious continuations and expose unsupported claims. This category can be further divided into four major subgroups: (1) *Entity & Relation-based Verification*, which grounds model outputs in structured entity-relation triples to ensure semantic and factual alignment; (2) *Factual Consistency Verification*, which directly evaluates generated text against reference knowledge or parallel generations for contradiction detection; (3) *Chain-of-Verification (CoVe)*, which introduces an additional reasoning step where models verify their own or other models' outputs through iterative questioning and evidence checking; and (4) *Sub-question Decomposition*, including its application in Visual Question Answering (VQA), which breaks complex queries into smaller, verifiable units to reduce reasoning errors and enforce consistency across intermediate answers. Collectively, these techniques highlight the role of consistency as a powerful constraint for hallucination mitigation, ensuring that model predictions are not only fluent but also logically and factually dependable.

- **Entity & Relation-based Verification** is a class of consistency-guided methods that detect and mitigate hallucinations by aligning generated outputs with structured factual representations of entities and their relations. The core idea is that hallucinations often manifest as fabricated entities or incorrect relational links between entities mentioned in the output. By leveraging external databases or structured triples, models can verify whether generated entities exist and whether the asserted relations are semantically and factually valid [42]. Wei et al. highlight that entity-relation verification not only improves factual consistency but also provides interpretable error signals by tracing hallucinations to missing or conflicting relations in the knowledge source. This strategy can be particularly effective in knowledge-intensive tasks such as biomedical text generation and scientific question answering, where relational accuracy is critical for maintaining credibility [42].

- **Consistency Verification** focuses on validating the internal and external coherence of model outputs by comparing generated content against trusted references, alternative reasoning paths, or previously established knowledge. This mechanism operates on the principle that factually reliable responses should remain stable across paraphrased prompts, knowledge sources, or verification passes. Wan et al. propose a knowledge-enhanced consistency framework that integrates external evidence with multi-view reasoning, ensuring that generated statements align with verified knowledge bases and reducing the persistence of spurious correlations [250]. Similarly, Harrington et al. demonstrate that enforcing consistency across multiple generations and model variants can effectively flag hallucinations, as divergence among outputs often signals low factual reliability [195]. By systematically re-examining outputs across dimensions of knowledge, reasoning, and context, consistency verification strengthens both factual grounding and robustness, making it a versatile strategy for hallucination mitigation in knowledge-intensive tasks.
- **Chain-of-Verification (CoVe)** enhances factual reliability by introducing a multi-step validation process that systematically checks intermediate reasoning steps before producing a final answer. Unlike single-pass generation, CoVe decomposes the response into smaller claims, each of which is individually verified against either retrieved evidence or consistency checks, thereby reducing the risk of unsubstantiated assertions. Liang et al. demonstrate that this layered verification pipeline not only improves factual grounding but also increases robustness in tasks requiring complex, multi-hop reasoning [172]. By embedding verification directly within the generation workflow, CoVe exemplifies how iterative reasoning and structured checking can serve as effective safeguards against hallucination.
- **Sub-question Decomposition (including VQA)** tackles hallucinations by decomposing complex queries into a series of simpler, verifiable sub-questions [251]. Each sub-question, often grounded via retrieval or external evidence, is addressed individually before synthesizing the responses into a final answer. This structured breakdown enhances factual precision and interpretability, as it allows validation at every reasoning step and reduces error propagation in multi-hop reasoning settings. Chang et al. propose a unified hallucination mitigation framework that leverages such decomposition to align intermediate outputs with factual signals at multiple reasoning stages [183]. Similarly, Pelican—a verification-centric LVLM framework—translates visual claims into logical sub-claims that are independently verified via executable checks, thereby significantly reducing hallucinations in VQA tasks [252]. These approaches demonstrate that decomposing a complex problem into verifiable components is a robust safeguard against hallucinated conclusions.

In sum, Knowledge-Infused Mitigation Methods mitigate hallucinations by anchoring model outputs in reliable external signals, whether through retrieval mechanisms, structured knowledge resources, or consistency verification. These methods complement intrinsic interventions by addressing the limitations of parametric knowledge and enabling factual grounding across diverse domains. However, while effective, they often operate as discrete modules targeting specific vulnerabilities. To achieve broader robustness, recent research has shifted toward *Meta- and Framework-based Methods*, which integrate heterogeneous strategies into unified architectures and adaptive pipelines. These higher-level frameworks provide holistic designs that extend beyond single interventions, ensuring that knowledge access, reasoning control, and verification interact seamlessly across the generation process.

3) Meta/Framework-based Methods: Meta- and Framework-based Methods constitute the third category of hallucination mitigation strategies, distinguished by their operation at a higher level of abstraction compared to direct technical adjustments or resource-grounded interventions. Rather than modifying token selection or anchoring outputs in external repositories, these methods establish overarching architectures and governance mechanisms that regulate, coordinate, or augment the generative process.

As illustrated in Fig. 7, this category can be divided into four major subgroups. *Data and learning strategies* emphasize standardization and pre-processing to improve input quality and supervision consistency, thereby enhancing downstream robustness. *Refusal and abstention-based methods* empower models to "say no" when confronted with queries that exceed knowledge boundaries or carry high hallucination risk. *Self-regulation mechanisms* incorporate self-evaluation and iterative feedback processes that enable models to internally verify, revise, or correct their outputs. Finally, *hybrid and architectural approaches*—including loop-based refinement, agent-based orchestration, and multi-LLM debate—build higher-order frameworks that integrate multiple perspectives or iterative decision cycles to systematically reduce hallucination propagation. Collectively, these methods highlight the transition from localized interventions toward comprehensive architectures, framing hallucination mitigation not as isolated corrections but as an integrated system of regulation and

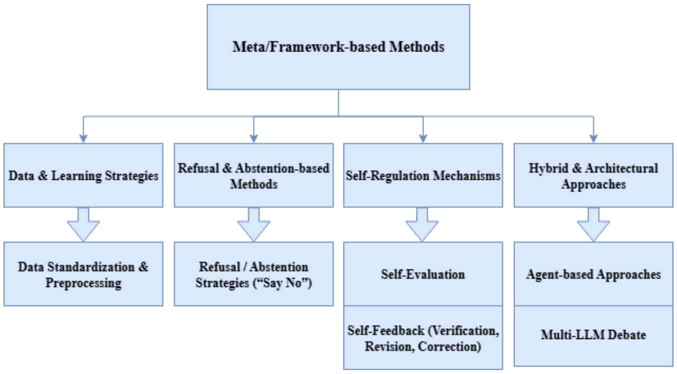

Fig. 7. Meta_Framework-based Methods

oversight.

a) Data and Learning Strategies: address hallucination mitigation by focusing on the quality, structure, and distributional properties of training and fine-tuning corpora. Unlike purely architectural or inference-level interventions, these methods target the upstream data pipeline, ensuring that the information supplied to large language models is both reliable and systematically curated. By enforcing consistency in formatting, reducing redundancy, and eliminating noisy or contradictory signals, such strategies enhance factual grounding while improving generalization across domains. Moreover, data-centric interventions operate synergistically with downstream methods, as models trained on standardized and well-preprocessed datasets are more amenable to self-regulation, external verification, and retrieval integration. Within this category, *data standardization and preprocessing* emerges as a foundational approach, establishing a structured and high-fidelity input base that supports both parameter optimization and post-hoc verification frameworks.

- **Data Standardization and Preprocessing** is a foundational strategy for hallucination mitigation that ensures training corpora are reliable, representative, and semantically consistent. The process begins with the removal of redundant and low-quality entries to prevent duplication and noise amplification, followed by automated and manual correction of typographical errors and normalization of linguistic formats [201]. Robust tokenization then segments text into standardized units, reducing variability in representation and improving the model's ability to capture meaningful patterns [253]. Filtering and annotation further refine datasets by eliminating biases, enriching contextual signals, and aligning inputs with domain-relevant knowledge, thereby supporting accurate downstream reasoning [254]. Beyond these conventional steps, advanced preprocessing strategies incorporate adaptive quality-control pipelines that detect domain-specific inconsistencies, enforce factual alignment, and systematically constrain error propagation at the source. By ensuring that the data ingested by the model is both clean and knowledge-grounded, preprocessing not only enhances the efficiency of learning but also serves as a preventive layer against hallucination-inducing artifacts [201].HalluciDoctor exemplifies such advanced preprocessing by detecting and removing hallucinations from training corpora through cross-checking with multiple models [255]. To further enhance robustness, it introduces counterfactual visual instruction expansion, which mitigates spurious correlations from long-tail multimodal co-occurrences. This combined strategy improves the factual reliability of training signals and strengthens the resilience of multimodal large language models against hallucinations [255].

b) Refusal and Abstention-based Methods: represent a distinct line of hallucination mitigation that focuses not on generating more content, but on deliberately withholding output when the model's confidence, reliability, or factual grounding cannot be guaranteed. Unlike strategies that attempt to correct or post-process potentially erroneous generations, refusal mechanisms proactively intervene in the decision process by equipping models with the ability to reject queries, abstain from uncertain predictions, or output explicit "don't know" responses. This paradigm shifts the emphasis from maximizing coverage of responses to prioritizing precision and safety, reducing the risk of confidently presented but incorrect outputs. Within this category, *Refusal / Abstention Strategies ("Say No")* constitute the primary instantiation, where models are trained or instructed to recognize high-risk prompts, detect factual uncertainty, and return cautious, non-committal answers instead of speculative generations.

- **Refusal / Abstention Strategies ("Say No")** represent a principled paradigm in hallucination mitigation by allowing language models to withhold an answer when faced with high uncertainty or potential toxicity. Instead of enforcing output generation under all circumstances, these approaches empower models to strategically "say no," thereby reducing the risk of factual inaccuracies or harmful outputs [58]. Technically, abstention can be operationalized through internal uncertainty quantification, where the model evaluates consistency across multiple sampled responses and declines to answer when divergence exceeds a threshold [56]. Recent studies further emphasize that refusal behaviors are not merely reactive but can be shaped during fine-tuning. For instance, HonestAI introduces specialized supervision to encourage abstention in ambiguous scenarios, demonstrating that abstention-finetuned models significantly reduce spurious generations without compromising overall utility [117]. Similarly, Kang et al. show that the way unfamiliar finetuning examples are supervised critically influences hallucination patterns, and strategically labeling such examples with abstention signals (e.g., "I don't know") enables models to generalize refusal behavior to unseen queries [165]. These findings highlight refusal not as a fallback mechanism but as a proactive alignment strategy, where abstention supervision during supervised fine-tuning (SFT) or reinforcement learning (RL) can calibrate model responses under epistemic uncertainty [165]. Collectively, refusal/abstention strategies reframe the mitigation problem by institutionalizing uncertainty-aware non-response, transforming "not answering" into a valid and safe operational choice.

- **Rails** methods apply predefined rules or guardrails to regulate model behavior during generation. By constraining outputs within safe and factual boundaries, these approaches prevent models from producing unsupported or unsafe content. Unlike purely probabilistic decoding strategies, rails-based methods enforce explicit structural safeguards, complementing abstention policies by ensuring reliability even when models attempt to generate uncertain responses [256]

c) Self-Regulation Mechanisms: constitute a class of hallucination mitigation strategies that embed autonomous evaluation and correction directly within the model. Rather than relying solely on external supervision, these approaches enable large language models (LLMs) to act as both generators and inspectors of their own outputs. Two main paradigms are commonly employed. *Self-evaluation* performs introspective checks on factuality, coherence, or plausibility before delivering a response, while *self-feedback mechanisms* iteratively refine outputs through verification, revision, and correction loops. Together, they foster a paradigm of "autonomous reliability," where models mitigate hallucinations by continuously diagnosing and improving their own generations.

- **Self-Evaluation** techniques empower large language models (LLMs) to assess the reliability of their own outputs before producing final responses. This mechanism is grounded in uncertainty estimation, confidence calibration, and familiarity assessment, which together allow models to regulate their generative behavior. For instance, zero-shot self-evaluation frameworks introduce *self-familiarity scores*, enabling the model to determine whether it has sufficient prior knowledge to answer a query or should abstain to avoid hallucination [257]. Similarly, self-consistency and self-checking methods prompt the model to compare multiple reasoning trajectories, suppressing answers with high internal disagreement [155]. Other approaches integrate self-assessment within abstention frameworks, where the model estimates its own uncertainty and refuses to answer if a predefined confidence threshold is not met [56]. Collectively, these methods introduce a layer of internal governance, reducing unsupported claims by aligning model responses with their measured confidence and familiarity.
- **Self-Feedback (Verification, Revision, Correction)** extends the capacity of LLMs by enabling them to iteratively assess and refine their own outputs through verification, revision, and correction cycles. This paradigm builds upon techniques such as *self-checking*, *sliding generation*, and *feedback-guided refinement*, which collectively reduce factual inconsistencies and semantic drift. For instance, Harrington et al. [195] demonstrate that self-verification pipelines can proactively flag potential hallucinations by comparing generated claims against contextual cues, while Lee et al. [258] highlight iterative correction loops for multimodal grounding. Ensemble-based self-feedback, such as the N-CRITICS framework, further strengthens reliability by incorporating multiple critics to evaluate and guide refinements, thereby mitigating both toxicity and factual hallucinations [259]. Complementarily, Ji et al. [143] emphasize the role of correction-oriented prompting strategies, where LLMs generate structured feedback that is recursively applied to adjust prior outputs. Together, these approaches illustrate how self-feedback transforms LLMs into dynamic, self-regulating systems capable of maintaining factual alignment and robustness without heavy reliance on human supervision.

d) Hybrid and Architectural Approaches: represent a higher-order category of hallucination mitigation that transcends isolated techniques by embedding self-regulation and external reasoning directly into the structural design of generation systems. Instead of limiting interventions to a single phase, these methods construct holistic architectures that integrate multiple mechanisms into collaborative frameworks. *Agent-based approaches* assign specialized roles to distinct components or agents (e.g., generator, verifier, retriever), enabling division of labor and collective reasoning within a coordinated system. Likewise, *multi-LLM debate strategies* leverage adversarial or cooperative exchanges among multiple models to stress-test factual claims and uncover inconsistencies, turning disagreement into a source of verification. Together, these approaches move hallucination mitigation beyond ad-hoc corrections toward structured, system-level architectures that enhance robustness, factual grounding, and interpretability across diverse tasks.

- **Agent-based Approaches** represent a paradigm where multiple autonomous agents—each with distinct roles such as reasoning, retrieval, or verification—collaboratively address complex queries, thereby enhancing factual reliability and interpretability. Unlike single-model pipelines, agent-based orchestration distributes responsibilities across specialized subsystems, enabling division of labor and error-checking among agents. For example, chain-based multi-agent frameworks demonstrate how a reasoning agent generates hypotheses while a verification agent cross-checks with external resources, significantly reducing unsupported claims [260]. Recent studies further emphasize interoperability, where shared protocols and APIs allow heterogeneous agents from different origins to integrate outputs seamlessly, fostering scalable and robust ecosystems [65], [261]. By combining individual specialization with collective intelligence, agent-based methods create resilient architectures for hallucination mitigation that mirror human collaborative problem-solving.

- **Multi-LLM Debate** is a collaborative paradigm in which multiple language models iteratively challenge and refine each other's outputs to improve factual grounding and reasoning robustness. Instead of relying on a single generative trajectory, debate frameworks distribute reasoning across different agents, each proposing candidate answers and counterarguments, followed by an arbiter or aggregation mechanism to consolidate the most consistent and evidence-backed conclusions [262]. Recent work demonstrates that such debate protocols significantly mitigate hallucinations by fostering diversity of reasoning paths and enabling error detection through cross-examination. For example, Yoffe et al. [159] introduced Debunc, where adversarial debates between LLMs expose unsupported claims and enforce verifiability before consensus is reached. Similarly, Sng et al. [27] proposed a novel debate-and-arbiter framework that integrates consistency checks with multi-round deliberations, showing improvements in both truthfulness and reliability of responses. By maintaining multiple hypotheses and leveraging structured arbitration, multi-LLM debate transforms generation into a dialectical process, yielding more cautious and trustworthy outputs.
- **Reason and Act (ReAct)** represents an early form of agentic design within a single model, where chain-of-thought reasoning is interleaved with concrete actions such as retrieval or tool use. By coupling reasoning traces with actionable operations, ReAct enables models to ground intermediate steps in observable outcomes rather than relying solely on parametric knowledge. This tight integration improves factual reliability and lays the conceptual foundation for more complex multi-agent systems, where reasoning and verification can be distributed across specialized agents [256]
- **Dialog-Enabled Resolving Agents (DERA)** extend debate-style paradigms by orchestrating multiple dialog agents that iteratively question, critique, and refine one another's responses. Through structured conversational exchanges, DERA frameworks filter out unsupported claims and converge on factually grounded outputs. By combining adversarial challenge with cooperative resolution, this approach strengthens reliability while maintaining interpretability in multi-agent settings [256]

Meta- and framework-based methods collectively represent a higher-order layer of hallucination mitigation, focusing less on isolated interventions and more on systemic architectures that embed robustness into the generative process. By leveraging strategies such as data standardization, abstention protocols, self-regulation, and hybrid frameworks—including iterative loops, agent collaboration, and multi-model debate—these approaches transcend single-point fixes and instead promote adaptive, scalable resilience against hallucination. Their unifying feature lies in establishing overarching infrastructures that integrate technical and resource-based mechanisms into coherent pipelines, ensuring not only immediate factual consistency but also long-term adaptability across diverse domains and tasks [183], [201].

Mechanism-based categorization, the first dimension of our taxonomy, highlights the breadth of strategies employed to mitigate hallucinations, ranging from technical interventions to knowledge-grounded methods and meta-frameworks. While this perspective clarifies the underlying principles that drive each approach, it does not fully capture the temporal dynamics of when and how these interventions are applied during the generation pipeline. To complement this mechanism-oriented view, the second dimension organizes hallucination mitigation methods along the phase-based classification, distinguishing interventions that occur before, during, and after generation. This shift in perspective enables a more process-centric understanding, revealing how diverse mechanisms can be systematically aligned with the stages of model operation, and how phase-specific as well as cross-phase strategies interact to form robust mitigation workflows.

B. Second Dimension: Phase-Based Classification

The temporal dimension of hallucination mitigation reveals a fundamental insight: *when* we intervene in the generation pipeline profoundly shapes both the effectiveness and the computational cost of our defenses. While the mechanism-based taxonomy in Section IV-A addresses *what* strategies we employ, this second dimension focuses on *when* these interventions occur across the model lifecycle and the generation workflow.

This temporal perspective is critical for several reasons. First, different phases offer distinct leverage points—pre-generation methods can prevent hallucinations at their source but require anticipating failure modes; in-generation techniques provide real-time control but impose latency penalties; post-generation approaches enable thorough verification but cannot recover from fundamental generation failures. Second, the phase determines integration complexity: pre-generation methods typically require one-time setup, in-generation methods often demand architectural modifications, whereas post-generation methods operate as modular filters. Third, computational budgets vary dramatically across phases, with pre-deployment fine-tuning affording extensive computation, real-time decoding requiring millisecond-scale decisions, and post-processing allowing selective deep verification.

We organize mitigation strategies into three sequential phases that mirror the natural flow of language generation: (1) **Pre-generation Phase Mitigation**, encompassing pre-deployment preparation (training, fine-tuning) and pre-inference configuration (prompting, retrieval setup) to establish factual guardrails before generation begins; (2) **In-generation Phase Mitigation**, comprising real-time interventions that dynamically steer the decoding process away from hallucination-prone trajectories; and (3) **Post-generation Phase Mitigation**, involving verification and revision mechanisms that audit completed outputs against factual sources.

To establish clear boundaries, we define phase membership based on the *primary point of intervention*:

- **Pre-generation**: Methods that complete their primary intervention before the first token is generated. This includes:
  - *Pre-deployment*: Training-time modifications (fine-tuning, instruction tuning, preference optimization)
  - *Pre-inference*: Runtime setup without token generation (prompt engineering, retrieval system configuration, confidence calibration)

  Example scenario: A medical QA system undergoes domain-specific fine-tuning and configures a medical knowledge retrieval system before deployment.
- **In-generation**: Methods that actively intervene during the token-by-token decoding process, characterized by:
  - Real-time decision making at each decoding step
  - Access to and modification of intermediate model states
  - Direct influence on token probability distributions

  Example scenario: During response generation, the system dynamically adjusts attention weights when detecting potential factual inconsistencies.
- **Post-generation**: Methods that operate on complete or substantially complete outputs, featuring:
  - Holistic analysis of generated content
  - External verification against ground truth sources
  - Ability to revise, filter, or reject entire responses

  Example scenario: A generated news summary undergoes fact-checking against a knowledge base before publication.

For methods that span multiple phases, we assign them based on where the *primary control mechanism* resides. Cross-phase interactions and feedback loops are addressed in our third dimension (Section IV-C). The following subsections detail representative methods within each phase, organized by technical approach.

1) Pre-generation Phase Mitigation: Pre-generation mitigation represents the most proactive stance against hallucinations—intervening before the model commits to any specific output trajectory. This phase encompasses both pre-deployment preparation (training, fine-tuning, data curation) and pre-inference configuration (prompting, retrieval setup, knowledge injection), unified by the goal of establishing strong factual guardrails prior to generation.

The strategic appeal of pre-generation methods lies in their preventive nature. Rather than correcting errors after they emerge, these approaches shape the model's knowledge representation and decision boundaries to make hallucinations less likely in the first place. For example, a legal document generator fine-tuned on verified case law is implicitly constrained against fabricating precedents, while pre-inference retrieval augmentation explicitly anchors responses in authoritative sources. This dual-layered defense—intrinsic knowledge alignment plus extrinsic factual grounding—exemplifies the complementary strategies available before generation.

Pre-generation methods can be organized into three families: (1) **Model-intrinsic Methods**, which modify parameters, representations, or prompts to improve factual reliability; (2) **Knowledge-Infused Methods**, which incorporate external sources to ground generation; and (3) **Meta-/Framework-based Methods**, which apply higher-level scaffolding, abstention, or preprocessing pipelines. Each offers distinct leverage points for reducing hallucinations at their source.

a) Model-intrinsic Methods: These methods act directly on the model or its inputs to calibrate internal behavior before token generation begins. Confidence-based Triggering (Interactive DualChecker) monitors uncertainty during distillation and refines prompts when confidence is low, improving factual alignment [28]. Atomic Consistency Preference Optimization (ACPO) leverages consistency across stochastic samples to construct preference pairs for self-supervised factual calibration without reliance on stronger models or external knowledge [207]. Contrastive learning approaches such as PURR fine-tune lightweight editors to denoise corrupted outputs with retrieved evidence, enhancing attribution without manual intervention [263]. Knowledge distillation methods—such as ZeroG, standard KD, and smoothed KD—transfer calibrated distributions from teacher to student models, aligning token probabilities with verified knowledge [85], [247], [248]. Prompt engineering further contributes by shaping task framing and input representation; examples include Visual Evidence Prompting (VEP) [35] and Counterfactual Inception [60], which ground outputs in verifiable cues or suppress spurious associations. In multimodal contexts, Multi-Frequency Perturbations (MFP) reduce visual hallucinations by perturbing both low- and high-frequency components of image features, weakening spurious correlations while preserving semantic fidelity [180].

b) Knowledge-Infused Methods: These methods enhance factual grounding by injecting or aligning external information prior to inference. Knowledge-graph approaches such as PGMR replace placeholders with accurate KG URIs retrieved via embedding similarity [249]. Retrieval-augmented generation (RAG) anchors outputs by incorporating domain-specific [232] or open-domain [236] passages into prompts. Knowledge injection (KI) techniques, including dynamic domain KI, directly integrate structured facts into prompts or model states to steer outputs toward trustworthy knowledge [29], [245]. Knowledge-Consistent Alignment (KCA) further ensures that alignment data does not contradict the model's intrinsic knowledge by pre-validating and filtering samples [250].

c) Meta-/Framework-based Methods: These strategies introduce higher-level scaffolding or systemic constraints before inference begins. Data standardization and preprocessing pipelines, exemplified by VIC, enforce reasoning-before-perception to reduce multimodal bias [253]. Self-regulation frameworks such as HRO iteratively refine outputs through Generate–Score–Improve loops under RLHF supervision [221]. Refusal and abstention mechanisms like Self-Familiarity score prompt familiarity and decline low-confidence queries, avoiding unsupported generations without retraining [257]. Beyond individual techniques, survey work [264] highlights trade-offs between gradient-based approaches (e.g., fine-tuning, RLHF, DPO) and non-gradient approaches (e.g., retrieval, heuristic prompting), situating hallucination mitigation within the broader landscape of LLM optimization. Architectural innovations such as heterogeneous model collaborations [265] further illustrate framework-level directions, orchestrating generators, classifiers, and contextualized models to improve controllability, reduce hallucinations, and extend to multimodal settings.

2) In-generation Phase Mitigation: In-generation mitigation represents the most technically sophisticated stage of hallucination prevention, operating precisely when language models are most vulnerable—during the token-by-token construction of their outputs. This phase is defined by real-time intervention, where millisecond-scale decisions must balance factual grounding against fluency, coherence, and latency constraints. Unlike pre-generation methods, which benefit from offline computation, or post-generation methods, which evaluate complete outputs, in-generation techniques must act instantaneously based on partial and evolving context. For example, in a medical AI system generating treatment recommendations, in-generation methods must detect when the model begins to fabricate a drug interaction and intervene before the error propagates.

These interventions can be grouped into three broad families: (1) **Model-intrinsic Methods**, which directly manipulate internal states, activations, or probability distributions to suppress hallucination-prone pathways; (2) **Knowledge-Infused Methods**, which dynamically retrieve and incorporate external sources to ground ongoing generation; and (3) **Meta-/Framework-based Methods**, which impose structural scaffolds or reasoning templates to maintain factual consistency—each with distinct trade-offs between precision, efficiency, and adaptability.

a) Model-intrinsic Methods: These methods intervene at the neural or decoding level, directly steering representations and token distributions in real time. They include:

Representation Steering: Methods such as TruthX [198] and Iter-AHMCL [37] identify a "truthful direction" in activation space and continuously adjust hidden states toward this subspace, reducing the probability of hallucination-prone trajectories while preserving coherence.

Search-Based Decoding: Rather than relying on greedy decoding, these approaches explore multiple candidate continuations. Lookback Lens [193] re-anchors attention to earlier context when drift is detected, while Monte Carlo Tree Search [173] builds a branching tree of continuations, pruning hallucination-inducing paths before they propagate.

Contrastive and Layer-Editing Approaches: Techniques that contrast model states or modify intermediate representations to suppress unreliable outputs. Freeze-CD [158] compares frozen and fine-tuned distributions; M3ID [266] aggregates multi-model consensus; multimodal methods such as HALC [43] and VCD [31] address modality-specific hallucinations. Two recent strategies extend this family: *Active Layer-Contrastive Decoding (ActLCD)* [206] dynamically activates layer-wise contrastive signals when drift is detected; and *Premature Layers Interpolation (PLI)* [199], a training-free, plug-and-play method that interpolates adjacent hidden states to refine semantic representations mid-decoding.

b) Knowledge-Infused Methods: These methods embed dynamic retrieval and factual guidance into the generative loop. A representative example is Self-Refinement-Enhanced Knowledge Graph Retrieval (Re-KGR) [240], which monitors attribution signals at the token and layer level. When hallucination risk is flagged, it selectively retrieves relevant triples from a knowledge graph and integrates them back into decoding, enabling on-the-fly corrections grounded in authoritative sources.

c) Meta-/Framework-based Methods: These approaches apply higher-level structural or procedural constraints throughout decoding. The Jodie prompt [267], for instance, reformats passages as attributed quotes and instructs the model to respond "according to" the designated source, privileging retrieved evidence over parametric memory. More generally, reasoning templates based on hierarchical or graph structures constrain generation paths and reinforce consistency across multi-step reasoning, embedding structured workflows directly into the decoding process without breaking autoregression.

3) Post-generation Phase Mitigation: Post-generation mitigation embodies a fundamental principle of reliable AI systems: *trust, but verify*. While pre- and in-generation methods work to prevent hallucinations, post-generation approaches acknowledge an uncomfortable truth—even the best preventive measures cannot guarantee perfect factual accuracy. This phase thus serves as a critical safety net, catching errors that slip through earlier defenses.

The strategic value of post-generation verification extends beyond error correction. By operating on complete outputs, these methods can leverage global coherence signals unavailable during incremental generation. For instance, a financial report generator might produce sentences that are individually plausible, but only post-generation analysis can detect that reported earnings across sections are mutually inconsistent. Moreover, post-generation methods can be upgraded without retraining the base model, allowing rapid adaptation to new verification standards or domain requirements.

Post-generation strategies follow a detect–analyze–revise pipeline: parse generated content to extract verifiable claims (entities, relations, numerical assertions); evaluate these against authoritative sources or internal consistency criteria; then revise problematic segments, request regeneration with constraints, or refuse to deliver the output when confidence is insufficient. This enables three complementary verification strategies: (1) **Model-intrinsic Methods** that analyze logical consistency and reasoning chains; (2) **Knowledge-Infused Methods** that validate claims against external databases and knowledge graphs; and (3) **Meta-/Framework-based Methods** that assess competence boundaries and implement principled abstention.

a) Model-intrinsic Methods: Rowen [184] adopts a "retrieve only when needed" paradigm: after an initial chain-of-thought answer, it applies multilingual semantic perturbation and cross-lingual consistency checks to detect potential hallucinations; only when inconsistencies are found does it activate retrieval augmentation for revision, thus avoiding unnecessary retrieval noise. Consistency-focused techniques further refine drafts by deleting entity–relation triples lacking external support, preserving a minimal factual scaffold derived from retrieved evidence [42]. Verified entities and relations are then encoded into a correction prompt to rewrite around grounded facts, followed by an additional consistency check.

b) Knowledge-Infused Mitigation Methods: Iterative Knowledge Graph Refinement: KGR [49] implements a verify–refine loop. It parses generated content into triples, queries knowledge graphs to verify each claim, and uses unsupported triples to trigger targeted regeneration with explicit constraints, ensuring global consistency. *Semantic Pruning and Reconstruction*: Sentence Trimming [146] aligns generated text with KG structure by pruning tokens that lack explicit support, leaving a verified skeleton as a template for conservative regeneration.

c) Meta-/Framework-based Methods: Multi-Model Arbitration: A debate framework [27] leverages model diversity: specialized models analyze the same output from different perspectives, and an arbiter with retrieval resolves disputes using authoritative sources. *Competence-Aware Abstention*: Meta-cognition training [268] learns to selectively respond based on self-assessed capability, using disagreement across model variants to model epistemic uncertainty and enabling refusal when confidence is low.

Phase Trade-offs and Design Considerations *Computational economics*: Pre-generation amortizes costs across queries but requires upfront investment. In-generation incurs per-token overhead (search-based methods add latency; lightweight contrastive methods are more efficient). Post-generation costs scale with output complexity—from milliseconds for simple checks to seconds for comprehensive KG validation. *Error coverage*: Pre-generation addresses systematic biases but may miss query-specific issues. In-generation detects local inconsistencies but may miss global incoherence. Post-generation catches cross-references and arithmetic inconsistencies but cannot fix deeply flawed reasoning chains. *Integration complexity*: Pre-generation integrates cleanly (fine-tuning, RAG). In-generation requires decoding-loop changes with careful engineering. Post-generation acts as a modular filter, enabling rapid domain customization without model changes.

Architectural Patterns and Best Practices Successful deployments typically follow one of three architectural patterns:

Pattern 1: Lightweight Real-time Stack. For consumer-facing applications, prioritizing latency: - Pre-generation: Instruction-tuned base model with confidence calibration - In-generation: Efficient contrastive decoding (e.g., single-layer contrast) - Post-generation: Selective verification only for low-confidence outputs Example: Virtual assistants that need sub-second response times

Pattern 2: Balanced Professional Stack. For professional tools, balancing accuracy and usability: - Pre-generation: Domain fine-tuning + retrieval-augmented generation - In-generation: Representation steering with moderate search depth - Post-generation: Fact-checking with user-facing confidence indicators Example: Legal research assistants requiring high accuracy with reasonable latency

Pattern 3: Maximum Reliability Stack. For safety-critical applications: - Pre-generation: Ensemble of specialized models + comprehensive knowledge base - In-generation: Multi-method intervention (steering + search + contrast) - Post-generation: Multi-stage verification with human-in-the-loop options Example: Medical diagnosis support systems where errors have severe consequences

Evolution and Future Directions The phase-based taxonomy reveals temporal trends in hallucination mitigation. Early work focused on post-generation detection, treating hallucination as an output quality issue. The field then shifted toward pre-generation prevention through better training. Current frontier research emphasizes in-generation intervention, seeking to catch hallucinations at their point of origin.

Looking forward, we anticipate tighter integration across phases. Emerging approaches include: (1) Post-generation feedback loops that update pre-generation retrieval indices; (2) In-generation methods that adaptively invoke post-generation verification mid-stream; (3) Unified frameworks that jointly optimize interventions across all phases. As foundation models improve, the emphasis may shift from heavy-handed intervention to subtle guidance, with hallucination mitigation becoming increasingly transparent to end users.

C. Third Dimension: Cross-Phase and Integrative Mitigation Approaches

The third dimension, Cross-Phase and Integrative Mitigation, advances a systems view of hallucination control organized around three pillars that operate in concert.

As shown in Fig. 9, multistage mitigation coordinates interventions across the pre, in, and post generations, propagating grounding, uncertainty, and corrective signals forward to constrain search and steer decoding and backward to recalibrate subsequent runs, thus achieving compounded reductions in hallucination risk. Multi-method fusion then orchestrates heterogeneous techniques (e.g., retrieval augmentation, parameter-efficient adaptation, structured prompting, constraint-aware decoding, verification and editing, knowledge-graph reasoning, uncertainty-aware detectors, etc.) so that complementary strengths align against distinct failure modes while balancing precision, coverage, and efficiency. Finally, loop- and iteration-centric optimization establishes closed-loop evaluation–correction cycles (e.g., self-checks, counterfactual re-decoding, feedback-guided revision, selective human-in-the-loop, etc.) to progressively stabilize factual reliability and update system priors.

Framed in this manner, hallucination mitigation emerges not as a discrete, phase-constrained intervention, but as a dynamic, signal-coupled pipeline in which guidance and correction are continuously propagated, integrated, and reused across time and method. The subsections that follow examine each pillar in turn, articulating their operative mechanisms, inter-phase linkages, methodological synergies, and implications for the design of resilient, end-to-end mitigation systems.

1) Multi-Stage Mitigation: refers to strategies that span multiple phases of the generation process—pre-generation, in-generation, and post-generation—so that guidance, correction, and verification mechanisms interact throughout the pipeline. Rather than confining mitigation to a single point of intervention, these approaches establish functional continuity across stages, enabling information (e.g., retrieval signals, uncertainty estimates, corrective feedback) to propagate and reinforce model behavior both prospectively and retrospectively. For instance, THaMES combines parameter-efficient fine-tuning (pre-generation), retrieval-augmented grounding (in-generation), and in-context verification prompting (post-generation), dynamically selecting among these paths based on task and model characteristics [172]. Similarly, Dentist orchestrates chain-of-thought prompting, reasoning classification, and multimodal verification in a looping diagnose–verify–revise cycle, coordinating distinct functions across stages [183]. The Sliding Generation with Self-Checks pipeline likewise operates at multiple levels: it structures the input into overlapping spans during decoding, and post-processes the output via consistency-based verification [195]. These examples illustrate how multi-stage methods align the architecture and logic of hallucination mitigation with the temporal structure of generation itself.

2) Multi-Method Fusion: integrates diverse hallucination mitigation techniques—such as fine-tuning, retrieval augmentation, knowledge injection, attention control, contrastive learning, verification, and semantic parsing—within unified systems to exploit their complementary strengths. These approaches target multiple hallucination failure modes simultaneously, including semantic drift, factual inconsistency, contextual irrelevance, and unverifiable outputs, thereby enhancing robustness across tasks and domains.

For instance, HALO injects Wikidata-derived factual knowledge into a base LLM and uses consistency-based gating to selectively invoke a stronger teacher model for high-risk cases [156]. MixCL applies contrastive fine-tuning to align model outputs with gold-standard responses while actively repelling hallucinated candidates [33]. Dynamic Federated Supplementation grounds generation in real-time search engine results, injecting corroborative snippets during decoding to reduce reliance on parametric memory [174]. A hybrid GNN+LLM model incorporates knowledge graph inference to compute belief scores for candidate facts and filter unsupported claims [254].

Other systems leverage representational optimization: one method fine-tunes attention maps to refocus decoding on context-relevant tokens, reducing drift from input context [201]. WikiSP performs semantic parsing over Wikidata to return verifiable, structured answers via SPARQL, clearly distinguishing factual outputs from educated guesses [269]. Another approach uses Direct Preference Optimization to train the model on topic-based preference pairs, suppressing hallucinated outputs by reinforcing alignment with accepted content clusters [47]. Collectively, these methods exemplify the modular and synergistic design of modern hallucination mitigation systems.

3) Loop-Based and Iterative Optimization: represents a class of techniques that reduce hallucinations through cycles of repeated generation, verification, and refinement. Instead of relying on single-pass outputs, these methods operationalize iterative feedback loops, where each round of model output is systematically re-evaluated, revised, or supplemented with external signals before finalization. The HADAS framework, for example, employs hallucination-diversity aware active learning, iteratively sampling and correcting outputs to refine predictions and ensure broader coverage of hallucination-prone cases [36]. Truth-oriented iterative refinement frameworks embed explicit factuality checks within generation cycles, enforcing alignment with verifiable evidence across multiple turns [270]. Unified multi-step pipelines dynamically integrate verification and correction into structured refinement loops [183]. In multimodal contexts, iterative mechanisms such as sliding generation and Volcano-based correction aggregate intermediate representations across multiple rounds to stabilize text–vision alignment and suppress hallucinations [258]. Multi-agent systems like DebUnc further extend the loop-based principle by enabling peer critique and confidence-sharing among LLMs, discounting unreliable responses while iteratively refining collective outputs [159]. Other methods, such as Induce-then-Contrast Decoding, leverage an auxiliary hallucination-prone model to define a penalty signal, iteratively suppressing unsupported outputs during generation [205]. Collectively, these loop-driven paradigms highlight the importance of self-corrective processes, positioning hallucination mitigation as an ongoing and dynamic refinement rather than a one-shot intervention [221], [259], [271].

The third dimension of our taxonomy, cross-phase and integrative mitigation, extends the analysis beyond isolated mechanisms or phases to a holistic, system-level perspective. It emphasizes how hallucination control can be strengthened through coordinated interventions that propagate signals across stages, the fusion of diverse methods targeting various failure modes, and iterative feedback loops that refine outputs over time. By foregrounding these integrative strategies, this dimension highlights hallucination mitigation not as a static correction but as a dynamic, interconnected pipeline where guidance, verification, and correction are continuously shared and reused. This perspective illuminates the synergies that arise when multi-stage, multi-method, and loop-based approaches are combined, offering a blueprint for designing resilient, end-to-end systems that adaptively manage hallucination risks.

D. Summary of Mitigation Methods

The three-dimensional taxonomy of hallucination mitigation—spanning mechanism-based, phase-based, and cross-phase/integrative perspectives—demonstrates the richness and complexity of current approaches. At the mechanism level, strategies range from fine-grained model-intrinsic interventions (e.g., prompt engineering, decoding control, parameter-efficient adaptation) to knowledge-infused methods that ground outputs in external sources, and finally to meta- and framework-based architectures that institutionalize abstention, self-regulation, and multi-agent collaboration. The phase-oriented dimension situates these mechanisms within the generative pipeline, clarifying how pre-generation prevention, in-generation steering, and post-generation verification each address distinct failure modes while incurring different computational and integration trade-offs. The integrative dimension highlights the need to transcend these silos by coordinating signals across stages, combining heterogeneous methods, and embedding iterative feedback loops, thereby aligning mitigation workflows with the temporal and systemic nature of language generation.

Taken together, these observations underscore two core insights. First, hallucination mitigation cannot be reduced to a single intervention point or methodological family; rather, it requires a layered and adaptive orchestration that balances efficiency, coverage, and robustness. Second, the most promising directions lie not in isolated techniques but in the design of holistic systems that continuously propagate grounding, verification, and corrective signals across time and modality. Building upon this synthesis, the next section proposes an *integrative hybrid framework* that operationalizes these insights into a unified pipeline for resilient hallucination control.

V. Proposal: An Integrative Hybrid Framework

Despite extensive research into hallucination mitigation techniques—including prompt engineering, decoding strategies, retrieval-augmented generation, fine-tuning, and multi-agent collaboration—no single method has achieved universal effectiveness across diverse tasks, domains, and modalities. The persistence of hallucinations indicates that existing approaches, while locally effective, remain constrained by systemic limitations and phase isolation. Table II summarizes the advantages and limitations of existing methods, highlighting their context-specific strengths alongside recurring systemic weaknesses. These observations provide the diagnostic basis for our proposed integrative hybrid framework, which seeks to transcend the boundaries of isolated methods and generation phases to achieve more robust, scalable, and adaptive hallucination mitigation.

A. Common Limitations Across Existing Methods

Analysis of current mitigation strategies reveals seven fundamental limitations that persist across methodological boundaries. While individual techniques such as prompt engineering, contrastive decoding, retrieval augmentation, and fine-tuning provide context-specific benefits, they share systemic weaknesses that undermine practical deployment. These weaknesses manifest as bounded effectiveness within narrow task settings, fragility under distributional shifts, and the emergence of new error modes at scale. Most critically, existing methods typically operate in isolation, targeting hallucinations at single generation phases without establishing cross-phase continuity or control.

a) Resource Intensity: Computational overhead severely constrains practical deployment. Contrastive decoding [39], multi-agent debate [262], and chain-of-verification [61] require multi-pass inference, ensemble coordination, or iterative verification cycles—all of which significantly increase latency and cost. Large-scale fine-tuning and RLHF [46] add further training burdens, making them prohibitive for real-time or resource-constrained settings.

b) Data Dependency: Many approaches critically depend on high-quality, domain-specific supervision. Instruction tuning and supervised fine-tuning require curated datasets that are costly to obtain and prone to annotator bias. Synthetic data risks inheriting hallucinations from teacher models, while human-in-the-loop supervision struggles with scalability and inter-annotator inconsistency.

c) Limited Generalization: Methods frequently exhibit weak transferability across domains and tasks. Fine-tuning tends to overfit narrow distributions, prompt engineering is brittle to phrasing variations, and retrieval-based approaches [48], [153] heavily rely on external knowledge quality and timeliness, failing when sources are noisy or outdated. Long-context utilization remains challenging due to attention dilution [106].

d) Phase Isolation: Current interventions target only isolated generation phases—such as pre-generation prompting, in-generation decoding control, or post-generation verification—without establishing cross-phase continuity. This fragmentation allows hallucinations to propagate unchecked across phase boundaries and prevents holistic correction.

e) Training Misalignment: Optimization objectives are often misaligned with factual accuracy. RLHF [46] and DPO [219] can induce reward hacking, verbosity bias, or excessive refusal behaviors that sacrifice informativeness for perceived safety. Knowledge distillation may transfer hallucinations from teacher to student, while standard training objectives still prioritize fluency over factual correctness.

f) Evaluation Inconsistency: Fragmented evaluation practices hinder reliable progress measurement. Metrics vary across studies, benchmark coverage remains narrow, and relationships among truthfulness, factuality, and informativeness are inconsistently defined [272]. This inconsistency obscures genuine methodological advances and risks overstating effectiveness.

g) System Vulnerabilities: Integration of external resources introduces new points of failure. RAG pipelines suffer from retrieval errors, coverage gaps, and latency issues [153], [231]. Tool-based systems face cascading breakdowns when individual components malfunction, while knowledge base drift and evolving task distributions undermine long-term reliability, particularly in dynamic domains where information changes rapidly.

Taken together, these seven limitations highlight enduring tensions—between efficiency and accuracy, between domain specialization and generalization, and between phase-specific interventions and holistic solutions. They underscore why hallucination, as an innate limitation of large language models, cannot be eliminated but only mitigated through adaptive, multi-layered, and cross-phase strategies. This recognition motivates the development of integrative frameworks that move beyond isolated techniques and balance factuality, efficiency, and generalizability in real-world deployments.

B. Proposal: An Integrative Hybrid Framework

The systematic analysis of existing approaches reveals a systemic limitation: while individual mitigation techniques demonstrate effectiveness within constrained settings, their isolated deployment fails to achieve the robustness required for real-world applications. This insufficiency stems from three core architectural problems. First, current methods operate within artificial phase boundaries that create information discontinuities, allowing hallucinations to emerge and propagate between disconnected stages. Second, single-method approaches exhibit inherent vulnerabilities that remain unaddressed when methods operate in isolation, creating systematic weak points that hallucinations exploit. Third, static mitigation strategies cannot adapt to the dynamic nature of hallucination patterns, which evolve across different contexts, domains, and model behaviors.

The recognition that hallucination represents an intrinsic property of large language models rather than an eliminable defect [81] necessitates a shift from elimination-focused approaches to systematic management strategies. This insight drives our framework design toward sustainable, adaptive control rather than unattainable elimination. To overcome the architectural weaknesses identified above, we map each problem to a corresponding design principle, forming the conceptual foundation of our hybrid framework.

a) Cross-Phase Continuity: Traditional approaches treat generation as discrete phases with hard boundaries between input preparation, token generation, and output verification. This segmentation creates information loss at phase transitions, where critical context about uncertainty, evidence quality, and constraint requirements fails to propagate. Our framework eliminates these discontinuities by establishing persistent information channels that carry constraints, uncertainty signals, and corrective feedback throughout the pipeline. This design directly addresses evidence that hallucinations often emerge not from isolated component failures but from information degradation at system boundaries [172], [195].

b) Multi-Method Synergy: Single-method approaches suffer from systematic blind spots where specific error types remain undetected and uncorrected. Retrieval-based methods fail when knowledge bases contain outdated information, while prompt-based approaches become ineffective when models encounter unfamiliar domains. Fine-tuning risks overfitting to narrow training distributions, and post-hoc verification cannot fully recover from unrecoverable generation errors. Our framework orchestrates complementary strategies such that each method's strengths compensate for others' weaknesses, creating robust error detection and correction capabilities.

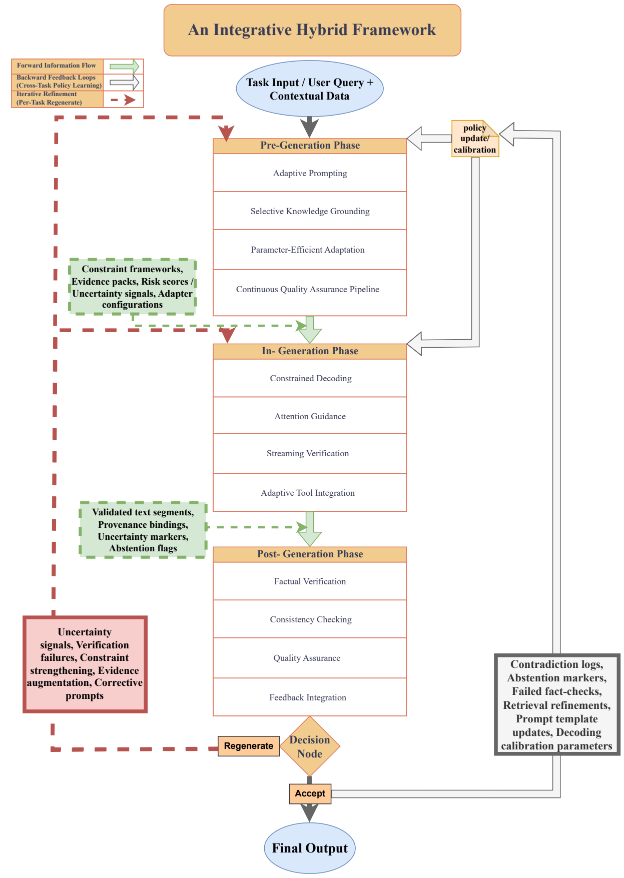

Fig. 8. Overview of the Integrative Hybrid Framework

Redundancy, here operationalized as resilience, ensures that no single failure mode can compromise overall system reliability [33], [156].

c) Iterative Refinement: Static mitigation approaches cannot adapt to the dynamic and context-dependent nature of hallucination patterns. Our framework reconceptualizes mitigation as a continuous optimization process where outputs undergo iterative evaluation, revision, and re-verification until reaching acceptable reliability thresholds. This transforms factual control from one-time correction to adaptive equilibrium maintenance, enabling the system to remain resilient under distributional shifts and novel task demands [143], [221].

1) Framework Architecture: The proposed framework is grounded in a systematic analysis of where and why hallucinations arise within current generation pipelines. Our investigation shows that vulnerabilities cluster around three computational stages, each shaped by distinct information-processing constraints and error-propagation mechanisms. This motivates a three-phase design that positions complementary mitigation strategies at points of maximum leverage while maintaining seamless information flow across stages.

The architectural design rests on several key insights into hallucination dynamics. First, preventive interventions at early stages are generally more effective and computationally efficient than corrective measures applied after generation. Second, token-level generation imposes strict strict real-time constraints, requiring lightweight mechanisms that operate within millisecond budgets. Third, post-generation verification enables comprehensive scrutiny through computationally intensive methods that would be infeasible during real-time decoding. Together, these insights motivate a layered, closed-loop architecture that integrates preventive, real-time, and corrective safeguards into a unified framework for hallucination mitigation, as illustrated in Fig. 8.

a) Pre-Generation Phase: The first phase implements proactive risk reduction under the principle that prevention is substantially more efficient than post-hoc correction. Analysis of hallucination dynamics shows that most errors originate from insufficient information, inadequate knowledge grounding, or misaligned model priors [8]. To address these root causes, the pre-generation phase integrates four complementary components:

- **Adaptive Prompting:** Traditional prompting strategies often apply uniform templates across tasks, disregarding variations in domain risk or query complexity. Our adaptive prompting mechanism dynamically calibrates prompt sophistication based on real-time risk assessment. For routine tasks, baseline prompts incorporate factuality reminders and few-shot demonstrations to maintain efficiency. By contrast, high-risk scenarios—identified through uncertainty estimation or domain classification—trigger enriched prompts that embed explicit verification directives (e.g., "verify claims against evidence before responding") and conditional abstention clauses (e.g., "if supporting evidence is absent, abstain from speculation"). This approach constrains hallucination onset by embedding behavioral safeguards while preserving fluency for low-risk queries.
  Such adaptivity is motivated by the observation that models exhibit heterogeneous hallucination propensities across domains and reasoning types. Medical queries, for instance, demand stricter evidential standards than general knowledge questions, while mathematical tasks benefit from explicit step-by-step reasoning scaffolds. Adaptive prompting operationalizes this differentiated guidance without necessitating separate retraining for each domain, thereby enhancing both reliability and scalability [24], [273].
- **Selective Knowledge Grounding:** Existing retrieval-augmented generation methods often suffer from unfiltered incorporation of evidence, which introduces noise, computational overhead, and even factual conflicts. To overcome this limitation, our framework employs risk-aware evidence retrieval that is selectively activated only when task assessment signals potential knowledge gaps. Retrieved content is then passed through a multi-stage filtering pipeline encompassing relevance scoring, contradiction detection against existing context, and quality assessment based on source authority and recency. Each retained evidence unit is enriched with explicit provenance metadata—including source identity, retrieval confidence, and temporal reliability—providing downstream modules with the ability to distinguish well-supported claims from weakly grounded assertions. By activating retrieval only when necessary and enforcing rigorous filtering, this design mitigates computational inefficiency while ensuring that knowledge augmentation contributes meaningful and verifiable support precisely where it is most needed.
- **Parameter-Efficient Adaptation:** Standard fine-tuning approaches require costly retraining and risk catastrophic forgetting of general capabilities [274]. To address these challenges, our framework employs domain-specific LoRA adapters, which provide lightweight factual biases while preserving the base model's general reasoning capacity. These adapters are trained on balanced datasets containing both factual instances and hallucination-prone samples, enabling them to internalize truth-grounded patterns while explicitly penalizing spurious continuations. In inference, adapter selection is governed by uncertainty-based routing that matches task characteristics with specialized knowledge requirements—for example, biomedical queries activate adapters trained on peer-reviewed literature, whereas technical domains invoke engineering-specific modules. Moreover, the training pipeline incorporates contrastive learning principles, reinforcing factual statements with positive signals while applying negative feedback to hallucination-prone outputs. This design enhances factual grounding, supports domain-specific generalization, and mitigates the tendency of models to produce fluent but incorrect information.
- **Continuous Quality Assurance Pipeline:** Static training approaches cannot adapt to evolving hallucination patterns or emerging domains. To address this, our framework incorporates a continuous quality assurance pipeline grounded in active learning [275]. The system continuously monitors performance across domains and automatically triggers adapter updates when accuracy metrics drop below predefined thresholds. In parallel, high-uncertainty cases—identified through measures of epistemic uncertainty such as dropout-based variance [276] or distributional confidence scoring [277]—are prioritized for expert annotation. By concentrating annotation resources on the most informative samples, the pipeline maximizes the utility of human expertise while avoiding the prohibitive costs of large-scale upfront supervision. This design ensures that the model remains adaptive to shifting domains and resilient to emerging hallucination patterns.

Beyond minimizing hallucination risk at its source, the pre-generation phase establishes a foundation of structured artifacts—refined prompts with embedded factuality constraints, curated evidence packs with provenance metadata, calibrated risk assessments, and domain-specific adapter configurations—that persist into subsequent stages. These artifacts function as constraint-bearing signals and reliability anchors, shaping decoding trajectories and verification processes downstream. By transforming input preparation into a structured channel for information continuity, the pre-generation phase ensures that factual constraints and uncertainty signals are not lost at phase transitions but instead actively guide the in-generation mechanisms that follow.

b) In-Generation Phase: The generation phase operates under unique constraints that distinguish it from pre- and post-processing stages. Token-level predictions must be produced within microsecond timescales while maintaining fluency and coherence, which limits the feasibility of heavy interventions. To address these challenges, our framework introduces a coordinated set of lightweight yet complementary mechanisms, constrained decoding, attention guidance, streaming verification, and selective tool integration, each targeting distinct vulnerabilities in real-time generation while collectively reinforcing factual reliability.

- **Constrained Decoding:** Standard decoding approaches often rely excessively on language-model probabilities, producing fluent but factually incorrect continuations when models follow statistical priors rather than evidential grounding. To address this, we adopt contrastive decoding principles [39], which down-weight tokens favored only by shallow priors while boosting alternatives supported by evidence prepared in the pre-generation phase. This adjustment directly targets the core mechanism underlying many hallucinations: the model's tendency to generate statistically likely but unsupported continuations. By reweighting probabilities, the framework steers outputs toward verifiable claims without sacrificing fluency. In parallel, for tokens flagged as high-risk by uncertainty estimation, the system employs uncertainty-aware beam search [278], maintaining multiple candidate continuations until downstream verification resolves ambiguity. This delayed-commitment strategy prevents premature fixation on hallucinated content and ensures that final token selection reflects both probabilistic plausibility and evidential grounding.
- **Attention Guidance:** Transformer attention mechanisms [279] play a decisive role in determining which input elements influence output, making their alignment critical for factual accuracy [280]. Yet, standard attention often diffuses across irrelevant context or overlooks fact-critical spans, leading to hallucinations despite the presence of correct evidence. To counter this, our framework introduces dynamic biasing that prioritizes entities, dates, numerical values, and other factual anchors identified during pre-generation processing. These bias signals direct attention toward fact-critical elements while preserving flexibility for natural language fluency. For long-context scenarios, we further adopt sliding-window attention strategies that maintain focus on recently referenced evidence and mitigate the "lost in the middle" effect [106], where information buried in extended sequences is neglected. Together, these mechanisms transform attention from a passive weighting scheme into an active safeguard that reinforces factual grounding during generation.
- **Streaming Verification:** Extended outputs pose unique risks because early errors often cascade into subsequent content, compounding factual inaccuracies. Traditional verification methods—either applied only after full generation or uniformly across all tokens—miss opportunities for timely correction and impose unnecessary computational burden. Our framework addresses this challenge through segmented generation, in which each content unit undergoes immediate factual validation before being committed to the final output.
  This design balances accuracy and efficiency by applying strict verification only to fact-bearing segments while allowing routine transitional content to proceed with lightweight checks. When inconsistencies or contradictions are detected, the system initiates immediate regeneration under enhanced constraints, preventing error propagation at its source. By embedding verification directly into the generation flow, this strategy offers a proactive safeguard that achieves higher factual reliability at lower computational cost than purely post-hoc correction methods.
- **Adaptive Tool Integration:** Certain queries demand capabilities that exceed the scope of parametric model knowledge, such as numerical calculation, entity–relation validation, or access to up-to-date information. Existing approaches either bypass external tools—thereby limiting reliability—or invoke them indiscriminately, which introduces latency and computational overhead. Our framework resolves this tension through uncertainty-triggered integration, selectively activating external tools only when model confidence falls below calibrated thresholds.
  The system maintains a modular toolkit comprising calculators for mathematical verification, knowledge graph queries for structured relation checks, and web search modules for recency-sensitive information retrieval. Each invocation is bound with explicit provenance metadata that links results to specific output tokens, thereby creating accountability trails that facilitate downstream verification and explanation.
  A lightweight orchestration controller manages tool selection in real time, balancing accuracy with computational budget while adapting to task-specific requirements. By learning from historical feedback, the controller refines activation policies over time, ensuring that interventions remain both efficient and targeted. This design transforms tool use from a blunt, uniform mechanism into a precision-guided safeguard, reducing hallucination risk while preserving system scalability.

Collectively, the in-generation phase produces structured outputs that extend beyond raw text, embedding layers of reliability metadata into the generation stream. These artifacts include validated text segments annotated with uncertainty scores, provenance bindings that explicitly trace claims to supporting evidence, abstention markers for unresolved content, and diagnostic metrics that capture model confidence and verification outcomes. Rather than treating generation as a self-contained step, this phase transforms it into a reliability-aware process whose outputs are deliberately structured for downstream scrutiny. These artifacts serve as inputs for the post-generation phase, where they undergo deeper evidential validation, logical consistency analysis, and corrective refinement. In this way, the in-generation phase not only constrains real-time generation but also prepares a robust foundation for the verification and repair processes that follow.

c) Post-Generation Phase: While the in-generation phase provides lightweight, real-time safeguards against hallucination, its interventions remain necessarily constrained by latency budgets and token-level granularity. As a result, many hallucinations that are locally plausible or statistically fluent can still escape detection during generation. The post-generation phase addresses this gap by treating model outputs not as final products but as provisional hypotheses requiring rigorous justification before release. Free from the strict latency constraints of decoding, this phase enables computationally intensive verification, consistency analysis, and repair strategies that systematically uncover subtle factual errors, logical contradictions, and unsupported claims. Positioned as the final line of defense, the post-generation phase not only strengthens reliability through multi-tiered verification pipelines but also generates structured feedback that propagates backward, transforming verification outcomes into actionable signals for continuous system improvement.

- **Factual Verification:** The first verification tier systematically evaluates each generated claim against supporting evidence, leveraging provenance metadata established in earlier phases and automated fact-checking methods [281], [282]. This layer addresses a core limitation of large-scale generation systems: human reviewers cannot feasibly validate every statement in real time. To operationalize verification, span-level entailment classifiers [283], [284] assess the logical relationship between generated text and retrieved evidence, distinguishing well-supported assertions from weakly grounded or contradictory claims. When available evidence is insufficient, the system initiates targeted retrieval to acquire additional, claim-specific sources, thereby avoiding the computational burden of exhaustive re-retrieval. Complementing this, entity–relation auditors validate structured factual content by cross-checking named entities, relational links, and temporal attributes against curated knowledge bases. This component is critical for detecting fabricated entities, implausible relationships, and subtle temporal inconsistencies that surface-level entailment checks might miss. Together, these mechanisms ensure that factual verification provides both breadth of coverage and fine-grained reliability, establishing a rigorous evidential foundation for subsequent consistency and quality assurance layers.
- **Consistency Checking:** Beyond factual correctness, high-quality outputs must maintain internal coherence across their entirety. This verification tier addresses the limitation that even factually accurate claims can undermine reliability when they appear contradictory or logically inconsistent. To operationalize this, the framework employs claim decomposition that breaks complex responses into atomic factual assertions, enabling systematic cross-referencing to detect contradictions between different segments of the output. Each decomposed claim is independently re-verified through alternative reasoning paths or paraphrased formulations, producing multiple perspectives on validity and exposing inconsistencies that single-pass checks may overlook.
  Uncertainty calibration further strengthens this process by assigning quantitative confidence scores to disputed or ambiguous claims, integrating factors such as evidence strength, source reliability, claim complexity, and historical performance on similar cases. When contradictions or inconsistencies are detected, the system applies surgical revision strategies that target only the problematic segments rather than regenerating entire outputs. These revisions are guided by evidence-anchored constrained generation, ensuring that corrections remain faithful to verified information while preserving the coherence and fluency of the overall response. Collectively, consistency checking ensures that generated content is not only factually accurate in isolation but also logically and semantically aligned as a whole.
- **Quality Assurance:** The final verification tier safeguards high-stakes scenarios where automated mechanisms may still fall short, ensuring that critical outputs meet stringent reliability requirements. This stage implements an escalated review pipeline that adapts verification intensity to both application demands and content sensitivity.
  Outputs flagged with persistent uncertainty in earlier tiers undergo specialized review using dedicated verifier models trained explicitly for accuracy assessment rather than general-purpose generation. Such models frequently demonstrate superior error-detection capabilities, offering an additional layer of scrutiny beyond standard verification pipelines. In domains where automated confidence thresholds remain unmet, the framework provides structured pathways for human expert involvement. Clear escalation criteria ensure that scarce human resources are reserved for cases where their judgment provides maximum value, such as legally binding, biomedical, or policy-critical outputs.
  Crucially, the quality assurance tier institutionalizes explicit abstention policies: when verification cannot establish adequate confidence, the system defaults to calibrated uncertainty expressions (e.g., "insufficient evidence to verify") rather than speculative claims. This prioritizes reliability over completeness, acknowledging that in many high-stakes contexts, avoiding errors is more important than maximizing coverage. Collectively, these

mechanisms ensure that even in adversarial or uncertain conditions, final outputs maintain integrity, accountability, and trustworthiness.

- **Feedback Integration:** Beyond functioning as a verification endpoint, the post-generation phase acts as a critical driver of continuous system adaptation. Verification outcomes are systematically propagated backward through automated policy-adjustment mechanisms, transforming detected failures into actionable improvements for earlier stages of the pipeline.

  Failed fact-checks prompt updates to prompt templates, embedding stronger factual constraints into future generations. Retrieval strategies are refined through analysis of verification failures linked to inadequate coverage or noisy evidence sources, while decoding parameters are recalibrated to mitigate systematic error patterns observed during verification. These feedback signals ensure that the framework does not merely identify errors but actively learns from them.

  By institutionalizing this continuous feedback loop, the system overcomes the static limitations of existing approaches. It progressively aligns prompts, retrieval modules, and decoding controllers with empirical performance data, enabling dynamic adaptation to evolving hallucination patterns, shifting domains, and emerging application requirements. In doing so, feedback integration transforms hallucination mitigation from a reactive correction mechanism into a self-improving process of sustained reliability.

In summary, the post-generation phase extends hallucination mitigation beyond real-time safeguards, offering rigorous verification, logical consistency checks, and adaptive quality assurance that collectively transform outputs into auditable, trustworthy artifacts. Crucially, its feedback integration ensures that verification outcomes do not remain terminal judgments but actively reshape earlier processes, recalibrating prompts, refining retrieval policies, and adjusting decoding strategies. In this way, the post-generation stage not only serves as the final line of defense but also as a generative source of improvement, feeding corrective signals back into the system. This reciprocal flow establishes the foundation for holistic phase integration and loop closure, where pre-, in-, and post-generation processes function as interdependent layers of a unified architecture rather than isolated modules.

d) Phase Integration and Loop Closure: The framework's effectiveness depends not on isolated modules but on the seamless integration of all phases into a unified, self-corrective system. Existing approaches often suffer from information discontinuities at phase boundaries, which allow hallucinations to propagate unchecked. Our integration strategy resolves this problem through three interconnected mechanisms that collectively establish a closed-loop architecture.

- **Forward Information Flow:** The integration design establishes persistent information channels that preserve and enrich critical context across all stages of the pipeline. In contrast to traditional pipelines, where information flows unidirectionally through lossy transformations, our framework ensures that factual constraints, uncertainty estimates, and evidential signals propagate seamlessly from input preparation to verification.

  In the pre-generation phase, constraint frameworks encoding factuality requirements, evidence availability, and risk scores are embedded into the downstream process through structured prompt modifications, targeted attention guidance, and decoding parameter adjustments. During generation, uncertainty estimates and evidence utilization patterns dynamically shape verification intensity in the post-generation phase: high-uncertainty segments trigger enhanced scrutiny, while well-supported content undergoes streamlined validation.

  This forward propagation eliminates the information loss that typically arises when outputs are analyzed in isolation from their creation context. By maintaining explicit connections to originating constraints and evidence, verification processes can perform more informed and targeted assessments of reliability, thereby aligning efficiency with factual rigor.
- **Backward Feedback Loops:** The framework incorporates systematic learning through automated policy adjustments driven by verification outcomes. Whereas traditional approaches treat generation and verification as disjoint processes, our design establishes a closed-loop adaptation cycle that enables progressive system enhancement over time.

  Failed fact-checks provide structured signals about which prompting strategies, retrieval patterns, or decoding heuristics correlate with reliability breakdowns. These signals are automatically translated into updates—refining prompt templates to embed stronger factual constraints, adjusting retrieval query formulations to target coverage gaps, and calibrating decoding parameters to mitigate systematic error patterns. Retrieval strategies, for instance, are adaptively optimized based on longitudinal analysis of evidence quality and coverage, while decoding calibration responds to observed correlations between uncertainty patterns and verification failures.

  By continuously transforming verification traces into actionable policy updates, the backward feedback mechanism overcomes the static nature of existing approaches. It enables the system to learn directly from real-world deployment performance, adapting dynamically to evolving hallucination patterns, shifting domain distributions, and emerging application requirements.
- **Iterative Refinement:**High-uncertainty scenarios are addressed through multi-pass processing that applies progressively enhanced constraints until outputs converge toward acceptable reliability thresholds. Instead of treating factual control as a binary success/failure outcome, this mechanism reframes it as a convergent optimization process.

  The refinement cycle begins with baseline generation under standard constraints. If verification detects uncertainty or factual inconsistencies, the system regenerates content under stronger constraints—such as tightened decoding thresholds, targeted evidence retrieval, abstention directives, or corrective prompting informed by detected failure modes. Each iteration integrates feedback from previous attempts, incrementally improving factual grounding and logical coherence. This approach reflects empirical findings that complex factual reasoning often requires multiple attempts to achieve stability. Rather than discarding uncertain outputs, the framework applies multi-pass local regeneration with progressively stricter constraints. This short-term refinement complements the global, long-term adjustments achieved through backward feedback loops, together ensuring both immediate reliability and continual system improvement.

The framework integrates preventive, real-time, and corrective strategies into a unified architecture that treats hallucination mitigation not as fragmented interventions but as an inherent, systemic property of the generation pipeline. The pre-generation phase establishes proactive safeguards and constraint-bearing artifacts; the in-generation phase operationalizes these signals through lightweight, real-time controls; and the post-generation phase subjects outputs to rigorous verification and repair. Finally, phase integration and loop closure ensure that information flows seamlessly forward across stages and that feedback flows backward to continuously refine earlier modules. This closed-loop design transforms hallucination control from fragmented, one-off fixes into an adaptive, self-improving process, providing a principled foundation for robust, scalable, and auditable model deployment.

TABLE I. STRUCTURED ARTIFACTS ACROSS PHASES OF MEDICAL QUERY PROCESSING

Phase	Artifacts Generated
Pre-Generation	Enhanced medical prompts with safety directives; FDA/clinical evidence with provenance metadata; high-stakes risk scores and domain classification; biomedical LoRA adapter configuration
In-Generation	Verified treatment segments with confidence scores; drug–document provenance bindings; real-time uncertainty markers; FDA query outputs with citations
Post-Generation	FDA-verified claims with detailed confidence scores; statistical uncertainty ranges contextualized by clinical guidelines; complete source citations; structured feedback for continuous learning

2) Illustrative Case Study: Medical Information Query: To concretely demonstrate the operation of the proposed hybrid framework, we present an illustrative case study in the medical domain—a high-stakes application area where hallucination risks must be systematically managed.

Query Scenario: A healthcare professional poses the question: "*What are the current FDA-approved treatment options for Stage IIIA non-small cell lung cancer in patients with EGFR mutations, and what are the typical response rates for each option?*" This query epitomizes the core challenges of hallucination mitigation: it requires current regulatory knowledge, specialized biomedical expertise, precise numerical reporting, and calibrated handling of uncertainty.

a) Pre-Generation Phase Implementation: The process begins with a comprehensive risk assessment that categorizes the query as high-stakes medical content, thereby activating enhanced safeguards. Domain classification routes the request through biomedical processing pathways, while uncertainty estimation highlights potential vulnerabilities associated with current regulatory status and precise statistical reporting—two areas where parametric knowledge alone may prove insufficient.

To mitigate these risks, the framework integrates four coordinated mechanisms. First, an enriched **adaptive prompt** embeds explicit directives to verify all claims against authoritative guidelines, express uncertainty when evidence is incomplete, and prioritize patient safety by recommending specialist consultation when appropriate. Few-shot exemplars further demonstrate cautious yet informative medical responses, discouraging speculation in sensitive contexts. Second, **selective knowledge grounding** retrieves evidence from curated biomedical repositories—including FDA approval records, oncology trial registries, and peer-reviewed clinical literature—and subjects it to multi-stage filtering for relevance, contradiction detection, and authority-based quality scoring. Each retained evidence unit is annotated with provenance metadata (e.g., source identity, retrieval confidence, and temporal validity), ensuring transparency and accountability. Third, **domain-specific LoRA adapters** trained on oncology corpora are dynamically loaded, incorporating contrastive training that distinguishes between FDA-approved therapies and investigational treatments. This provides reliable factual priors without compromising general reasoning capacity. Finally, **a continuous quality assurance pipeline** monitors performance signals and flags high-uncertainty cases for potential expert escalation, ensuring that annotation resources and human review are directed to scenarios where automated safeguards may be inadequate.

Collectively, the pre-generation phase outputs a structured artifact bundle comprising enriched prompts with embedded factuality constraints, curated evidence packs with provenance metadata, calibrated risk assessments that trigger downstream verification, and domain-specific adapter configurations optimized for oncology content. These artifacts establish reliability-bearing signals that persist into subsequent phases, ensuring that factual constraints and risk indicators are preserved across the generation pipeline.

b) In-Generation Phase Implementation: Token generation proceeds under strict real-time constraints, guided by artifacts established during the pre-generation phase, with continuous monitoring to enforce medical accuracy and calibrated uncertainty expression.

To achieve this, the framework integrates four complementary mechanisms. First, **constrained decoding** applies contrastive principles that down-weight tokens favored solely by statistical priors while amplifying alternatives supported by retrieved medical evidence. For example, when generating treatment names, the system preferentially selects FDA-approved terminology over linguistically likely but potentially incorrect alternatives. Second, **attention guidance** introduces dynamic biasing toward fact-critical entities identified during preprocessing—such as specific drug names (e.g., osimertinib, erlotinib), regulatory designations ("FDA-approved"), efficacy statistics, and clinical staging terminology—thereby preventing diffusion across irrelevant context and ensuring focus on medically salient information. Third, **streaming verification** segments output into smaller units that undergo immediate factual validation; for instance, the claim "osimertinib is FDA-approved for EGFR-positive NSCLC" is cross-checked against FDA databases before continuation. Statistical assertions that cannot be immediately confirmed are annotated with explicit uncertainty markers. Finally, **adaptive tool integration** is triggered when uncertainty thresholds exceed calibrated limits, invoking external resources such as FDA drug databases for regulatory updates and clinical trial registries for response rate verification. All retrieved results are bound to specific tokens via provenance metadata, creating transparent accountability links between claims and supporting evidence.

The in-generation phase thus produces structured outputs that extend beyond raw text, including validated treatment segments annotated with confidence scores, provenance bindings linking claims to authoritative sources, uncertainty markers highlighting content requiring cautious interpretation (e.g., "response rates may vary based on patient characteristics"), and real-time performance metrics tracking medical accuracy. These artifacts serve as reliability anchors for the post-generation phase, enabling more comprehensive verification and repair.

c) Post-Generation Phase Implementation: The post-generation phase functions as the framework's final safeguard, treating outputs as provisional hypotheses requiring rigorous justification, where outputs produced under real-time constraints are treated as provisional hypotheses requiring rigorous justification. Free from latency limitations, this phase applies computationally intensive methods to uncover subtle factual errors, logical contradictions, and unsupported claims that may evade earlier stages. Its primary role is to ensure that responses in high-stakes domains, such as medicine, meet stringent standards of reliability and accountability.

First, **factual verification** systematically evaluates each medical claim against authoritative sources: drug approval statements are cross-checked with FDA databases, efficacy statistics are validated against peer-reviewed clinical trial publications, and treatment protocols are compared with current clinical guidelines. Claims lacking adequate evidentiary support are flagged for revision or removal. Second, **consistency checking** ensures logical and semantic coherence across the response. Reported response rates are cross-referenced to maintain coherent ranges, treatment sequencing information is examined for internal logic, and uncertainty expressions are validated to confirm appropriate scope and specificity. This process prevents subtle contradictions that could undermine trustworthiness even when individual claims appear factually accurate. Third, **quality assurance escalation** addresses residual uncertainty in high-stakes cases. Outputs are reviewed by specialized verifier models trained exclusively for medical accuracy assessment, which provide a deeper layer of scrutiny beyond general-purpose language models. Statistical claims that exceed uncertainty thresholds are flagged for potential expert review, with predefined escalation criteria guiding when consultation with human medical professionals is warranted. This ensures that scarce human resources are applied only where their expertise provides maximum value. Finally, **feedback integration** converts verification outcomes into structured signals for system refinement. Successful fact-checks strengthen retrieval strategies for regulatory and clinical databases, while failures in statistical verification trigger adjustments in data retrieval protocols. Performance metrics from post-generation evaluation further inform ongoing biomedical adapter training. By systematically feeding verification outcomes back into earlier phases, the framework evolves into a self-improving system that continuously enhances reliability in medical domains.

Through these layered mechanisms, the post-generation phase consolidates the framework's reliability. It not only corrects residual errors but also institutionalizes a learning loop that propagates improvements back into earlier stages. In doing so, it converts hallucination mitigation from a static correction process into an adaptive, self-improving cycle, thereby ensuring that outputs remain trustworthy, auditable, and responsive to evolving domain demands.

d) Outcome: The final response produced by the framework is a validated set of FDA-approved EGFR-targeted therapies for Stage IIIA NSCLC, each explicitly linked via provenance metadata to authoritative FDA documentation and peer-reviewed clinical trial evidence. Associated efficacy information is conveyed through bounded response-rate ranges (e.g., 60–80% in clinical trial populations, with variation across patient subgroups), supplemented with explicit uncertainty markers to prevent overgeneralization. Content lacking sufficient evidentiary support is systematically flagged with abstention, ensuring that unverifiable claims are not presented as factual.

In contrast to baseline LLM outputs, which frequently conflate investigational with approved therapies, misrepresent regulatory status, or provide outdated statistics without citations, the hybrid framework demonstrates clear advantages. By integrating provenance trails, calibrated confidence scores, and uncertainty-aware disclaimers, it achieves not only factual accuracy but also transparency and accountability in the reasoning process. Table I summarizes the structured artifacts generated at each phase.

C. Expected Benefits

The proposed hybrid framework provides substantial improvements over existing approaches through five interconnected advantages that directly counter the fundamental limitations observed in current hallucination mitigation strategies.

These benefits collectively transform mitigation from fragmented interventions into a coherent, closed-loop capability.

a) Robust Error Prevention Through Defense in Depth: A primary benefit lies in eliminating the single points of failure that undermine existing methods. Traditional approaches often depend on isolated techniques, leaving them vulnerable when specific assumptions are violated. The proposed framework distributes safeguards across phases, ensuring that deficiencies in one component are compensated by others. When retrieval-based evidence is incomplete or misaligned, prompt constraints and decoding-level interventions maintain control; when real-time decoding constraints preclude sophisticated checks, pre-generation preparation and post-generation verification provide complementary coverage. This layered redundancy design enhances resilience, particularly under adversarial or high-stakes conditions, by reducing the likelihood that hallucinations can bypass all protective mechanisms simultaneously.

b) Computational Efficiency Through Adaptive Resource Allocation: Uniform application of intensive mitigation methods is computationally prohibitive and limits scalability. The framework addresses this challenge through uncertainty-aware resource allocation that activates costly components—such as database queries or expert review—only when risk assessments identify high-stakes or ambiguous cases. Routine, low-risk queries are processed with streamlined safeguards, while complex scenarios receive enhanced verification. This selective allocation enables superior trade-offs between accuracy and efficiency and ensures scalability in large-scale deployments where the majority of queries do not warrant exhaustive scrutiny.

c) Adaptive Learning Through Continuous Improvement: Static methods cannot adapt to evolving hallucination patterns or shifting domain requirements. The framework incorporates continuous feedback integration, transforming verification outcomes into updates for prompts, retrieval strategies, and decoding parameters. This enables incremental refinement without the need for costly retraining cycles. The adaptive learning capability is particularly valuable in dynamic domains such as healthcare or finance, where factual knowledge evolves rapidly. By adjusting system behavior based on deployment experience, the framework maintains effectiveness under changing conditions and progressively strengthens its reliability profile.

d) Transparency and Auditability for High-Stakes Applications: High-stakes contexts require not only accurate outputs but also verifiable and explainable decision processes. The framework ensures transparency through complete provenance tracking, linking every factual claim to supporting sources, and explicit uncertainty quantification that distinguishes between verified, uncertain, and unverifiable content. These mechanisms facilitate informed human oversight, systematic error analysis, and compliance with regulatory standards. Detailed audit trails further support continuous quality improvement by exposing error patterns that can be systematically addressed, making the framework particularly suitable for domains where accountability and trustworthiness are paramount.

e) Modular Extensibility for Future Development: The architecture is designed with modularity that accommodates technological advances and domain evolution. New detection algorithms, mitigation strategies, evaluation metrics, and knowledge bases can be integrated without structural redesign, preserving system stability while enabling incremental enhancement. This design ensures long-term sustainability, allowing the framework to remain relevant as hallucination research and application requirements evolve. Extensibility also applies to deployment configurations and evaluation protocols, enabling the system to adapt to emerging standards and practices without disruptive re-engineering.

In summary, these five advantages provide a systematic response to the seven limitations identified in existing methods: resource intensity, data dependency, limited generalization, phase isolation, training misalignment, evaluation inconsistency, and system vulnerabilities. Although complete elimination of hallucinations remains unattainable due to the inherent characteristics of generative models, the framework establishes a principled foundation for managing them at acceptable reliability levels. By combining robust error prevention, computational efficiency, adaptive learning, transparency, and extensibility, it enables scalable deployment of language models in critical real-world scenarios where factual accuracy is indispensable. A detailed mapping between the seven limitations and the framework's corresponding benefits is provided in Table III, offering a structured view of how systemic weaknesses are systematically transformed into strengths.

D. Summary of the Integrative Hybrid Framework

The integrative hybrid framework advances hallucination mitigation by addressing the systemic weaknesses of existing methods—resource intensity, data dependency, limited generalization, phase isolation, training misalignment, evaluation inconsistency, and system vulnerabilities—through cross-phase continuity, multi-method synergy, and iterative refinement. Its layered design combines preventive, real-time, and corrective safeguards within a closed-loop architecture that not only constrains hallucination onset but also enforces rigorous verification and adaptive learning across deployment cycles. The illustrative medical case study further demonstrates how these mechanisms translate into transparent, auditable, and trustworthy outputs in high-stakes contexts, while the expected benefits highlight its scalability, resilience, and long-term sustainability.

VI. Discussion

This paper consolidates a fragmented body of hallucination research into a structured three-dimensional taxonomy, highlighting the interplay between mechanisms, phases, and integrative strategies. The proposed framework not only clarifies where existing methods converge and diverge but also reveals underexplored intersections that hold promise for advancing robust hallucination mitigation. By tracing the lifecycle of hallucinations and situating detection as the diagnostic backbone of mitigation, our analysis underscores that effective solutions must extend beyond isolated fixes toward system-level resilience. Importantly, the integrative hybrid framework illustrates how cross-phase continuity, multi-method fusion, and iterative refinement can transform scattered techniques into coherent pipelines, paving the way for trustworthy deployment in high-stakes contexts. Nevertheless, despite these contributions, several limitations remain, and new research avenues are urgently required.

A. Limitations

The limitations of this work can be grouped into two categories: those stemming from the current state of mitigation and detection methods, and those inherent to the scope and design of this study itself.

Current hallucination mitigation techniques, though increasingly diverse, remain constrained in several respects. Many approaches are *resource-intensive*, particularly fine-tuning, reinforcement learning, and retrieval-augmented systems, which limit scalability in real-time or resource-constrained environments. Others are *data-dependent*, requiring curated, domain-specific corpora that are not always readily available, thereby hindering transferability across contexts. Detection mechanisms also face a persistent *precision–recall trade-off*: high-precision detectors risk missing subtle hallucinations, whereas high-recall detectors often generate excessive false alarms, leading to overcorrection and diminished generative flexibility. Moreover, most strategies are *optimized in isolation*—either at pre-, in-, or post-generation stages—without addressing cross-phase continuity, resulting in fragmented pipelines. Generalizability is another concern: methods validated on open-domain LLMs frequently degrade in domain-specific, high-stakes, or multilingual tasks where hallucinations pose disproportionate risks. Finally, the lack of *standardized evaluation protocols* exacerbates these challenges, as existing benchmarks capture only partial facets of hallucination and prevent fair, reproducible comparisons across studies.

1) Limitations of This Study: Beyond the limitations inherent to existing methods, this survey itself has several constraints. First, it is largely *conceptual and taxonomic*: while the three-dimensional framework clarifies the landscape, it has not been empirically validated through large-scale experiments. Second, the study is subject to *coverage bias*, given the rapid evolution of the field; emerging directions such as long-context memory architectures, symbolic–neural hybrids, or adaptive retrieval remain underexplored. Third, although the importance of evaluation is emphasized, this work does not provide *concrete guidelines or protocols* for systematically employing benchmarks and metrics, leaving a gap between conceptual synthesis and practical applicability. Fourth, the taxonomy faces *granularity and boundary issues*, as some hybrid methods resist neat categorization and straddle multiple dimensions, which may create ambiguity in application. Fifth, the analysis is *limited in modality and interaction scope*: while LVLMs are included, dynamic multimodal contexts such as interactive video reasoning, embodied agents, and multi-turn dialogues are insufficiently addressed. Sixth, the study does not incorporate a *longitudinal perspective*, overlooking how hallucination patterns evolve across model iterations or in continual learning settings. Finally, while strong emphasis is placed on technical strategies, comparatively less attention is devoted to *socio-technical and governance factors*, including human–AI interaction design, verification workflows, and regulatory implications, which are critical for real-world deployment.

Taken together, these limitations suggest that this work should be regarded as a conceptual foundation and organizing lens, rather than as a definitive or exhaustive solution to hallucination mitigation.

B. Future Works

Building on these limitations, future research should advance in several interrelated directions.

1) Standardized and Actionable Evaluation Protocols: To address the lack of systematic evaluation, future work must establish what we term a *task–metric–protocol triad*: benchmarks that capture task diversity, metrics that integrate factuality, uncertainty, and provenance, and protocols that enforce comparability through unified reporting and version control. This would transform evaluation from a measurement tool into a guiding framework for method development.

2) Large-Scale Empirical Validation: While the current taxonomy offers theoretical clarity, it requires comprehensive empirical validation across domains, modalities, and benchmarks. Developing open reference implementations and testing pipelines will be crucial for moving from conceptual synthesis to practical reliability.

3) Adaptive Hybrid Pipelines: Future systems should explore *adaptive orchestration* of mitigation methods, where detection signals regulate interventions across pre-, in-, and post-generation phases. Such cross-phase pipelines would optimize trade-offs between latency, accuracy, and robustness, ensuring factuality without excessive computational cost.

4) Efficiency-Constrained Mitigation: Scalability challenges demand the design of lightweight and resource-aware solutions—including parameter-efficient tuning, selective retrieval, and training-free interventions—that balance factual alignment with deployability in real-world and real-time settings.

5) Generalization to Domain-Specific and Multilingual Contexts: Future research must explicitly test and adapt methods for high-stakes domains (e.g., medicine, law), underrepresented languages, and multimodal reasoning tasks. Ensuring equitable robustness across such contexts is critical for the societal relevance of hallucination research.

6) Refinement of Taxonomic Boundaries: As hybrid methods increasingly blur categorical lines, future work should pursue meta-framework refinement that embraces fluid boundaries, enabling flexible classification and clearer guidance for method selection in practice.

7) Interactive, Multimodal, and Longitudinal Expansion: Hallucination mitigation must extend beyond static LLMs to *interactive, multi-turn, and multimodal systems*, such as video-grounded reasoning and embodied agents. Equally important is adopting a *longitudinal perspective*, tracking how hallucination behaviors evolve across model versions or in continual learning pipelines. These contexts introduce temporal and contextual complexities that demand novel strategies.

8) Human-Centered and Governance-Aware Design: Future pipelines should embed human-in-the-loop safeguards, interpretable interfaces, and uncertainty calibration, while aligning with governance frameworks for auditability and accountability. This socio-technical integration will be essential for trustworthy real-world deployment.

9) Cost–Benefit and Trade-Off Analysis: Finally, future studies should incorporate systematic cost–benefit analyses, quantifying how different mitigation strategies affect latency, computational load, factuality, and user experience. Such evaluations will guide the design of efficient yet effective pipelines.

C. Concluding Remarks

In summary, this discussion highlights that hallucination mitigation must be conceived not as a collection of isolated fixes but as an evolving ecosystem of methods, benchmarks, and governance practices. By situating mitigation within a three-dimensional taxonomy and proposing an integrative hybrid framework, this study provides a conceptual foundation that can orient both research and practice. Yet the very limitations identified here—ranging from empirical validation gaps and evaluation inconsistencies to scalability, multimodality, and socio-technical concerns—underscore that the journey toward trustworthy generative AI remains far from complete. The future directions outlined above thus represent not only incremental refinements but also an invitation to reimagine hallucination research as a longitudinal, interdisciplinary, and human-centered endeavor. In doing so, the field can move toward systems that are not only technically robust but also socially responsible, ultimately bridging the gap between algorithmic innovation and real-world reliability.

VII. Conclusion

This survey offers a comprehensive and systematized account of hallucination mitigation in large language and vision–language models, repositioning the problem from isolated technical defects to a broader architectural challenge. By advancing a three-dimensional taxonomy—spanning intervention mechanisms, generation phases, and integrative strategies—the study provides a unifying lens through which the field's fragmented techniques can be meaningfully aligned, evaluated, and extended. This taxonomy not only clarifies where existing methods operate but also illuminates the underexplored intersections across phases and modalities, offering principled guidance for building coherent and robust mitigation pipelines.

Crucially, the paper identifies detection not as a peripheral task but as a structural enabler of effective mitigation. Fact-checking, consistency verification, uncertainty estimation, and dual-model feedback are reframed as diagnostic inputs that regulate intervention across pre-, in-, and post-generation phases. However, the current landscape still lacks cohesive detection–mitigation pipelines capable of adaptive refinement and low-overhead integration, particularly under real-world constraints of latency, interpretability, and cross-domain generalization.

To address these challenges, the proposed integrative hybrid framework outlines a cross-phase architecture that enables interaction between preventive, reactive, and corrective strategies. Demonstrated through a high-stakes medical case, this framework exemplifies how layered interventions—guided by detection signals and operationalized through loop-based control—can substantively improve factuality, transparency, and reliability in downstream applications.

Finally, the survey foregrounds the pressing need for evaluation standardization. It calls for benchmark protocols that are sensitive to task structure, modality differences, and abstention behaviors, thereby bridging methodological innovation and practical trustworthiness. As generative models enter increasingly sensitive domains, such standardization becomes critical not only for scientific reproducibility but also for governance, auditability, and alignment with human-centric values.

In sum, this work reorients hallucination mitigation from an assortment of techniques into a structured discipline—anchored in taxonomy, operationalized through integration, and aimed at enabling reliable deployment. It charts a research agenda that prioritizes cross-phase coordination, modularity, and accountability, offering both a theoretical foundation and a practical blueprint for future advances in trustworthy generative AI.

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APPENDIX

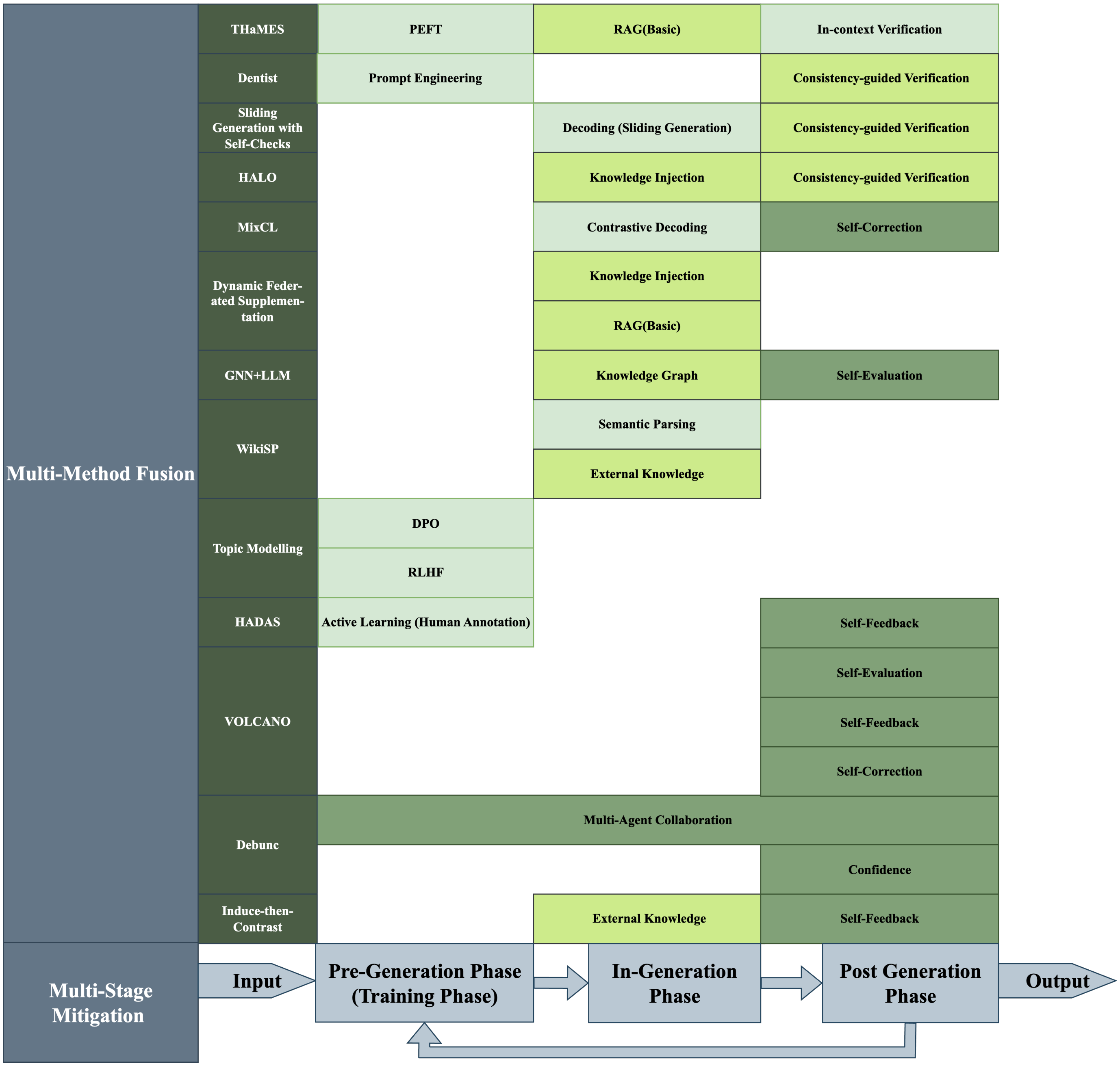

Fig. 9. Cross-Phase and Integrative Mitigation Approaches Summary

TABLE II. Pros & Cons of Exsisting Methods

Method		Pros	Cons
Prompt & ICL	Prompt Engineering	simple to implement; no retraining needed [285]	over-regularize reasoning; limited generalization; brittle to prompt phrasing [170], [286], [287]
	few-shot & zero-shot	flexible adaptation; handles low-resource settings [141], [185]	sensitive to example choice; unstable [141], [185]
Decoding	Sliding Window	handles long context; efficient memory use [193], [195]	reliance on training data; computational load because of iterative refinements [193], [195]
	Tree	structured reasoning; actionable problem solving; modular flexibility [173], [196]	computational complexity; resource heavy [196]
	Attention	better context selection [201]	dependency on quality data; vulnerable to modality priors [163], [201]
	Contrastive Decoding	most are training-free; no reliance on external knowledge [31]	Computationally expensive; needs paired models
	Counterfactual Reasoning	explores alternatives; improves factuality [60]	hard to scale; computational load [60]
	Token Correction via FFN	modular and lightweight correction layer; reduces local errors [179], [192]	limited to token-level errors; no deep reasoning [179]
Fine-Tuning	DPO	aligns with preferences directly; less manual effort [164]	sensitive to preference quality; limited scalability [164]
	RLHF	human preference alignment; improves trust [46], [219]–[221]	expensive; biased by annotators [46], [219]–[221]
	SFT	effective task adaptation; simple pipeline [198], [214]	risk overfitting; limited generalization [198], [214]
	PEFT	efficient [223]	still task-specific; requires careful dataset construction [172], [223]
	Instruction Tuning	improves alignment with tasks; desensitize models to negative prompts [212], [213]	requires large labeled datasets; costly fine-tuning; challenges in generalizability [30]
Active Learning	Human Annotation	high-quality labels; reliable [36]	labor-intensive; expensive [36]
	Automated Annotation	scalable; reduces human effort	noisy labels degrade model
RAG	Basic	simple pipeline [231], [232]	hinges on retrieval quality [231], [232]
	Adaptive	dynamic retrieval; saves resource [184]	complex computation before retrieval [184]
Knowledge	Knowledge Distillation	efficient knowledge transfer [247], [248]	may amplify teacher's hallucinations [247], [248]
	Knowledge Graphs	structured knowledge; strong factual grounding [249]	limited coverage; costly graph construction [254]
	Knowledge Injection	directly embeds facts; lightweight [156]	risk of conflicts; brittle to updates [156]
Divide and Conquer	Consistency Verification	detects contradictions; fits multiple stages	adds latency
	Chain-of-Verification (CoVe)	stepwise validation; boosts reasoning accuracy [61]	costly; may still fail on ambiguous queries [61]
	Subquestion VQA	handles complex queries via decomposition [183]	error propagation from substeps; dependency on quality subquestion generation [183]
Data Standardization & Pre-processing		reduces noise in data [201]	not sufficient alone
Inducing/Adversarial learning (desensitization)		improves robustness; desensitizes to traps [212], [213]	training overhead; adversarial leakage risk [225]
Refusal/Abstention-based Methods (Say No Strategy)		improves safety	over-refusal harms usability
Self	Self-Evaluation	encourages introspection; improves reliability	still self-referential; may reinforce errors [62]
	Self-Feedback (Verification, Revision, Correction)	iterative refinement; boosts factual accuracy	computationally heavy
Multi-LLM Collaboration	Agent	specialization across agents; modular [261]	coordination overhead; inconsistent outputs; requires careful tool selection [66], [261]
	Debate	reliable discussion [27]	may converge to wrong consensus

TABLE III. MAPPING BETWEEN COMMON LIMITATIONS OF EXISTING METHODS AND BENEFITS OF THE PROPOSED HYBRID FRAMEWORK

Common Limitation	**Corresponding Framework Benefit**
Resource Intensity High computational cost of multi-pass inference, ensemble methods, or fine-tuning prevents practical deployment.	**Adaptive Resource Allocation** Uncertainty-aware design activates costly modules (e.g., FDA queries, expert review) only for high-risk cases, while routine queries proceed with lightweight safeguards.
Data Dependency Dependence on expensive, domain-specific supervision; difficulty scaling annotated data.	**Modular Adaptation** Parameter-efficient adapters (e.g., LoRA) and continuous QA pipelines reduce reliance on large curated datasets, enabling targeted retraining only on failure cases.
Limited Generalization Weak transferability across domains and brittleness under distributional shifts.	**Cross-Phase Continuity** Provenance-rich evidence and uncertainty signals persist across phases, reducing brittleness and supporting robustness under domain shifts.
Phase Isolation Interventions confined to single phases allow hallucinations to propagate unchecked.	**Closed-Loop Integration** Forward information flow and backward feedback loops ensure constraints, uncertainty signals, and evidential anchors propagate across all phases.
Training Misalignment Optimization objectives (e.g., RLHF) emphasize fluency or refusal over factuality.	**Feedback-Driven Alignment** Verification outcomes recalibrate prompts, retrieval filters, and decoding heuristics, aligning model behavior with factual accuracy rather than proxy objectives.
Evaluation Inconsistency Fragmented benchmarks and unclear definitions of truthfulness/factuality hinder progress.	**Transparency and Auditability** Complete provenance tracking and explicit uncertainty quantification provide consistent evaluation bases and systematic error analysis.
System Vulnerabilities External resource errors, retrieval failures, or tool malfunctions cause cascading breakdowns.	**Redundancy Across Phases** Multiple safeguards (retrieval, constrained decoding, streaming verification, post-hoc fact-checking) prevent single-point failures from propagating into hallucinations.